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**A three-body waltz: numerical search for recurring
dynamics in the three-body problem**

Soutenue par
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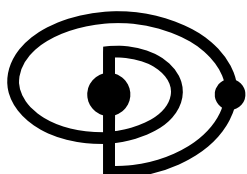
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Résumé

La mécanique céleste décrit les trajectoires dans l'espace des objets célestes, des satellites artificiels et astéroïdes jusqu'aux galaxies. Dans beaucoup de cas, à courte échelle de temps les trajectoires peuvent être approximées par des orbites quasi-périodiques, justifiant un intérêt particulier pour la dynamique récurrente. Dans cette thèse, nous présentons des nouveaux outils numériques pour étudier la dynamique récurrente :

- `TaylorInterface.jl`, une librairie open-source pour utiliser l'intégrateur numérique `taylor` depuis le langage de programmation Julia;
- un algorithme de recherche d'orbites périodiques levant les dégénérescences locales;
- des outils d'étude de la stabilité et des bifurcations des familles d'orbites quasi-périodiques.

En utilisant ces méthodes, nous trouvons la famille verticale d'orbites quasi-périodiques émergeant de l'équilibre relatif de Lagrange dans le problème à trois corps, que nous appelons $\mathcal{VF}_L$. Nous montrons que cette famille conserve une partie stable au-delà de la limite de Gascheau-Routh. Nous apportons des preuves numériques pour la conjecture généralisée de Marchal sur le lien entre $\mathcal{VF}_L$ et la famille P_{12} à masses égales, en montrant qu'elles bifurquent à une inclination mutuelle d'environ 91° . Nous trouvons deux autres bifurcations, non étudiées jusqu'à présent, le long de P_{12} .



Aucune IA générative n'a été utilisée durant cette thèse, que cela soit pour la recherche, la programmation ou l'écriture du manuscrit.

Abstract

Celestial mechanics describe the trajectories in space of celestial objects, from artificial satellites and asteroids to galaxies. Recurring dynamics are of special interest, as a significant amount of body trajectories can be approximated at short time scales by slowly evolving quasi-periodic orbits. In this thesis, we present new numerical tools to study recurring dynamics:

- `TaylorInterface.jl`, a package to call the `taylor` numerical integrator from the Julia programming language, bundled in an open-source package;
- a search algorithm to find periodic orbits and lift local degeneracies through sections.
- tools to study the stability, and bifurcations of families of periodic orbits.

We apply these tools to find the vertical family of quasi-periodic orbits emerging from the Lagrange relative equilibrium of the three-body problem, we call $\mathcal{VF}_{\mathcal{L}}$. We show this family retains a stable part at medium inclination beyond Gascheau-Routh's mass limit. We shine light upon Marchal's generalised conjecture linking $\mathcal{VF}_{\mathcal{L}}$ to the P_{12} family in the equal masses case, showing they bifurcate at a mutual inclination near 91° . We find two other bifurcations along P_{12} , currently unstudied.



No generative AI was used in this PhD, neither for the research, coding or writing of the manuscript.

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Chapter A

Introduction

« Ne tends jamais l'oreille aux musiques des sphères,
N'arrête pas tes yeux sur ces coursiers brûlants :
Rien n'est pour les humains dans la haute atmosphère,
Crois-en mon noir vertige et mon corps pantelant. »

— Comtesse de Noailles, *Les espaces infinis*, in [Les Forces éternelles](#), 1920

A-1 Celestial bodies and their movements: celestial mechanics

Humanity has looked at the stars from the moment it laid foot on earth; various civilisations created astronomical observatories, observed the regularity of star and planetary motions and deduced analytical rules to predict them. From the Babylonians' discovery of the lunar eclipse recurrences and the greek Antikythera mechanism to the Kepler laws of planetary motion, a long tradition of observational astronomy was fundamentally reshaped by the work of Isaac Newton. By proposing gravitation as a universal physical law, from the ground to the sky, explaining and describing analytically the planetary motions, Newton confirmed Kepler's results as a consequence of the gravitational two-body problem, in which two bodies are moving in a vacuum, only affected by their mutual gravity – all this being published in his *Philosophiæ Naturalis Principia Mathematica* in 1687.

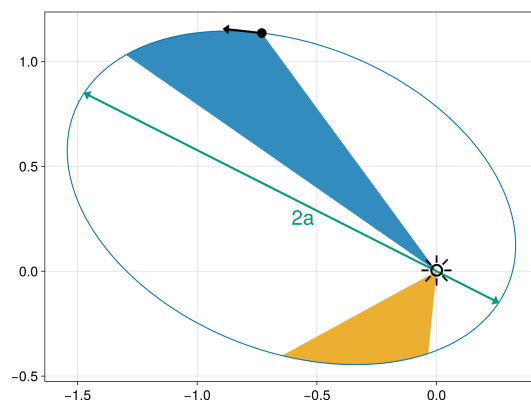


Figure 1: Left: the Antikythera mechanism, found in a shipwreck in 1901 near the greek island of Αντικύθηρα (hence the name), is the oldest known mechanism with gears (dating to), and has often been dubbed “the first computer”. It could predict the positions of several celestial bodies as well as solar and lunar eclipses. Right: the Kepler laws: celestial bodies orbiting around the Sun describe ellipses; the line Sun-body always sweeps an equal area in an equal amount of time ; and their period T and semi-major axis a are linked by $T^2/a^3 = \text{cst}$.

The three-body problem, however, proved to be way more of a challenge, being actively studied up to this day. 80 years after Newton, in 1762 and 1772 respectively, Leonhard Euler and Joseph-Louis Lagrange provided the first examples of periodic solutions: the colinear and equilateral triangle-shaped configurations, known as the Euler and Lagrange points today [Euler \(1764\)](#), [Lagrange \(1772\)](#). The question of these orbits' stability was only solved in the following century: [Liouville \(1842\)](#) proved the instability of the colinear configurations for any triplet of masses, and the following year [Gascheau \(1843\)](#) showed that the triangle-shaped configurations' stability depended on the mass values.

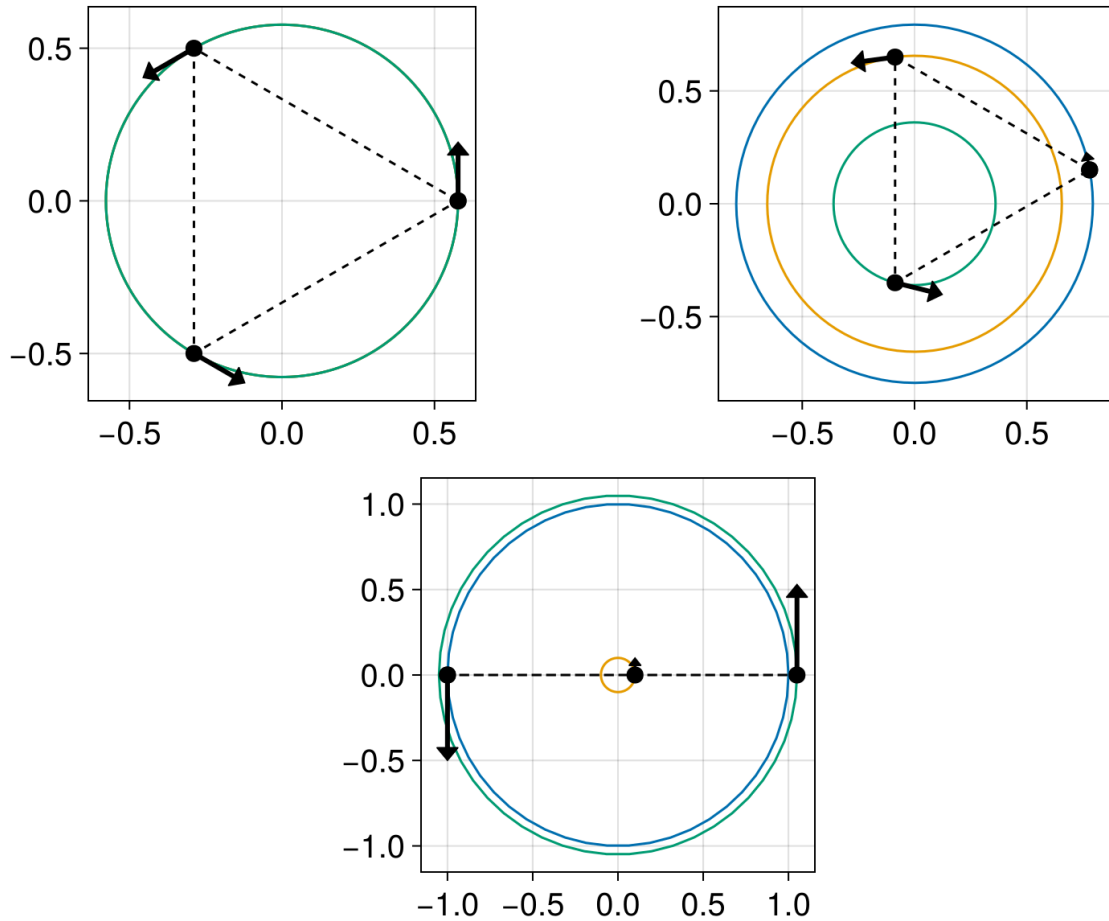


Figure 2: Top: the Lagrange relative equilibrium in the equal (left) and unequal (right) masses case. In the equal masses case, the three circles are identical: the orbit is called a choreography. Bottom: the Euler relative equilibrium.

Finally, the work of Henri Poincaré – especially its “New methods of Celestial Mechanics” ([Poincaré \(1892\)](#)) – kicked off a new way to work on physical differential systems, by focusing on qualitative long-term behaviour instead of exact specific solutions. This was the birth of the mathematical theory of *dynamical systems*. In fact, a study group for this specific book was the starting point of the “Astronomy and Dynamical Systems” group, at the intersection between astrophysics, celestial mechanics and dynamical systems [Chenciner et al. \(2002\)](#); in this group was I given the chance to spend my PhD, whose results you will find in the following pages.

A-2 Investigating recurring dynamics

Dynamical systems can present a large variety of properties of interest to the mathematician; one of the most crucial ones, especially linked to celestial mechanics, is the study of behaviours repeating in time – in other words, recurring dynamics.

The simplest case is the one of a system eternally repeating the same behaviour; for the three-body problem, the three masses would, after a finite amount of time, come back to their initial positions and velocities. We call this a *periodic orbit*, and at short time scales most of the orbits in the Solar System can be approximated by periodic orbits: the Moon’s rotation around the Earth, Earth’s rotation around the Sun, and so on.

However, because of the influence of other massive bodies, these orbits are never exactly repeating, often having multiple periods; we call them *quasi-periodic*. Most motions in the Solar System are in this case: they have multiple periods, and will never come back exactly to a previous state (position and velocity) relative to the Sun.

Investigating such orbits allows us to understand the local dynamics of the system: even though single orbit is only a one-dimensional path in a state space that can be e.g. 18-dimensional for the three-body problem, its properties (periods, stability, etc.) give us precious insight into the surrounding area of the state space. As an example, the stability of the orbits we will see in [Chapter F](#) governs the whole region around it, and could explain the lack of observed small bodies with durably high inclinations (eg. not in a transition phase) in co-orbital regions of the Solar System. This is what motivates us to look for quasi-periodic orbits and investigate their properties.

A-3 Numerical search of recurring dynamics, and a peculiar example

The thesis’ first idea, given by Jacques Fejoz, was to develop a numerical toolbox to investigate recurring phenomena, being as general as possible in the applicable dynamical systems; the three-body problem could serve as a first practical case. The secondary goal was to check the abilities of a relatively new programming language, Julia, which aimed to combine the speed of low-level languages (like C) and the ease of use of higher-level languages (like Python) (see [D-5.1](#) for more details).

The first step was to choose an integrator; after a phase of testing, we chose the `taylor` integrator, a performant high-order integrator... outputting code in C. Thus, I spent the first months building a bridge to use `taylor` from Julia: `TaylorInterface.jl`, an open-source package available on [GitHub](#)¹.

At this time, Philippe Robutel was interested in a peculiar family of quasi-periodic orbits emerging from the Lagrange relative equilibrium; he was trying to generalise the results of [Marchal \(2009\)](#) (see below for more information), which are in the restricted case, to the planetary case. However, by doing an analytical approach which simplified the problem, he

¹<https://www.github.com/Alseidon/TaylorInterface.jl>

was stuck on the family’s stability. This seemed like the perfect case in which to apply a numerical local search of periodic orbits. (a prototype of the algorithm described in [D-5.3](#))

After finding this family at low masses, we started conjecturing a link between $\mathcal{VF}_\mathcal{L}$ and P_{12} , a highly symmetrical family of the equal masses cases, studied among others by Jacques Fejoz. We then discovered Marchal had already made this conjecture in [Marchal \(2009\)](#) (see below for the statement)! However, while P_{12} leads to the Eight – a remarkable periodic orbit –, $\mathcal{VF}_\mathcal{L}$ ends at double collisions for masses $(1 - 2\varepsilon, \varepsilon, \varepsilon)$. Thus the final goal of the thesis, to shine light upon this question – and the search for bifurcations along P_{12} .

A-4 A conjecture by Marchal

The Lagrange triangle (shown in [Figure 2](#)) is not only a periodic orbit, but, seen in the frame rotating with the masses, it is a relative equilibrium: it never changes shape. From this singular orbit emerge, depending on the mass ratios, two or four families of periodic orbits in this rotating frame – meaning they are quasi-periodic (in the general case) in the rest frame. Three of these are planar, remaining in the triangle’s plane. The last one, the vertical family (we will call it $\mathcal{VF}_\mathcal{L}$, for Vertical Family of Lagrange), has the masses shifted normally from the plane and moving in all three dimensions.

Although the vertical family’s existence near the Lagrange orbit has been known for a long time (see e.g. [Moulton \(1920\)](#) for a method to compute the first orbits along the family), it hasn’t been much studied apart from a couple of special cases. Most notably, when all masses are equal, this family – called P_{12} in this case, because of its order 12 symmetry group – leads to another remarkable periodic orbit where all masses follow the same eight-shaped figure (see [Figure 3](#)) – hence its name, the Eight [Chenciner and Montgomery \(2000\)](#).

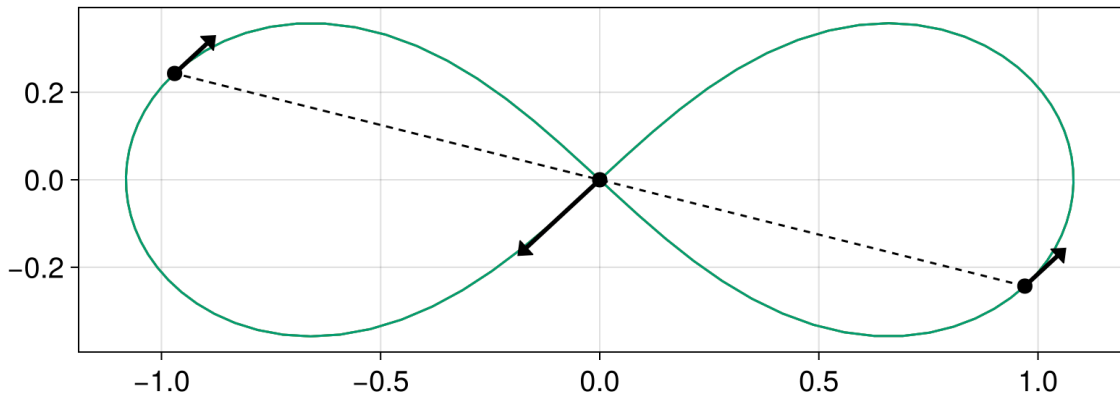


Figure 3: The “Eight” solution. All three orbits follow the same path (we call this a “choreography”), and the total angular momentum is zero.

When the Eight figure was presented, Christian Marchal suspected a link to P_{12} (see [Marchal \(2001\)](#)); this conjecture was only proved in [Calleja et al. \(2024\)](#) with a computer-assisted

proof. In a later article ([Marchal \(2009\)](#)), he studied $\mathcal{VF}_\mathcal{L}$ in the restricted case, when one of the masses is zero. In it, he made the following statement:²

These periodic orbits belong to a three parameter family of similar periodic orbits of the general three body problem, two parameters being the mass ratios of the three bodies and the third a generalization of the inclination i . This family of periodic orbits always begins at the circular Lagrangian motion of the three masses of interest and “escapes” in the normal direction. The one parameter family P_{12} of [Marchal \(2001\)](#) is another limit case of this three parameter family: the case of three equal masses.

We call this Marchal’s generalised conjecture.

A-5 Objective and plan

Our objectives are:

- to develop numerical tools for recurring dynamics, starting with the search for periodic orbits;
- as a first application, to find the family $\mathcal{VF}_\mathcal{L}$ in the general case and investigate its properties, especially stability;
- to find the link between $\mathcal{VF}_\mathcal{L}$ and P_{12} .

In order to do this, we first introduce more formally in [Chapter B](#) the three-body problem and some results about coordinate systems and the Lagrange triangle’s properties that we will use in the following. We then sum up the state of the art about $\mathcal{VF}_\mathcal{L}$ in [Chapter C](#), in the restricted and the equal masses cases. [Chapter D](#) contains the numerical methods we use for integration and root-finding, and the Julia pipeline they are integrated in.

We present our findings about $\mathcal{VF}_\mathcal{L}$ ’s general properties in [Chapter E](#): the preliminary results of Philippe Robutel in the average planetary case, and numerical results in the general case. In [Chapter F](#), we discuss $\mathcal{VF}_\mathcal{L}$ ’s stability and its consequences on the co-orbital region. Finally, in [Chapter G](#) we bring numerical proof for Marchal’s generalised conjecture, as $\mathcal{VF}_\mathcal{L}$ and P_{12} are identical up to a bifurcation; we also show P_{12} ’s other bifurcations.

²Note that, in his work, Marchal calls the orbits periodic even though they are such only in a rotating frame; see [G-1](#) for a more detailed discussion.

Chapter B

Some aspects of the three-body problem

But this third sphere gave “emptiness” life. The three spheres, given initial movements, went through complex, seemingly never-repeating movements. [...] The three spheres continued to dance in my dream, a patternless, never-repeating dance.

— Liu Cixin. *The Three-Body Problem* (2014)

B-1 Presentation

Let us consider three masses $m_j, j \in (0, 1, 2)$ drifting in a vacuum, only affected by their mutual gravity. At any time, the j -th body will have an acceleration of

$$\ddot{\mathbf{q}}_j(t) = \sum_{i \neq j} \mathcal{G} m_i \frac{\mathbf{q}_i - \mathbf{q}_j}{\|\mathbf{q}_i - \mathbf{q}_j\|^3} \quad (1)$$

where $\mathcal{G}$ is a constant and $\mathbf{q}_i$ the position of the i -th body. Writing $\mathbf{p}_i$ the momentum of the i -th body, the system’s Hamiltonian is:

$$H(\mathbf{p}, \mathbf{q}) = \sum_{i=0}^2 \left(\frac{\mathbf{p}_i^2}{2} - \sum_{j=i+1}^2 \frac{\mathcal{G} m_i m_j}{\|\mathbf{q}_j - \mathbf{q}_i\|} \right) \quad (2)$$

B-1.1 Specific solutions

As mentioned in [A-1](#), albeit being defined by Newton in 1687, the three-body problem first explicit solutions found only 80 years later, by [Euler \(1764\)](#) and [Lagrange \(1772\)](#); these are valid for all non-negative mass values. In Euler’s solutions, the three bodies are colinear, whereas for Lagrange’s solution, they form an equilateral triangle. In both cases, the starting configuration rotates around the center of mass, keeping its original form. Euler’s configurations are unstable no matter the mass values (see [Liouville \(1842\)](#)); Lagrange’s triangle stability condition is given by Gascheau:

The movement is stable or unstable, depending on whether the ratio of the square of the masses’ sum and of the two-by-two products of these masses is higher or lower than 27.

— Gascheau (1843)

This amounts to one mass dominating over the other two: see [Figure 4](#).

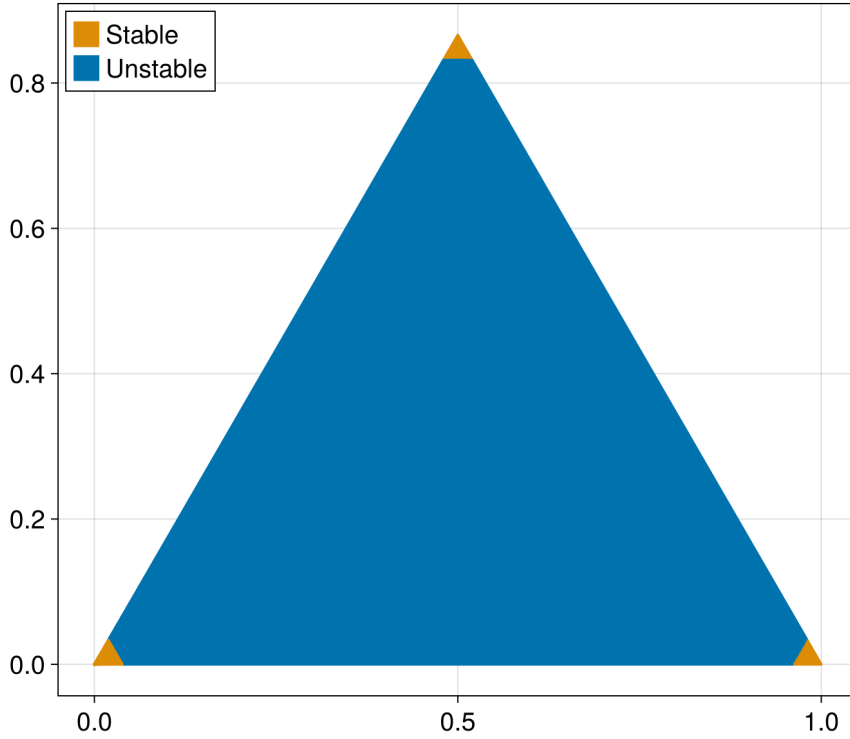


Figure 4: Domain of linear stability of the Lagrange relative equilibrium, depending on the mass ratios. A set of masses (m_0, m_1, m_2) is linked to a point P in the mass triangle $V_0V_1V_2$ through $\vec{P} = \sum_{i=0}^2 \frac{m_i}{M} \vec{V}_i$, with $M = m_0 + m_1 + m_2$. An edge corresponds to the restricted case (one zero mass) and a vertex to two zero masses; the barycenter is the equal masses case. This illustrates the Gascheau condition: one mass must dominate over the other two in order for Lagrange’s triangle to be stable.

When the Lagrange triangle is stable, its *monodromy matrix* has only eigenvalues of norm 1: the orbit is *elliptic* in the sense of stability. When accounting for symmetries and invariances of the problem, four elliptic directions remain, corresponding to four families of quasi-periodic orbits [Meyer and Hall \(1992\)](#) according to the Lyapunov central theorem. The first two families disappear when Lagrange becomes unstable (hyperbolic-elliptic): the “circular” family has the planets librate around the triangle position, and the “anti-Lagrange” family sees the planets describe precessing ellipses of increasing eccentricity as we follow the family, conjectured to end up at collisions. [Giuppone et al. \(2010\)](#), [Hadjidemetriou and Voyatzis \(2011\)](#), [Robutel and Pousse \(2013\)](#)

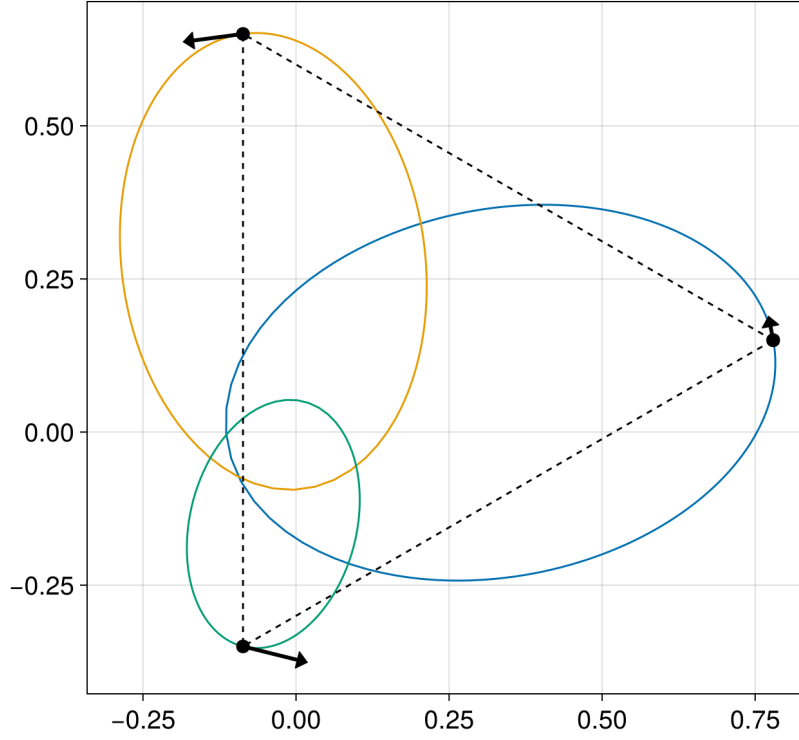


Figure 5: Elliptic, or self-similar, Lagrange orbit. In a rotating frame, the triangle now pulsates.

Two other families remain beyond the Gascheau limit. The first is the self-similar family, or “elliptic” Lagrange: the orbits resemble the Lagrange case, but the bodies follow ellipses instead of circles – making the triangle “pulsate” as it rotate. For its stability, see [Danby \(1964\)](#), [Bennett \(1965\)](#) for the restricted case, and [Roberts \(2002a\)](#), [Nauenberg \(2002\)](#), [Hu et al. \(2014\)](#) for the general case.

The second family is the “vertical family”, and the only spatial one (with non-planar movement). We will discuss it in special cases in [C-1](#) (restricted problem) and [C-2](#) (equal masses); it hasn’t been studied much outside of these.

B-2 Coordinate systems

B-2.1 Canonical heliocentric coordinates

The three-body problem is invariant by translation of the barycenter’s position or momentum; to take advantage of this, we introduce the canonical heliocentric coordinates $(\mathbf{r}, \tilde{\mathbf{r}})$ [Laskar and Robutel \(1995\)](#), defined as such:

$$\begin{cases} \mathbf{r}_0 = \sum_{i=0}^2 \eta_i \mathbf{q}_i \\ \mathbf{r}_1 = \mathbf{q}_1 - \mathbf{q}_0 \\ \mathbf{r}_2 = \mathbf{q}_2 - \mathbf{q}_0 \end{cases}, \begin{cases} \tilde{\mathbf{r}}_0 = \sum_{i=0}^2 \mathbf{p}_i \\ \tilde{\mathbf{r}}_1 = \mathbf{p}_1 - \eta_1 \tilde{\mathbf{r}}_0 \\ \tilde{\mathbf{r}}_2 = \mathbf{p}_2 - \eta_2 \tilde{\mathbf{r}}_0 \end{cases}$$

(3)

with $\eta_i = \frac{m_i}{M}$, $M = m_0 + m_1 + m_2$

Most notably, the barycenter's position is $\mathbf{r}_0$ and its momentum is $\tilde{\mathbf{r}}_0$; as they remain constant, we can ignore them (by setting them to 0) and reduce the problem to six degrees of freedom (twelve dimensions). Note that, for $i \neq 0$, the momenta here are those of the i -th body in the *barycentric* frame (where the barycenter is fixed at the origin), whereas their positions are in the *heliocentric* frame (where the first body is fixed at the origin); this means that the time derivative of $\mathbf{r}_{i \neq 0}$ will not, in general, be tangent to $\tilde{\mathbf{r}}_i$.

In these coordinates, the Hamiltonian is

$$H(\mathbf{r}, \tilde{\mathbf{r}}) = \frac{\tilde{\mathbf{r}}_0^2}{2M} + \underbrace{\sum_{i=1}^2 \left(\frac{\tilde{\mathbf{r}}_i^2}{2\beta_i} - \frac{\mathcal{G}m_0m_i}{\|\mathbf{r}_i\|} \right)}_{\text{Two-body problem}} + \underbrace{\frac{\tilde{\mathbf{r}}_1 \cdot \tilde{\mathbf{r}}_2}{m_0} - \frac{\mathcal{G}m_1m_2}{\|\mathbf{r}_2 - \mathbf{r}_1\|}}_{\text{Perturbation}} \quad (4)$$

where $\beta_i = \frac{m_0m_i}{m_0 + m_i}$

where we usually drop the term in $\tilde{\mathbf{r}}_0$ by setting it to 0.

This Hamiltonian can be seen as the sum of two two-body problems with the mass m_0 and each of the bodies, perturbed by an interaction term. This is especially interesting in the *planetary* case, where one mass (m_0) is dominant over the other two: we take masses $(m_0, \varepsilon m_1, \varepsilon m_2)$ with $\varepsilon \ll 1$ and m_i of order 1. In heliocentric canonical coordinates and rescaling both action variables and time by a factor ε (see eg. [Robutel et al. \(2016\)](#)), the Hamiltonian reads:

$$H(\mathbf{r}, \tilde{\mathbf{r}}) = H_K(\mathbf{r}, \tilde{\mathbf{r}}) + \varepsilon H_P(\mathbf{r}, \tilde{\mathbf{r}}) \text{ with}$$

$$\begin{cases} H_K(\mathbf{r}, \tilde{\mathbf{r}}) = \sum_{i=1}^2 \left(\frac{\tilde{\mathbf{r}}_i^2}{2\beta_j} - \frac{\mu_j \beta_j}{\|\mathbf{r}_i\|} \right) \text{ and } \mu_j = \mathcal{G}(m_0 + \varepsilon m_j) \\ H_P(\mathbf{r}, \tilde{\mathbf{r}}) = \frac{\tilde{\mathbf{r}}_1 \cdot \tilde{\mathbf{r}}_2}{m_0} - \mathcal{G} \frac{m_1 m_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \end{cases} \quad (5)$$

In the special case where one of the masses is exactly zero, we call this the *restricted* problem; the two massive bodies describe ellipses in an unperturbed two-body problem, and we follow the test particle's orbit under their influence.

B-2.2 Poincaré-like coordinates

We now introduce symplectic Poincaré-like coordinates related to the invariant plane – the plane perpendicular to the total angular momentum and containing the most massive body (see [Figure 6](#)). They are defined as:

$$\begin{aligned} \Lambda_j &= \beta_j \sqrt{\mu_j a_j}, \tilde{\lambda}_j = M_j + \omega_j \\ x_j &= \sqrt{\Lambda_j - G_j} \exp(i\omega_j), \tilde{x}_j = -i\bar{x} \\ \Psi_1 &= G_1 \cos I_1 + G_2 \cos I_2, \psi_1 = \frac{\Omega_1 + \Omega_2}{2} \\ \Psi_2 &= G_1 \cos I_1 - G_2 \cos I_2, \psi_2 = \frac{\Omega_1 - \Omega_2}{2} \end{aligned} \quad (6)$$

$$\text{with } G_j = \Lambda_j \sqrt{1 - e_j^2} \quad (6)$$

where, for $j \in \{1, 2\}$ the planet number, a_j is its semi-major axis, e_j its eccentricity, I_j its inclination, M_j its mean anomaly, ω_j the argument of its pericenter, and Ω_j the longitude of its ascending node. $\bar{x}$ represents the complex conjugated of x and $i = \sqrt{-1}$.

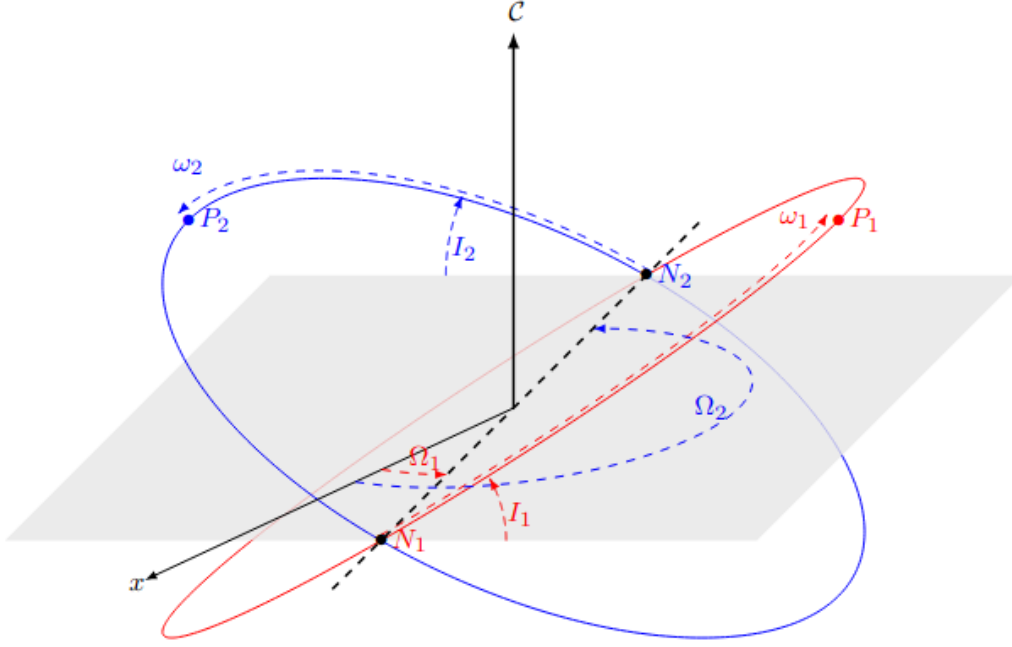


Figure 6: Coordinate system linked to the invariant plane (perpendicular to the total angular momentum and containing the most massive body).

These coordinates are especially useful in perturbative cases, where one might develop in powers of a given orbital element: Λ_j are proportional to the square roots of semi-major axes, and $x_j, \tilde{x}_j$ are proportional to the eccentricities (at first order).

B-2.3 Jacobi's reduction

The invariance of the problem under rotation around the axis of the total angular momentum allows us to reduce the number of degrees of freedom by eliminating inclinations and ascending nodes – in other words, we perform the Jacobi reduction [Robutel \(1995\)](#), [Mastroianni and Efthymiopoulos \(2023\)](#).

As in this reference frame the ascending nodes of the two planets are in opposition, $\psi_2 = \frac{\pi}{2}$ and its conjugated variable Ψ_2 are ignorable. Similarly, the action Ψ_1 being equal to the modulus of the total angular momentum, ψ_1 is also ignorable. As a result, the Hamiltonian depends only on 8 of the variables expressed in the coordinates of [Equation 6](#): $(\Lambda_j, \tilde{\lambda}_j, x_j, \tilde{x}_j)_{j=1,2}$ – plus an additional parameter $\Psi_1 = \mathcal{C}$. Since these latter variables do not involve the angles Ω_j but only M_j and ω_j , using these coordinates amounts to considering the trajectories in a reference frame rotating with the common line of the nodes of the planets. This rotating frame will henceforth be referred to as the draconic frame.

The ascending nodes being in opposition, the mutual inclination J between the two planets is equal to the sum of the individual inclinations: $J = I_1 + I_2$. Thus, the modulus of the total angular momentum and the mutual inclination are related by the following expression:

$$\mathcal{C}^2 = G_1^2 + G_2^2 + 2G_1G_2 \cos J. \quad (7)$$

Let us finally note that, although the inclinations and longitudes of the nodes have been eliminated by Jacobi's reduction, it is still possible to retrieve them using:

$$\mathcal{C} \cos I_1 = G_1 + G_2 \cos J, \quad \mathcal{C} \cos I_2 = G_2 + G_1 \cos J. \quad (8)$$

Similarly, the time derivative of the ascending nodes' longitudes are given by:

$$\dot{\Omega}_1 = \dot{\Omega}_2 = \frac{\partial H}{\partial \mathcal{C}}, \quad (9)$$

where H is the Hamiltonian of [Equation 5](#) expressed in the variables given by [Equation 6](#).

B-2.4 Hill coordinates

In the Poincaré coordinates used before, the Hamiltonian and its associated vector field cannot be explicitly expressed without truncating in eccentricity. To avoid this limitation, we must introduce coordinates that allow an explicit expression of the Hamiltonian and are adapted to the Jacobi reduction³.

The Hill variables [Laskar \(2017, and references therein\)](#), given by $(r_j, w_j, \Omega_j, R_j, G_j, G \cos I_j)_{j=1,2}$ satisfy this double constraint. The draconic true anomalies w_j , longitudes of the nodes Ω_j , angular momenta G_j and vertical angular momenta $G_j \cos I_j$ (I_j being the inclinations) of each planets have already been introduced in [B-2.2](#), while r_j and R_j are respectively the distance between the bodies of masses m_j and m_0 , and the norm of their linear momentum in the barycentric frame. Similarly to [B-2.3](#), we perform the Jacobi reduction and eliminate the nodes and vertical momenta, but keep the angular momentum $\mathcal{C}$ as a parameter. As a consequence, 8 dimensions remain corresponding to the coordinates: $(r_j, w_j, R_j, G_j)_{j=1,2}$ to which the parameter $\mathcal{C}$ has to be added. We can express the positions and momenta in the canonical heliocentric coordinates as:

$$\mathbf{r}_j = R_z(\Omega_j)R_x(I_j)R_z(w_j) \begin{pmatrix} r_j \\ 0 \\ 0 \end{pmatrix}, \quad \tilde{\mathbf{r}}_j = R_z(\Omega_j)R_x(I_j)R_z(w_j) \begin{pmatrix} R_j \\ G_j \\ \frac{G_j}{r_j} \\ 0 \end{pmatrix} \quad (10)$$

where for $\theta \in \mathbb{R}$, $R_u(\theta)$ is the rotation matrix of angle θ along axis u :

³It is possible to perform the Jacobi reduction eg. in cartesian coordinates, see eg. [Hadjidemetriou and Voyatzis \(2011\)](#) in the planar case; however, the resulting Lagrangian can be cumbersome to use.

$$\begin{aligned}
R_x(\theta) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \\
R_z(\theta) &= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned} \tag{11}$$

The planetary Hamiltonian of Equation 5 can now be written in Hill variables. As $\tilde{\mathbf{r}}_j^2 = R_j^2 + \frac{G_j^2}{r_j^2}$ and $\mathbf{r}_j^2 = r_j^2$, its Keplerian part H_K is trivially translated in Hill coordinates. In order to express the perturbative part in terms of Hill variables, it suffices to note that the dot product $\tilde{\mathbf{r}}_1 \cdot \tilde{\mathbf{r}}_2$ can be written as:

$$\tilde{\mathbf{r}}_1 \cdot \tilde{\mathbf{r}}_2 = \left(R_x(w_1) \begin{pmatrix} R_1 \\ \frac{G_1}{r_1} \\ 0 \end{pmatrix} \right) \cdot \left(R_x(-I_1) R_z(\Omega_2 - \Omega_1) R_x(I_2) R_x(w_2) \begin{pmatrix} R_2 \\ \frac{G_2}{r_2} \\ 0 \end{pmatrix} \right) \tag{12}$$

As $\Omega_2 - \Omega_1 = \pi$, the mutual inclination J is the sum of the two individual inclinations. Thus, a final straightforward calculation establishes the relation

$$R_x(-I_1) R_z(\pi) R_x(I_2) = R_z(\pi) R_x(J) \tag{13}$$

which leads to the final expression:

$$\begin{aligned}
\tilde{\mathbf{r}}_1 \cdot \tilde{\mathbf{r}}_2 &= \left(\cos w_1 \tilde{r}_1 + \sin w_1 \frac{G_1}{r_1} \right) \left(-\cos w_2 \tilde{r}_2 + \sin w_2 \frac{G_2}{r_2} \right) \\
&+ \cos J \left(\sin w_1 \tilde{r}_1 - \cos w_1 \frac{G_1}{r_1} \right) \left(\cos w_2 \frac{G_2}{r_2} - \sin w_2 \tilde{r}_2 \right)
\end{aligned} \tag{14}$$

Using the same transformation for the dot product $\mathbf{r}_1 \cdot \mathbf{r}_2$, the planetary Hamiltonian is expressed in Hill variables as follows:

$$\begin{aligned}
H_C(r_j, w_j, \Omega_j, R_j, G_j) &= \sum_{i=1}^2 \left(\frac{m_0 + \varepsilon m_i}{2m_0 m_i} \left(R_i^2 + \frac{G_j^2}{r_j^2} \right) - \mathcal{G} \frac{m_0 m_i}{r_i} \right) \\
&+ \frac{\varepsilon}{m_0} \left[-\cos w_1 \cos w_2 R_1 R_2 - \sin w_1 \sin w_2 \frac{G_1 G_2}{r_1 r_2} \right. \\
&\quad \left. + \cos w_1 \sin w_2 \frac{G_2 R_1}{r_2} + \sin w_1 \cos w_2 \frac{G_1 R_2}{r_1} \right. \\
&\quad \left. + \cos J \left(-\sin w_1 \sin w_2 R_1 R_2 - \cos w_1 \cos w_2 \frac{G_1 G_2}{r_1 r_2} \right. \right. \\
&\quad \left. \left. - \sin w_1 \cos w_2 \frac{G_2 R_1}{r_2} - \cos w_1 \sin w_2 \frac{G_1 R_2}{r_1} \right) \right] \\
&- \varepsilon \mathcal{G} m_1 m_2 (r_1^2 + r_2^2 - 2r_1 r_2 \cos w_1 \cos w_2 - 2r_1 r_2 \sin w_1 \sin w_2 \cos J)^{-\frac{1}{2}}
\end{aligned} \tag{15}$$

with

$$\cos J = \frac{C^2 - G_1^2 - G_2^2}{2G_1 G_2} \quad (16)$$

Note that, even though the longitudes of the nodes do not appear explicitly in the expression of the Hamiltonian, we can recover the node precession (the two node precessions are equal, as $\Omega_1 = \Omega_2 + \pi$) by integrating Equation 9 along a trajectory $\mathbf{x}(t)$ by:

$$\forall t, \Omega(t) = \Omega(0) + \int_0^t \frac{\partial}{\partial \mathcal{C}}(H_{\mathcal{C}}(\mathbf{x}(\tau))) d\tau \quad (17)$$

B-2.4.1 Limit cases

The Jacobi reduction has two limit cases: the planar case ($J = 0$) and the restricted case ($m_j = 0$ for one j).

In the planar case, the nodal line – the intersection of the orbital plane and the invariant plane – is ill-defined, as the planes are identical. However, no singularity appears in the Hamiltonian, nor in the associated vector field; therefore, we can still use the Hill variables in this case. Let us remember that in this case the only relevant angle is the true longitude $\Omega_j + w_j$.

The restricted limit, where one of the masses is zero, is not as smooth: the Jacobi reduction simply cannot be used there, as the zero-mass body isn't constrained by the direction of the total angular momentum. Thus, we restrict ourselves to low, but non-zero masses.

B-2.4.2 The Lagrange relative equilibrium

As indicated above, Jacobi reduction still allows us to consider coplanar motions and, in particular, Lagrange's relative equilibria. Thus, in Hill coordinates, the quantities $(r_j, R_j, G_j, w_1 - w_2)$ are constants given by:

$$\begin{aligned} R_1 &= -\frac{\sqrt{3}\omega\rho\epsilon m_1 m_2}{2(m_0 + m_1 + m_2)}, R_2 = \frac{\sqrt{3}\omega\rho\epsilon m_1 m_2}{2(m_0 + m_1 + m_2)}, r_1 = \rho, r_2 = \rho, \\ G_1 &= \frac{\omega\rho^2 m_1 (2m_0 + m_2)}{2(m_0 + m_1 + m_2)}, G_2 = \frac{\omega\rho^2 m_2 (2m_0 + m_1)}{2(m_0 + m_1 + m_2)}, w_1 - w_2 = \frac{5\pi}{3} \end{aligned} \quad (18)$$

where ρ is the radius of the circles described by the two planets, which can be considered a scale factor, and ω is the planetary mean motion. These two quantities are, of course, not independent but linked by Kepler's third law. Contrary to their difference, the true anomalies w_1, w_2 are not constant. Indeed, they satisfy the first order differential equation:

$$\dot{w}_j = \omega \left(a + b \sin \left(2w_j + (-1)^j \frac{\pi}{3} \right) + c \cos \left(2w_j + (-1)^j \frac{\pi}{3} \right) \right) \quad (19)$$

where a, b, c are constants depending only on the three masses (see appendix III-1) that satisfy the relation: $a^2 - b^2 - c^2 = 1$. Thanks to this last relation, it can be shown that the solution to this differential equation takes the form: $w_j(t) = \omega t + F(\omega t)$, where F is 2π -

periodic. Although the w_j are pulsating (due to $F(\omega t)$), the physical angles to consider in the planar case are the $\Omega_j + w_j$, which can be shown to be exactly equal to ωt .

The spectral stability of the relative equilibrium described above can be deduced from the eigenvalues of the monodromy matrix of its variational equation. The corresponding spectrum is given by:

$$\begin{pmatrix} 1 \\ \exp\left(\pm \frac{i\omega}{\sqrt{2}} \sqrt{1 - \sqrt{1 - \frac{27p}{\sigma^2}}}\right) \\ \exp\left(\pm \frac{i\omega}{\sqrt{2}} \sqrt{1 + \sqrt{1 - \frac{27p}{\sigma^2}}}\right) \end{pmatrix} \quad (20)$$

where $\sigma = m_0 + m_1 + m_2$ and $p = m_0 m_1 + m_0 m_2 + m_1 m_2$. The eigenvalue 1 is of multiplicity 4: two because of the Hamiltonian's time invariance, and two because of the homographic family. We can also find the Gascheau criterion here: if $\frac{27p}{\sigma^2}$ is greater than 1, the root becomes complex, and the four non-1 eigenvalues aren't of modulus 1 anymore; the Lagrange relative equilibrium goes from being elliptical to hyperbolic-elliptic and thus linearly unstable.

B-3 Averaging

In perturbative cases of the three-body problem (e.g. when one mass is very small, or far away compared to the other two), variables evolve over different time-scales. Averaging over fast-varying variables (angles) can then allow us to transform into an *average* problem.

For instance, far away from resonances, we can average the Hamiltonian over the anomalies $\tilde{\lambda}_j$ of the Poincaré coordinates; this is the *secular* problem, with only slow time scales. Near semi-major axes resonances, we can average over one angle only (see eg. [E-1.1.2](#), or resonant normal form theory [Meyer and Hall \(1992\)](#)). This amounts to consider the ellipses and their deformations, instead of following the bodies' instantaneous eclipse along them.

Chapter C

State of the art: Lagrange’s vertical family, $\mathcal{VF}_{\mathcal{L}}$

La famille... C’est compliqué.

— *Clair Obscur : Expedition 33*; directed by Guillaume Broche,
Sandfall Interactive/Kepler Interactive, 2025

C-1 The restricted case

C-1.1 Marchal 09: the family in the average restricted problem

While researching the orbital behaviour of Trojan asteroids (asteroid near the triangular Lagrange points) of Jupiter, [Marchal \(2009\)](#) described a spatial family of periodic orbits in the frame rotating with Jupiter, for the average restricted three-body problem. This family links L_4 to L_5 ; at high inclination (around 145.46°), it meets with a similar vertical family emerging from the colinear Euler relative equilibrium, L_3 (we call it $\mathcal{VF}_{\mathcal{E}}$, for vertical family of Euler); the meeting point (the bifurcation) is illustrated in left of [Figure 7](#). They appear to be linearly stable in the restricted problem, although Marchal only considers the direction of longitudes.

In this paper, Marchal says that the orbits emerging from L_3 “belong to the family of symmetric periodic orbits discovered by [Zagouras and Markellos \(1985\)](#)”; we believe he actually meant to quote [Zagouras and Kazantzis \(1979\)](#). This pointed us to look at [Zagouras \(1985\)](#), which already describes the L_4 to L_5 family in the restricted problem, extending the results of chapter IX of [Moulton \(1920\)](#), written by Thomas Buck! This family has thus been described for a century, and its end (connection of L_4 and L_5 , crossing with the family from L_3) for fourty years. [Figure 8](#) illustrates the orbits from Buck’s and Zagouras’ findings.

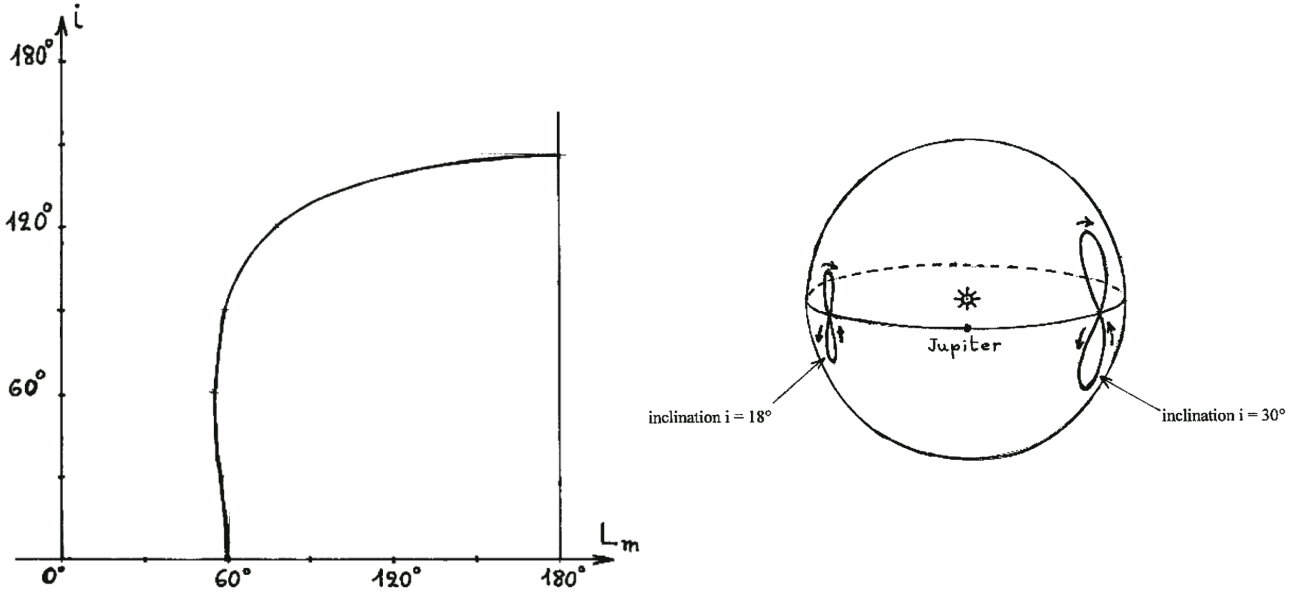


Figure 7: Left: Marchal's vertical family in the (difference of mean longitudes, inclination) plane. Right: two example orbits in the frame rotating with Jupiter. Figures 3 and 4 of [Marchal \(2009\)](#)

C-1.2 Zagouras 85: an algorithm to approximately find the family

In [Zagouras \(1985\)](#), a method to find a parametric expansion for the orbits of $\mathcal{VF}_L$ is described; the same method was used, albeit to a lesser order, in [Moulton \(1920\)](#). Everything is done in the restricted problem, in the barycentric frame rotating with the massive bodies (also called the primaries), with L_4 as the origin.

We introduce a new time variable $\tau = 1 + \sigma$; then, we develop the coordinates x, y, z and the time scale σ in powers of a small parameter ε . Then, we inject the series in the equations of motion and equate the coefficients of a given power of ε . For each power, we obtain the same ordinary differential equation with only the right member changing, depending on the lower powers; thus, we can recursively compute the coefficients of x, y, z and τ .

This expansion is obtained up to order 4 in ε , and used as a start to numerically continue the family with a predictor corrector algorithm (described in Zagouras 1982, unavailable). Like Marchal, Zagouras finds that the bifurcation between $\mathcal{VF}_L$ and $\mathcal{VF}_\varepsilon$ has been studied for instance in [Bray and Goudas \(1967\)](#), [Zagouras and Kazantzis \(1979\)](#). He also shows the family is linearly stable up until a given inclination, where it becomes unstable. Finally, at the stability change point, they find a bifurcating family that goes to collisions.

The analytical methods used here could also be used in the non-restricted problem to have an initial guess for the family near the Lagrange triangle; however, the significant increase in the dimension number, as well as the heavy degeneracy of the differential system make the task significantly more difficult.

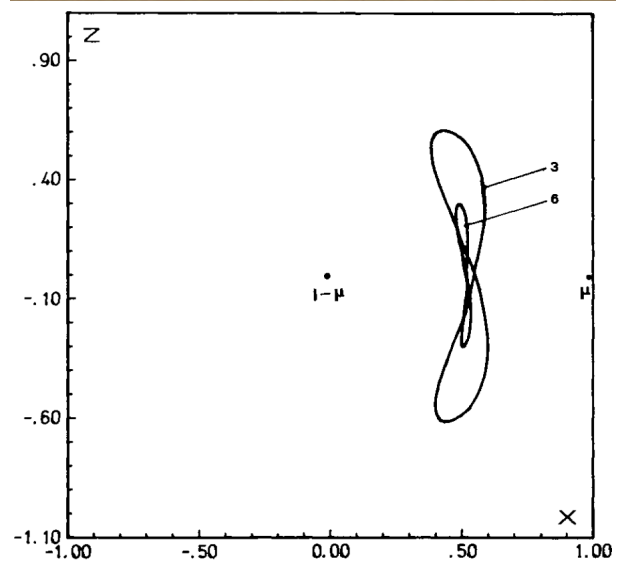
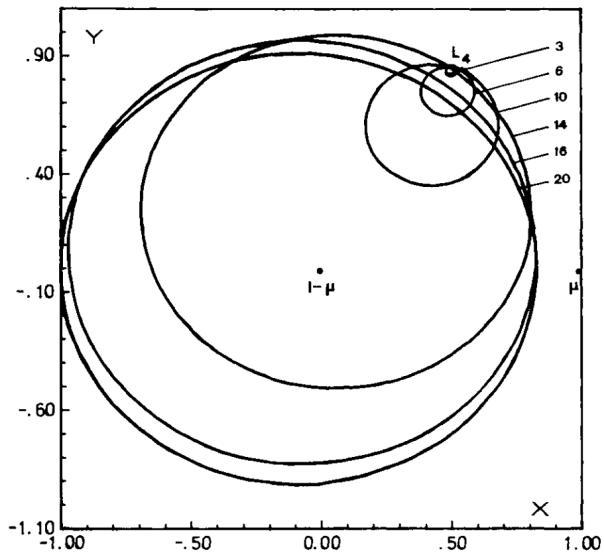
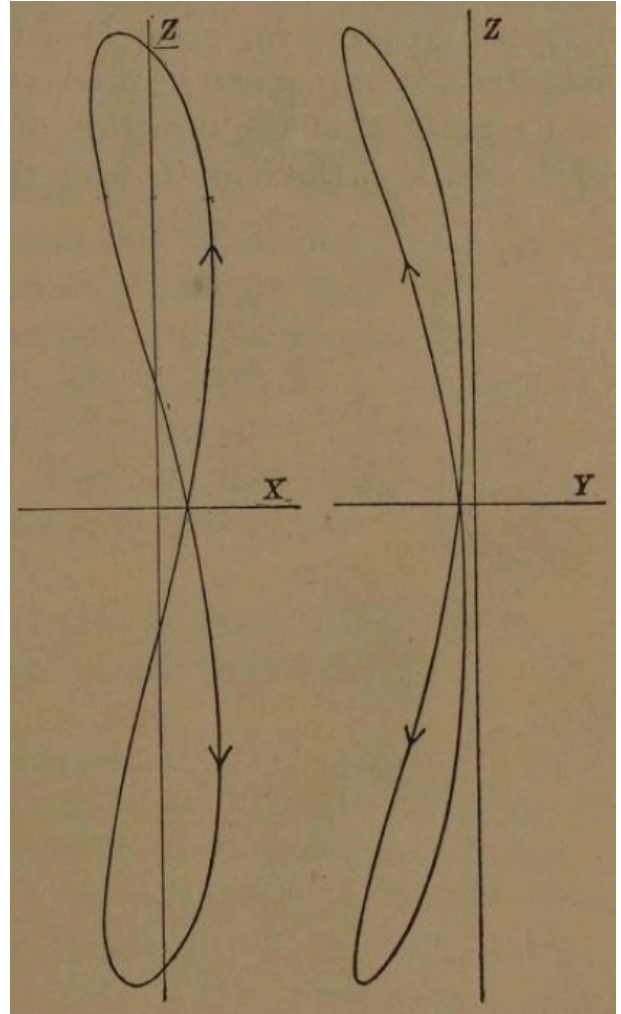
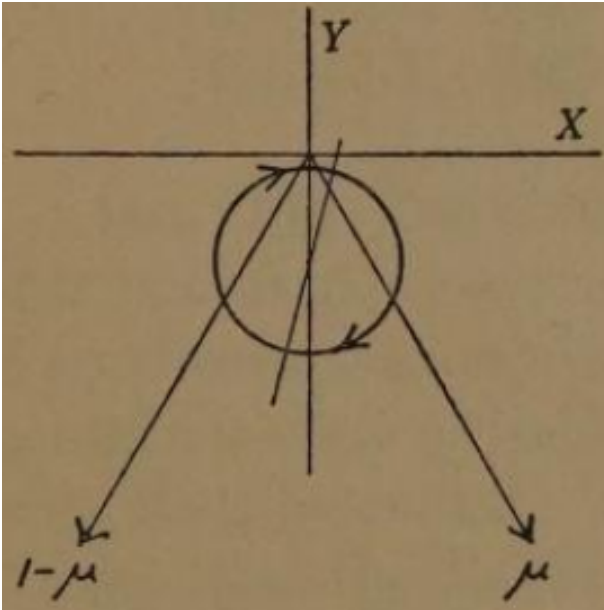


Figure 8: Top: figures 5 to 7 of chapter IX, by Thomas Buck, of [Moulton \(1920\)](#). Bottom: figures 2a and 2b of [Zagouras \(1985\)](#). Both show approximated example orbits of $\mathcal{VF}_{\mathcal{L}}$.

C-2 The equal masses case: P_{12}

C-2.1 Marchal 90: describing the family's symmetries near Lagrange

In [Marchal \(1990\)](#), Christian Marchal presented “space-time symmetries”: configurations that would lead to identical orbits (up to trivial transformations) with normal or opposed initial velocities. There are four types of such cases, shown in [Figure 9](#) (see [Marchal \(1990\)](#) or [Marchal \(2001\)](#) for exact descriptions).

Although he attributed them to Poincaré, and the last two are in [Poincaré \(1884\)](#), he couldn't find a source for the first two.

With them, we can construct periodic solutions; indeed, if we link two symmetric configurations S_1 and S_2 by a trajectory, at times 0 and θ :

- S_1 's symmetry gives us the orbit for times $[-\theta, \theta]$;
- S_2 's symmetry then gives us the orbit for times $[-\theta, 3\theta]$;

And so on, until we construct the whole orbit. [Poincaré \(1884\)](#) was the first to use this method; Marchal used it to find a maximally symmetric family of quasi-periodic orbits (periodic in a rotating frame); it presented the maximal amount of space-time symmetries per period possible in the three-body problem, twelve. This family was later called P_{12} , because of its symmetry group of order 12.

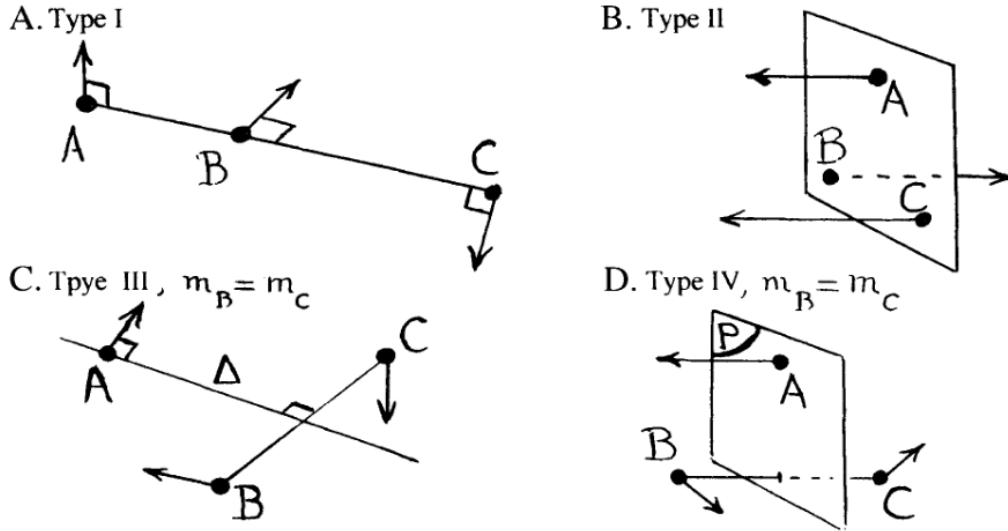


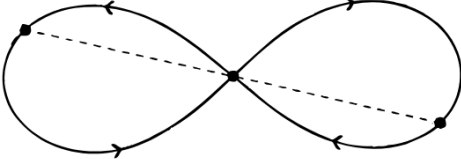
Figure 9: The four types of space-time symmetries. Source: figure 3 of [Marchal \(2001\)](#), reproducing a figure of [Marchal \(1990\)](#).

C-2.2 The Eight figure and Marchal's conjecture

In [Chenciner and Montgomery \(2000\)](#), Chenciner and Montgomery presented a remarkable orbit of the three-body problem with equal masses: the three bodies follow the same planar eight-shaped orbit, hence the name “Eight” – [Moore \(1993\)](#) actually numerically found this orbit a few years before. This solution has a zero total angular momentum and is highly

symmetrical: in fact, its order 12 symmetry group was the same as the one of P_{12} (see Figure 10 for the symmetry group).

EIGHT ($\exists$ equal masses in $\mathbb{R}^2$)



$$G = \{g_1, g_2 \mid g_1^6 = g_2^2 = 1, g_1 g_2 = g_2 g_1^{-1}\}$$

$$\cong D_6$$

	g_1	g_2
τ	$t \mapsto t + T/6$	$t \mapsto -t$
σ	$\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}$	$\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array}$
ϑ	$(x, y) \mapsto (x, y)$	$(x, y) \mapsto (-x, -y)$

Figure 10: Symmetry group of the Eight. Source: figure 2 of [Chenciner \(2006\)](#).

Christian Marchal, according to [Fejoz \(2010\)](#), was quick to notice this fact and conjectured that the Eight was part of the P_{12} family, a conjecture formalised in [Marchal \(2001\)](#). Although numerical evidence backed this conjecture (see eg. [Chenciner and Féjoz \(2009\)](#)), the formal computer-assisted proof was only achieved twenty years later, in [Calleja et al. \(2024\)](#).

The Eight's stability was investigated by Kapela and Simó; they proved it was linearly stable in [Kapela and Simó \(2007\)](#), and even KAM stable in [Kapela and Simó \(2017\)](#).

[Doedel et al. \(2003\)](#) showed continuations of the Eight for unequal masses (two equal masses, one different), finding a continuation up to the restricted problem (with bifurcations along the way) with two equal masses – which isn't the problem [Marchal \(2009\)](#) studied, where the mass ratio was 10^{-3} . As far as we know, this is the only continuation of the Eight for unequal masses.

Chapter D

Numerical methods: searching for a periodic needle in a dynamical haystack

A computer lets you make more mistakes faster than any other invention with the possible exceptions of handguns and Tequila.”

— Mitch Ratcliffe

D-1 Framing the problem

In this section, we give an overview of the mathematical problem we approach numerically ; this is in no way meant to be exhaustive on the matters of dynamical system and recurring dynamics.

D-1.1 Recurring dynamics

Let us consider a continuous-time dynamical system described by an ordinary differential equation (ODE) on a manifold M :

$$\dot{x}(t) = f(x(t)) \quad (21)$$

We will call $\Phi_t : x(0) \mapsto x(t)$ its local flow at a time $t \in \mathbb{R}$. A solution of [Equation 21](#) $x : A \rightarrow M$ (with A an interval of $\mathbb{R}$ containing 0) is called a *trajectory*, and its image in M is an *orbit*.

Recurring dynamics focus on trajectories which lead the system “close to” previous states. The simplest such case is of course an initial state for which the system will, after a non-zero amount of time, strictly come back to:

$$\exists T > 0, x(T) = x(0) \quad (22)$$

Then, if the system is *autonomous* (independent of time), we have

$$\forall t \in \mathbb{R}, x(t + T) = x(t) \quad (23)$$

This means the system follows a closed loop in the space of its possible configurations, with a period T ; then, there exists a function $p : \mathbb{T} \rightarrow M$ such as

$$\forall t \in \mathbb{R}, x(t) = p(t * \omega), \omega = \frac{2\pi}{T} \in \mathbb{R} \quad (24)$$

ω is the *frequency* of the system (physicists would call it the *pulsation*, because of the 2π factor). x is a *periodic orbit*.

In a more general case, we can have multiple frequencies, and we can write the orbit as

$$\forall t \in \mathbb{R}, x(t) = p(t * \omega), \omega \in \mathbb{R}^k, p : \mathbb{T}^k \rightarrow M \quad (25)$$

If the frequencies in ω are solutions of a Diophantine equation: $\exists v \in \mathbb{Z}^k \setminus \{0\}, \omega \cdot v = 0$, this amounts to a single frequency, and thus a periodic orbit. If that isn't the case, however, x is called a *quasi-periodic* orbit: it will describe a trajectory dense in a subspace of the configuration space, *without ever* coming back to its original state (albeit coming arbitrarily close to it). In the limit case where ω is in $\mathbb{T}^{\mathbb{N}}$, thus being an infinite sequence, the orbit is called *almost periodic*.

We will focus on *periodic orbits*: as they are fixed points of Φ_T , this is our way of numerically looking for them⁴. The three main issues we have to solve are:

- computing the function Φ_T : the challenge of numerical integration, treated in [D-2](#);
- finding fixed points of Φ_T , which are roots of $\Phi_T - I$: methods for root-finding will be seen in [D-3](#);
- solving the variational equations to study the stability of the orbits: we will see this in [D-2.4](#)

D-2 Numerical integration

D-2.1 The initial value problem

The initial value problem consists, given an initial condition x_0 , the final time T and the function f , in computing Φ_T . In other words, we want to compute $x(T)$ such as:

$$\begin{cases} x(0) = x_0 \\ \dot{x}(t) = f(x(t)) \quad \forall t \in [0, T] \end{cases} \quad (26)$$

D-2.2 Taylor integration

D-2.2.1 The method

In the Taylor method, we approximate $x(t)$ by its Taylor development at successive time steps t_i ⁵:

$$x(t) = x(t_i) + \sum_{k=1}^m \frac{1}{k!} \frac{d^{(k)}x}{dt^k}(t_i) (t - t_i)^k + \mathcal{O}((t - t_i)^{k+1}) \quad (27)$$

The time derivatives of $x(t_i)$ can be computed with [Equation 26](#), using a generalisation of Faà di Bruno's formula for higher order derivatives of composite functions, with an intermediate vector argument – see eg. [Mishkov \(2000\)](#). We get:

$$\forall k \in \llbracket 1, m \rrbracket, j \in \llbracket 1, n \rrbracket, \quad (28)$$

⁴Later, we will search for quasi-periodic orbits in the same way, because the orbits of interest are periodic in a rotating frame. By using the equations of motion in said rotating frame, we can use the same methods

⁵Of course, this requires that x and f be sufficiently smooth. Moreover, the Taylor method isn't suited for so-called stiff ODEs (for instance, if a high-order derivative is unexpectedly high).

$$\begin{aligned}
 \frac{d^{(k)}x_j}{dt^k}(t_i) &= \frac{d^{(k-1)}(f_j \circ \mathbf{x})}{dt^{k-1}}(t_i) \\
 &= \sum_0 \sum_1 \cdots \sum_{k-1} \left[\frac{(k-1)!}{\prod_{a=1}^{k-1} (a!)^{r_a} \prod_{a=1}^{k-1} \prod_{b=1}^n q_{ab}! \prod_{b=1}^n \partial x_i^{p_b}} \frac{\partial^{(R)} f_j}{\partial x_i^{p_b}}(\mathbf{x}(t_i)) \right. \\
 &\quad \left. \times \prod_{a=1}^{k-1} \prod_{b=1}^n \left(\frac{d^{(a)}x_b}{dt^a}(t_i) \right)^{q_{ab}} \right]
 \end{aligned} \tag{28}$$

where the sums are over all non-negative solutions of the Diophantine equations:

$$\begin{cases} \sum_0 \rightarrow \sum_{a=1}^{k-1} ar_a = k-1 \\ \sum_{a \neq 0} \rightarrow \sum_{b=1}^n q_{ab} = r_a \end{cases} \tag{29}$$

and the orders of the derivatives are:

$$\begin{cases} \forall b \in [1, n], p_b = \sum_{a=1}^{k-1} q_{ab} \\ R = \sum_{a=1}^{k-1} r_a = \sum_{b=1}^n p_b \end{cases} \tag{30}$$

As derivatives of order k only depend on derivatives of order $k-1$ or lower, we can recursively compute derivatives of arbitrarily high order (up to numerical precision).

Injecting Equation 28 into Equation 27, we have an approximation of $\mathbf{x}(t)$ near time t_i , provided we know $\mathbf{x}(t_i)$. We also can have an estimation of the error we make by neglecting the rest $\mathcal{O}((t-t_i)^{k+1})$, if we compute the $(k+1)^{\text{th}}$ derivative of $\mathbf{x}(t)$. We can now choose the next time-step t_{i+1} such that the error is below a certain criterion, and repeat until we reach the final time T .

The precision can be controlled with two parameters: the maximum derivation order k , and the step size $t_{i+1} - t_i$; combined with the estimation of the error, this allows for refined error control mechanisms, suitable for high-precision integration.

For instance, in the `taylor` implementation we will discuss later, the strategy is described in [Jorba and Zou \(2005\)](#): to give an overview, they try to maximise the step size and minimise the order (to achieve maximum speed), using a property of [Simó \(2001\)](#) that gives the optimal parameters given the convergence radius of the Taylor polynomial, and how fast the coefficients decrease with order. The key part then becomes estimating accurately these two new values, which we will not discuss here.

D-2.2.2 Comparison with Runge-Kutta

Despite its capabilities, the Taylor method has seen less use than other, more popular methods – most notably the widely used Runge-Kutta methods, which find the next step by evaluating the integrand at multiple in-between steps. A major reason is the implementation complexity, as the method needs high-order derivatives; moreover, because the method is recursive, the automatic differentiation engine needs to be closely combined with the integrator, allowing re-use of the intermediate coefficients – see [Tan et al. \(2026\)](#) for more details.

The main advantage compared to Runge-Kutta methods in our case is the superior efficiency at high precision; see [Jorba and Zou \(2005\)](#) for some benchmarks. In the same paper, Jorba shows a possible reason: in order to maximise speed and minimise the number of arithmetic operations, the main parameter to tweak is the order of derivation, not the step size. Thus, even adaptative step-size methods would fall short when the required precision increases drastically – in the case of the N body problem, this can be needed e.g. with close approaches.

D-2.3 Differentiation

A major question for implementations of this method is how to compute the partial derivatives of a vector function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Schematically, there are three ways of computing derivatives of a function without writing them explicitly: numerical differentiation, symbolic differentiation and automatic differentiation.

Numerical differentiation is the easiest one. We approximate the derivatives as the slope of the function graph's tangent:

$$\frac{\partial f_j}{\partial x_i}(x) \approx \frac{1}{\delta}(f_j(x + \delta e_j) - f_j(x)) \quad (31)$$

for a sufficiently small δ , and writing $f = (f_1, \dots, f_m)$, $e_j = (\delta_{ij})_{i \in \llbracket 1, n \rrbracket}$. The crucial part is the choice of dx : too big and we might miss small-scale variations, too small and numerical errors might appear, in the division by δ or in evaluating $x + \delta e_j$ when x is big.

Symbolic differentiation is what a human would do: it reads the function's definition as a mathematical expression, of which it computes the formal derivative. The first problem is inefficiency: the derivative expression can explode in size, and be inefficient to compute – too many non-simplified mathematical operations possibly leading to numerical evaluation errors. The second problem is more conceptual: not all computer programs easily lead to mathematical expressions, especially when control flows appear (e.g. for and while loops, if statements).

Automatic differentiation, or AD, operates in a way similar to symbolic differentiation, but instead of a mathematical expressions, it sees the function as a series of arithmetic operations and elementary functions: additions, subtractions, multiplications, divisions, trigonometric functions, exponential, etc. It computes the derivative by repeatedly applying the chain rule as it evaluates the function.

Because AD is “theoretically” correct (compared to numerical differentiation) and leads to more efficient expressions than symbolic differentiation, it is best suited to our needs.

Let’s take first order automatic differentiation of f at a point $x \in \mathbb{R}^n$ as an example: we replace our vector of numbers x with :

$$\forall j \in \llbracket 1, n \rrbracket, x_j = x_j^0 + x_j^1 \varepsilon_j \quad (32)$$

where x_j^k are numerical coefficients and ε_j are infinitesimals. Each numerical operation in the function modifies the coefficients: for instance, the function $x \mapsto \cos(x_1 + x_2)$ would lead to

$$\cos(x_1 + x_2) = \cos(x_1^0 + x_2^0) - x_1^1 \sin(x_1^0 + x_2^0) \varepsilon_1 - x_2^1 \sin(x_1^0 + x_2^0) \varepsilon_2 \quad (33)$$

The final output $y = f(x)$ will then be:

$$\forall j \in \llbracket 1, m \rrbracket, y_j = y_j^0 + \sum_{l=1}^n y_{j,l}^1 \varepsilon_l \quad (34)$$

The coefficients $y_{j,l}^1$ form a $m \times n$ matrix, the Jacobian of f . This works beyond first order derivative: if we completed Equation 33 and Equation 34 with higher orders of ε_l , the coefficients of order k would give us the derivatives at this order.

Note that here, we are using what is called *forward mode* AD, meaning we start the differentiation from the input x and propagate the differentiation *forward*. When the *output’s* number of dimensions is significantly lower than the input’s – like in machine learning –, *reverse mode* differentiation is more efficient: the chain rule is propagated *backwards*, from the output.

D-2.4 Assessing stability

D-2.4.1 Linear stability: characteristic coefficients

Beyond its use in the Newton-Raphson algorithm, the computation of the Jacobian of Φ_T (where T is the period), which is the solution of the first-order variational equation, can also give us insights about the orbit’s stability. Following Meyer and Hall (1992)’s nomenclature, we call it the *monodromy matrix*, and its eigenvalues are the *characteristic exponents* of the orbit. They come as quadruplets $(\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1})$ with $\lambda \in \mathbb{C}^*$. If all eigenvalues are of modulus 1, the orbit is elliptic and thus linearly stable; otherwise, it is unstable, and either hyperbolic (if none of the eigenvalues have an modulus of 1) or hyperbolic-elliptic (if some do).

D-2.4.2 Beyond linearity: long-term and analytical stability

Studying linear stability has its limits: a linearly unstable orbit can’t be stable, but linear stability doesn’t guarantee complete stability. To investigate, further tools are needed.

For instance, we can construct higher order normal forms in the orbit’s neighbourhood. We can also cite KAM theory, which allows proofs of existence of stable tori (of quasi-

periodic orbits) under perturbations of a system, if said perturbations are small enough. The conditions for analytical KAM theory can however prove very constrained: for the case of the Solar System, Michel Hénon estimated that for the system to be stable, the mass ratios should be inferior to 10^{-320} ! (see [Laskar \(2014\)](#))

Otherwise, if using numerical methods, several indicators of stability (and chaos) exist: Nekhoroshev estimates, around Lyapunov exponents, or linked to KAM theory, like frequency analysis [Laskar \(1999\)](#).

D-2.4.3 Integrating the variational equations

Automatic differentiation can also be used to compute the derivatives of Φ_T , by using the initial values of [Equation 32](#) instead of x_0 for Φ_T , and propagating them in the integration; this gives us the monodromy matrix (which is just the first-order derivative), which will be needed in the following parts, and could also compute higher order derivatives.

In the general case (where the coefficients $x_{i,j}^l$ can have different initial values from $\delta_{l,1}\delta_{i,j}$), *jet transport* is the flow of variational equations (of some order) of the initial ODE. Its phase space is the space of (Taylor) polynomials.

However, this has a computational cost. In the following sections, as we only need the first-order derivative in 9 dimensions, the computation time is multiplied by a factor of circa 10.

D-2.5 Implementation: `taylor`

`taylor`, first presented in [Jorba and Zou \(2005\)](#), is a software which reads a set of equations in a C-adjacent format and outputs highly efficient C code to integrate the corresponding ODE with the Taylor method. It is also capable of integrating the variational equations ([Gimeno et al. \(2022\)](#)) up to an arbitrary order for jet transport.

We will showcase a brief overview; the full documentation can be found in the [PDF manual](#)⁶. First, we need to write in a file `equations.txt`:

⁶<https://github.com/joang/taylor2-dist/blob/main/src/manual/manual.pdf>

```

1  const1 = 3.14;
2  const2 = 42.;
3
4  var1 = 2*x + const1;
5
6  diff(x, t) = px;
7  diff(px, t) = var1 + const2 * x * x;
8
9  jet x, px symbols 2 deg 1;

```

Listing 1: Example syntax for the taylor equation files. This example was taken from the [documentation](#)⁷ of TaylorInterface.jl, written by the author. The code first defines constants and variables; the state variables are declared through their time derivatives `diff(x, t)`. Finally, we declare that the two variables should have jet transport of degree 1.

Then, we can compile the file to C code. We can then ask taylor to produce a valid C file:

```

user@machine:~$ taylor -o code.c equations.txt
user@machine:~$ █

```

Compiling and running the file would look like this:

```

user@machine:~$ gcc code.c -lm
user@machine:~$ ./a.out
Enter Initial xx[0]: 1.
Enter Initial xx[1]: 0.
Enter start time: 0.
Enter stop time: 0.1
Enter absolute error tolerance: 0.1e-5
Enter relative error tolerance: 0.1e-5
1 0 0
1.015084027816637 1.200743783742144 0.0252397488068032
1.060392373730684 2.452268939926622 0.0501599633443915
1.136842424384952 3.818858010796434 0.0746562355514192
1.242398382185734 5.320264280331545 0.09786144538432054
1.253938096840076 5.472244331254157 0.1
user@machine:~$ █

```

Of course, entering values manually isn't the usual way of using taylor; the taylor stepper and its associated header can be written in specific files:

```

user@machine:~$ taylor -name example -o code.c -jet -step equations.txt
user@machine:~$ taylor -name example -o code.c -header equations.txt
user@machine:~$ █

```

The first and second lines create the files `code.c` and `code.h` respectively, with a stepper function called `taylor_step_example`. We add the `-jet` flag because we specified in `equations.txt` that we wanted jet transport.

We can now use these files anywhere we like, by linking them to the rest of our code. We can even ask taylor to generate a simple main function for this, with

⁷<https://research-alex.prieur.eu/TaylorInterface.jl/dev/quickstart/>

```

user@machine:~$ taylor -name example -o main.c main_only equations.txt
user@machine:~$ gcc main.c code.c -lm
user@machine:~$ █

```

The main way to call the taylor-generated code is through the function `taylor_step_auto`, shown in [Listing 2](#) – note that the explanations for each parameter are simplified. For instance, the step control flag [Listing 2-7](#) can be used to loosen the control conditions, provide a user defined control, or no control at all.

```

1  // MY_FLOAT is the numerical type chosen by the user, usually
   double
2  // In this case, MY_JET would be double*
3  int taylor_step_auto(
4      MY_FLOAT *ti,          // start time
5      MY_FLOAT *x,          // input/output of the state variables
6      int      dir,          // time direction to integrate towards
7      int      step_ctl,     // step control to use, usually 2
8      double   log10abserr,  // log10 of absolute error required
9      double   log10relerr,  // log10 of relative error required
10     MY_FLOAT *endtime,     // end time
11     MY_FLOAT *ht,          // outputs the time step used if not null
12     int      *order,       // outputs the degree used if not null
13     MY_JET   *jetInOut     // input/output of jet variables
14 )
15 // returns 0 if succeeded, 1 if succeeded and reached the endtime,
16 // -1 if an error occurred

```

Listing 2: Prototype of the main function for the numerical integrator. More details can be found in the taylor manual and in [Jorba and Zou \(2005\)](#).

Depending on the features used, other functions will need to be manually called. Most notably, jet transport, if used, needs initialisation through `taylor_initialize_jet_library`, and jet variables can be initialised with `taylor_make_identity_jets` if needed. Some numeric types also require initialisation, for instance with the MPFR library for arbitrary precision.

Note that, apart from the Julia interface presented below in [D-5.2](#), taylor offers Python and FORTRAN bridges.

D-3 Root-finding

D-3.1 Newton-Raphson method

The multi-dimensional Newton-Raphson method is a widely used method to find zeros of functions numerically. Its idea is very simple: iteratively approach the zero by approximating the function with an affine line.

The algorithm is shown in Figure 11 for a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

```

1 function newton-raphson( $f, f', \varepsilon, x_0$ )
2    $x \leftarrow x_0$ 
3   while  $|f(x)| > \varepsilon$ 
4      $h \leftarrow \frac{f(x)}{f'(x)}$ 
5      $x \leftarrow x - h$ 
6   end while
7   return  $x$ 
8 end
    
```

Figure 11: Newton-Raphson algorithm in pseudocode.

When f is a vector field (mapping $\mathbb{R}^n$ to $\mathbb{R}^n$), we replace line 4 of Figure 11 with $h \leftarrow (Df)^{-1}(x).f(x)$, meaning we have to solve the linear system

$$Df(x).h = f(x) \quad (35)$$

for h . This can be done in a number of ways; we choose a QR decomposition.

The algorithm of Figure 11 isn't directly usable in practice, as it has no failsafe condition to ensure it won't run forever. In practice, we propose two conditions which end prematurely the main loop and throw a failure:

- a maximum number of loop iterations – the Newton-Raphson converges exponentially, and shouldn't need above a few dozens iterations;
- a maximum norm for the current guess x – if the algorithm escapes “far away”, there is no chance of it coming back.

D-3.2 Adding sections

D-3.2.1 Recovering local uniqueness

The Newton-Raphson method only works, as most other root-finding algorithms do, if the roots we are looking for are unique. This is never the case with periodic orbits; first of all, if $x(0)$ is a root of $\Phi_T - I$, then so are $x(t)$ for all real times. Moreover, dynamical systems often have symmetries or invariances that increase the kernel dimension; for instance, in the three-body problem, shifting all positions by a constant vector Δq doesn't affect the dynamics. Thus, if x_0 is a root, so is $x + \Delta q$. The same phenomenon occurs with velocities.

To recover local uniqueness for the roots, we add *sections* $\sigma = (\sigma_1, \dots, \sigma_s) : \mathbb{R}^n \rightarrow \mathbb{R}^s$, $s \in \mathbb{N} \setminus \{0\}$, to the function: instead of searching for roots of $\Phi_T - I : \mathbb{R}^n \mapsto \mathbb{R}^n$, we search for roots of

$$\begin{pmatrix} \Phi_T - I \\ \sigma \end{pmatrix} : \mathbb{R}^n \mapsto \mathbb{R}^{n+s} \quad (36)$$

When looking for a periodic orbit, we need to isolate a point along the orbit: the corresponding section is also called the Poincaré section.

When we will look at families of orbits, we also need to select a given orbit along the family: this also can be done through means of a section, if the orbits all have the same period along the family. The section parameter can be challenging to find, especially if there is a turning point. Techniques, such as pseudo-arc-length continuation, exist to circumvent such problems; however, we didn't need them for the results presented in this article.

D-3.2.2 Sections versus changes of coordinates

An alternative way to recover local uniqueness is by changing coordinates. For instance, in the three-body problem, because of the spatial translation invariance, we can either add the position of the barycenter $\sum_{i=0}^2 m_i \mathbf{q}_i$ as a section, or change coordinates by dropping the first body's position, and replacing it in the equations with $-\sum_{i=1}^2 m_i \mathbf{q}_i$.

Because changes of coordinate allow us to drop dimensions, they can be very tempting numerically – allowing to easily speed up computations. However, they are not always easy to find, and coordinates can become difficult to interpret. Sections, on the other hand, marginally increase the computational cost, but are usually quite easy to define.

For this reason, we adopt the following approach: for general invariances of the problem, we choose adequate changes of coordinates; for anything more specific, we prefer adding sections to the function – such as the family section or the Poincaré section, mentioned in the previous section.

D-3.2.3 Implementation

The only numerical part that is affected by sections is the Newton-Raphson algorithm, specifically the solving of [Equation 35](#). In fact, because the QR decomposition also works for non-square matrices, we just have to compute the Jacobian of the sections:

$$Df = \begin{pmatrix} D(\Phi_T - I) \\ D\sigma \end{pmatrix} \quad (37)$$

This can be done directly in Julia, with automatic differentiation.

D-4 Bifurcations

D-4.1 The problem

The family of interest can intersect with another family, locally increasing its dimension. It is highly improbable to numerically attain the intersection by chance; however, without sufficient care the sections can intersect both families and thus two close zeros can be found by the root-finding algorithm.

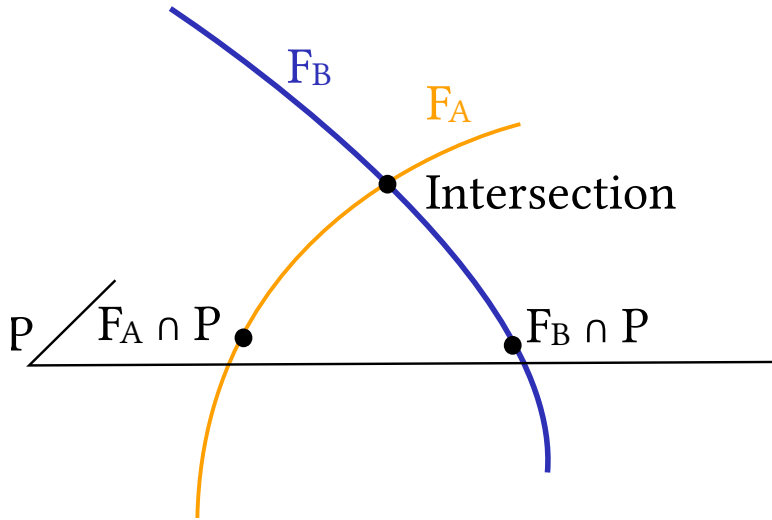


Figure 12: When a bifurcation between two families F_A and F_B appears, problems can arise beyond the intersection: a section P could intersect both families, leaving two possible zeros for the root-finding algorithm.

D-4.2 Detecting bifurcations

To detect bifurcations, we look at the evolution of eigenvalues along the family. If the distance of some eigenvalues to 1 drops down to numerical error near a given value of the family parameter, this is an indicator of a bifurcation.

Note that for an eigenvalue of multiplicity k and a numerical error η , the error on the eigenvalues will be of order $\sqrt[k]{\eta}$. For instance, for standard double precision numbers of error 2^{-53} , an eigenvalue of multiplicity 2 would only be precise up to $2^{-26.5} \approx 1.05 \cdot 10^{-8}$. If the bifurcation's dimension (and thus the eigenvalue's multiplicity) increases further, arbitrary precision is likely to be required.

D-4.3 Following a bifurcating family

Let's say we detected an n -dimensional bifurcation along our m -dimensional family; although we know the direction in which to look for the original family, the crossing family could come from any direction in the $n + m$ -dimensional subspace formed by corresponding eigenvectors.

In the cases encountered in the following (in [Chapter G](#)), instead of searching in this subspace, we made use of prior knowledge to find the bifurcating family: we provided the algorithm with an initial guess sufficiently far from the bifurcation, continuing the search from there.

D-5 Implementing in a new language: Julia

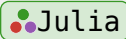
D-5.1 Why a new language?

A secondary goal of the thesis was to put the Julia programming language to the test in celestial mechanics, checking its abilities compared to other options. It has already been used in neighbouring domains: for instance, the Marchal conjecture's computer-assisted proof

of [Calleja et al. \(2024\)](#) was written in Julia, and the package `OptimalControl.jl` [Caillaud et al. \(2026\)](#) is a state-of-the-art implementation in optimal control. However, changing language is a significant investment, and the advantages ought to be significant. We shall see in the following Julia’s characteristics, promises and the challenges it yet faces.

D-5.1.1 A new language with new promises

Julia is a relatively new programming language, with the 1.0 version being released in 2017 [Bezanson et al. \(2017\)](#). It was conceived as a generic language, mixing high performance (close to C) and user-friendliness (like Python or MATLAB, see [Listing 3](#)). It is open-source and community-developed.



```

1  v(x) = cos(2π * x) # one-line function definition
2  #=
3  dissipative_std_map((x, y)) returns the dissipative standard map at
4  the point (x, y). This is a multi-line comment.
5  =#
6  function dissipative_std_map(input) # multi-line function definition
7      # we can use Unicode characters, not only ASCII ones
8      ε = 0.98; λ = 0.9; μ = 0.1
9      x, y = input # split input in variables x, y
10     y_prime = y + ε * v(x) + λ * (x - μ)
11     return (x + y_prime, y_prime) # return a tuple
12 end
13
14 # grid is a 51*21 matrix
15 grid = [
16     (x, y) for x in LinRange(0., 1., 51), y in LinRange(-2., 2., 21)]
17 # Map the function on the matrix
18 first_iteration = map(dissipative_std_map, grid)

```

Listing 3: Example of Julia code.

As a newcomer and a modern language, Julia brings many promises to the table, which we will examine in the following subsections. However, on top of the cost of changing for a new language, Julia also has downsides which shall be considered.

D-5.1.2 Combining speed and ease of use: just-in-time compilation

Languages like C or FORTRAN are *compiled*⁸: the *compiler* needs the whole code at once to output an executable, made of machine code — this is “ahead of time” compilation. To be run, the whole code is needed, and the compilation happens in advance; this means small adjustments to the code need a full recompilation — but advanced optimisation can happen.

⁸Technically, a *language* (like C) is distinct from its *implementations* (like `clang` or `gcc`), and the *implementations* are compilers or interpreters. Usually though, a language is designed with a specific implementation method in mind, catered to its use case.

On the other end, Python, MATLAB and others are *interpreted*: an *interpreter* reads each statement (a variable assignment, a function declaration...) and executes it before continuing. Here, no trace of the machine code remains after the statement is executed; there is no compilation overhead, allowing user-friendlier coding workflows like the Python terminal (formally called a REPL, or Read-Eval-Print Loop) or code notebooks. However, the produced code is much slower, as fewer optimisations are possible and interpretation happens at every statement.

Hybrid approaches, trying to get the best of both worlds, exist: such is the case of “just-in-time” compilation. The idea there is to read the code statement by statement, just like an interpreter, and to *compile* it to machine code just before execution; if the same statement happens again later, the stored machine code can be simply reused. In the case of Julia, *functions* are compiled at their *first call point*. This allows for different optimisations compared to ahead-of-time compilation; for instance, a function can be written in a generic way, without specific types in mind, but be compiled at the last moment, when types are known, to still get high computation speed.

D-5.1.3 Interest for scientific computing

D-5.1.3.1 Writing fast code, quickly

Prototyping, an essential part of any code development, is made easier in Julia by a multitude of small helpers – first of all thanks to the available coding environments. There are official Julia extensions for Visual Studio, Vim, Emacs and others popular IDEs⁹. Jupyter notebooks are also available for Julia; a very popular alternative for Julia users is `Pluto.jl`. Pluto notebooks add, among other things, reactivity to the cells: if a variable `b` depends on a variable `a` and the value of `a` changes, `b` will be re-computed with the new value of `a`; this means the notebook has no hidden part, (mostly)¹⁰ preventing it from achieving hardly reproducible states.

Julia being fast by design, following a few simple design rules¹¹ allows high efficiency from the get go – although, as for any language, optimisation can always go a step further. Multiple tools exist for helping with profiling (`Profile` in the standard library), benchmarking (`BenchmarkTools.jl`), library development (`Revise.jl` to propagate definition changes).

Multi-threading is in the core library, and Julia has an extensive GPU programming environment ([Besard et al. \(2019a\)](#), [Besard et al. \(2019b\)](#), [Faingnaert et al. \(2022\)](#)) for all major GPU models. An impressive example is `KernelAbstractions.jl`, which allows to write GPU-ready code independently of the specific GPU it will be run on.

⁹Integrated Development Environment.

¹⁰Because detection of side effects is a hard problem in informatics, one can easily circumvent the change detection algorithm in Pluto; the goal is to catch obvious changes, such as `a = 4.2 → a = 6.7` or definition changes.

¹¹A common starting place for code optimisation in Julia is [this documentation page](#).

D-5.1.3.2 A competitive library for scientific computing

Despite its young age, Julia features a significant ecosystem in most scientific domains, making it easy for a newcomer to find their usual tools already implemented. In some domains, the state of the art is even the Julia implementation, most notably in scientific machine learning, ODE¹² solving or automatic differentiation (Enzyme.jl).

More often than not, the problem then becomes to find which of the existing implementations best suits their needs, or if the package is still maintained. To try and establish standards for specific domains, people have organised into communities (see [Figure 13](#)) – facilitating the path of new users and increasing interoperability of packages.

D-5.1.3.3 Reproducibility

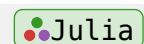
A major threat to scientific ethics is the lack of reproducibility of previous results. With code, the first step is of course to publish the algorithms with the results, but this rarely is enough. The chain of reproducibility can break at many different levels: in-language packages, language versions, operating systems or even microprocessor architectures can become unavaible or hardly obtainable by future researchers.

Although there is only so much one can do to prevent this, a significant part of the problem can be solved, or at least eased, at the programming language level: here, the code was executed with a certain *version* of the language, and most likely with certain *versions* of several external packages. This is the *environment* in which the code was executed.

In Julia, the environment information is stored in two files: `Project.toml` and `Manifest.toml`. The first one contains the names and versions of packages the user requested specifically; the last contains the Julia version as well as all packages effectively pulled for the environment (packages and their dependencies).

Initializing and populating an environment is as simple as [Listing 4](#). The resulting environment is simply two text files: see [Listing 5](#).

```
1 using Pkg
2 Pkg.activate(".")
3 Pkg.add("Random")
4 run(`ls`) # Project.toml Manifest.toml
```



Listing 4: Code example for creating an environment. One can also add packages in the REPL interactively, more quickly (see [here](#)).

The reproducibility of the environment, however, depends on being able to recover, in the future, the Julia executable and packages used. The Julia versions are all currently available on the main [Julia website](#), as well as on the [GitHub repository](#); the [juliaup](#) utility allows for easy handling of multiple Julia versions, maintaining and updates. The packages used, however, can be hosted on a diversity of platforms; Julia uses *registries* to determine which package name and version corresponds to which source code. The main one is the General

¹²Ordinary Differential Equation

Registry, hosted on a [GitHub repository](#) were this registry – or any registry used to create the environment – to become unavailable, the code would be much harder to execute in the same conditions. Most notably, the dependence of a majority of Julia’s backend on a private service, GitHub, has been a cause of concern in the community and for the language.

```

1 [deps]
2 Random = "9a3f8284-a2c9-5f02-9a11-845980a1fd5c"

```

↓

```

1 # This file is machine-generated - editing it directly
  is not advised
2
3 julia_version = "1.12.6"
4 manifest_format = "2.0"
5 project_hash = "9e64cf17f9522d20edabe6f2b4ec85252943fcae"
6
7 [[deps.Random]]
8 deps = ["SHA"]
9 uuid = "9a3f8284-a2c9-5f02-9a11-845980a1fd5c"
10 version = "1.11.0"
11
12 [[deps.SHA]]
13 uuid = "ea8e919c-243c-51af-8825-aaa63cd721ce"
14 version = "0.7.0"

```

Listing 5: Project.toml (top) and generated Manifest.toml (bottom) for an environment created with Listing 4. The environment was created on May 13th, 2026 with Julia 1.12.6, but can be reproduced as long as a Julia executable (currently at <https://www.julialang.org>) and the general Julia registry (currently at <https://github.com/JuliaRegistries/General>) stay available.

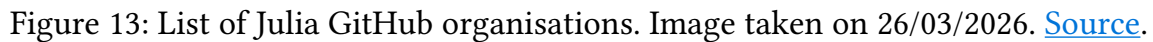
D-5.1.4 Caveats

D-5.1.4.1 Missing features: the example of standalone executables

Julia, like any programming language, isn’t exempt of faults. Most of all, it does show its young age in a few areas: to cite a common example, because standalone executables aren’t the main “intended” way to use Julia, producing small ones proves to be a challenge at the time of writing. A toy module containing a single function which prints “Hello world” currently takes 1.7 Mo.

Julia is, however, being actively developed, and this particular case has started being tackled on in the last couple of years. Because Julia is compiled “just in time”, to produce a standalone executable, previously the whole Julia compiler was being shipped together with the code, resulting in 1 Go minimal files; the trim option allowed this drop of three orders of magnitude, and was shipped as experimental with [Julia 1.12 \(october 2025\)](#).

With nine years after the 1.0 release, breaking changes have become increasingly rarer, showing Julia’s progressive maturity. Moreover, in specific domains, communities have been forming to guarantee stability in specific areas (see [Figure 13](#)).



The julian community, albeit small, partially counteracts these effects by being especially active online (on the main forum, <https://discourse.julialang.org>) and with annual worldwide conferences (“JuliaCon”, and “JuliaCon Local” for smaller side events).

D-5.2 Numerical integration with TaylorInterface.jl

Because the majority of computing work is done in Julia, we'd like to leave the Julia environment as little as possible, while still using `taylor` (from D-2.5). To do this, we developed the package `TaylorInterface.jl`, which is open-source and available on <https://github.com/Alseidon/TaylorInterface.jl>. The package allows interaction with `taylor` from within Julia. The only outside part remaining is writing the equations in a text file, with the `taylor` syntax. Afterwards, `TaylorInterface.jl` handles the building and compilation of the binary with the `TaylorGenerator` structure, and the calling of the resulting shared library with the `TaylorHandler` structure.

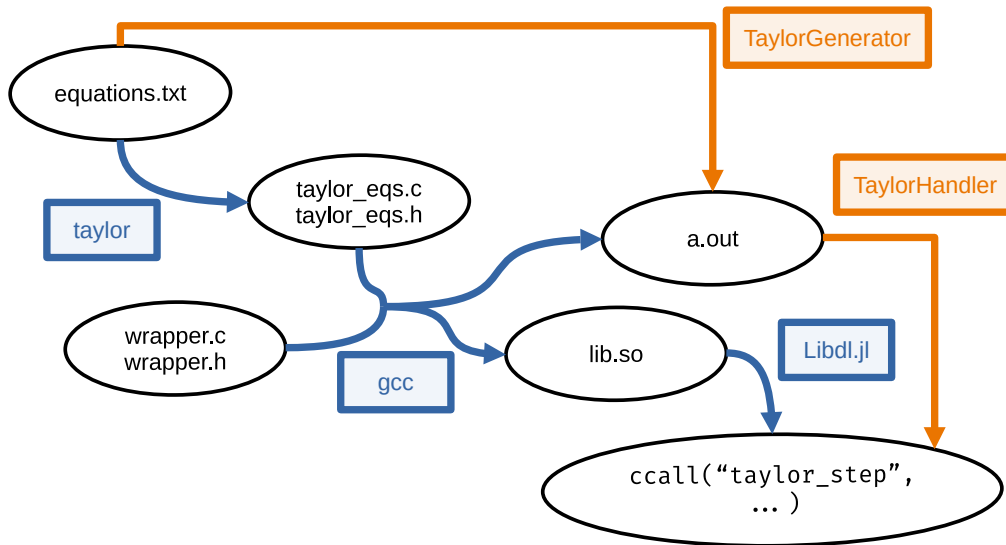


Figure 14: Pipeline using `taylor` with Julia, and the shortcut allowed by `TaylorInterface.jl`.

Let's take the previous example Listing 1 and suppose it is in a file `equations.txt`, in the working directory; Listing 6 is the Julia equivalent of the previous section. Most notably, the line Listing 6-8 corresponds to the following bash commands:

```

user@machine:~$ taylor -name auto -headername taylor-my_eqs.h -o src/taylor-
my_eqs.c -jet -step -jlib jet_1 -jet_helper equations.txt
user@machine:~$ taylor -name auto -o src/taylor-my_eqs.h -jlib jet_1 -
header equations.txt
user@machine:~$ gcc -O2 -fPIC -Wno-unused-result -c src/wrapper-my_eqs.c -o src/
wrapper-my_eqs.o
user@machine:~$ gcc -O2 -fPIC -Wno-unused-result -c src/taylor-my_eqs.c -o src/
taylor-my_eqs.o -lm
user@machine:~$ gcc -O2 -fPIC -Wno-unused-result -shared src/wrapper-my_eqs.o src/
taylor-my_eqs.o -o lib.so -lm
user@machine:~$ █
  
```

Because we want to use the C code generated by `taylor` in Julia, we ask the C compiler, `gcc`, to generate a shared library `lib.so` (with the `-fPIC` option) instead of a standalone executable. This library links function names to pointers, pointing the actual executable code.

```

1  using TaylorInterface
2  generator = TaylorGenerator(
3      "my_eqs",          # name to append to the build directory
4      "equations.txt",  # file containing the equations
5      "."                # directory in which to put the build files.
6  )
7  # Create the build directory at ./taylor_my_eqs
8  generate_dir(generator, true)
9  # Get the corresponding TaylorHandler structure
10 handler = get_handler(generator, false);
11 # Open the shared library
12 open_lib(handler)
13 # ...do stuff...
14 x = [0., 1.]
15 y = similar(x) # initializes an array without assigning values
16 t = 0.1
17 df = zeros(4) # multi-dimensional arrays need to be passed as vectors
18 flow!(handler, x, y, t, df) # result is stored in y and df
19 close_lib(handler)

```

Listing 6: Minimal example using the taylor software via TaylorInterface.jl.

The structure `TaylorHandler` then simplifies the manipulation of the shared library `lib.so` via the `Libdl.jl` Julia package. There are three ways to use the `TaylorHandler` to interact with `taylor`:

- `TaylorInterface.jl` defines a wrapper function `flow(han, x, t)` and its in-place equivalent `flow!(han, x, y, t, df)` (necessary when jet transport is used) that calls the corresponding C function;
- The handler contains a dictionary of symbols `handler.symbols` that are pointers to the corresponding C functions, for the time-step function and the flow function. For instance, the previous flow function makes this call:

```

1  ccall(
2      han.symbols["flow"],
3      Cint, # type of the output
4      (Cdouble, Ptr{Cdouble}, Ptr{Cdouble}, Ptr{Cvoid}), #type of inputs
5      t, x, y, df # inputs
6  )

```

Listing 7: `ccall` example with a `TaylorHandler`.

- All symbols available in the dynamic library are available via the `handler.lib` pointer. In the rare case one might want to access specific functions of `taylor`, they would simply

replace [Listing 7-2](#) with `Libdl.dlsym(handler.lib, "my_func")` (after a line using `Libdl`, since it needs to be directly used there).

In [Appendix II](#) we show a few benchmarks comparing `TaylorInterface.jl` with other integration packages `TaylorIntegration.jl`.

D-5.2.1 Possible improvements

`TaylorInterface.jl` still has several limitations compared to the `taylor` software: most notable, it only supports double precision real numbers and doesn't leverage the parallelisation abilities of `taylor`-generated code.

Moreover, the user still needs to exit Julia in order to write the equation file; one could imagine leveraging the meta-programming abilities of Julia to convert native functions to the `taylor` syntax (the Julia compiler is bootstrapped – it is written in Julia itself, allowing easy manipulation of Julia expressions).

Furthermore, by using a C backend, the package comes back to the two language problem which Julia tried to solve, and loses on interoperability with other Julia packages – such as automatic differentiation, testing along the integration, etc. As we mentioned earlier, `TaylorIntegration.jl` is a pure Julia package inspired by `taylor`, despite some shortcomings; other implementations of the Taylor method are ongoing; see eg. [Tan et al. \(2026\)](#) for a Taylor integrator with symbolic-numeric computing.

As for us, work on a pure Julia implementation of `taylor` is underway, hoping to take advantage of the thriving work on automatic differentiation in Julia: see eg. `TaylorDiff.jl` [Tan \(2022\)](#) for Taylor-mode AD and `Diffactor.jl` [\(2026\)](#) for source-code transformation (meaning the derivative will be evaluated as if the programmer wrote it directly).

D-5.3 In practice: pipeline when searching for a family of periodic orbits

D-5.3.1 First implementation

Let's say we are searching for a family of periodic orbits, of period T , in a dynamical system. We need at the very least:

- The equations of motions f
- The period (or periods)
- An initial guess for the first point
- A Poincaré section to select a point on the orbit
- If the period is identical along the family, a section to select an orbit on the family

We can now:

- write the equations of motion in the format required by `taylor` in a file `system.eqs`, and use them from `TaylorInterface.jl`;
- write the section(s) in Julia;
- combine the above with a Julia implementation of the Newton-Raphson algorithm to follow the family, as shown in [Figure 15](#).

```

1 function family-search( $\Phi_T, x_0, \text{array\_parameter}, \text{sections}$ )
2    $x_{\text{ini}} \leftarrow x_0$ 
3    $f \leftarrow \{(x, \text{param}) \mapsto \text{concat}(\Phi_T(x) - x, \text{sections}(x, \text{param}))\}$ 
4    $\text{array\_results} \leftarrow []$ 
5   for  $p$  in  $\text{array\_parameter}$ 
6      $f_p \leftarrow \{x \mapsto f(x, p)\}$ 
7      $x_{\text{res}} \leftarrow \text{newton-raphson}(f_p, x_{\text{ini}})$ 
8      $\text{append}(\text{array\_results}, x_{\text{res}})$ 
9      $x_{\text{ini}} \leftarrow x_{\text{res}}$ 
10  end for
11  return  $\text{array\_results}$ 
12 end

```

Figure 15: Pseudo-code for searching a family of periodic orbits, of period T . The algorithm is easily tweakable to allow for varying periods. Note that, contrary to [Figure 11](#), newton-raphson takes a single function input (which outputs the result and its derivative), not both f and f' .

D-5.3.2 Improving search in case of Newton-Raphson failure

When the successive points in the family are too far away from each other, using the previous one as an initial guess for the next can fail. If the Newton-Raphson search fails at a given parameter p_i , as long as it is not the first one, we can add an intermediate search, at the parameter $\frac{p_{i-1} + p_i}{2}$. We thus replace the main for loop of [Figure 15](#) with [Figure 16](#).

```

1   $n_{\text{iters}} \leftarrow 0$ 
2   $i_p \leftarrow 1$ 
3  while  $i_p < \text{length}(\text{array\_parameter})$ 
4       $f_p \leftarrow \{x \mapsto f(x, \text{array\_parameter}[i_p])\}$ 
5      try
6           $x_{\text{res}} \leftarrow \text{newton-raphson}(f_p, x_{\text{ini}})$ 
7           $\text{append}(\text{array\_results}, x_{\text{res}})$ 
8           $x_{\text{ini}} \leftarrow x_{\text{res}}$ 
9           $i_p \leftarrow i_p + 1$ 
10     catch
11          $\text{append}(\text{array\_parameter}, \frac{\text{array\_parameter}[i_p] + \text{array\_parameter}[i_p - 1]}{2})$ 
12     end try
13      $n_{\text{iters}} \leftarrow n_{\text{iters}} + 1$ 
14     if  $n_{\text{iters}} > \text{max\_iters}$ 
15         error
16     end if
17 end for

```

Figure 16: Catching Newton-Raphson failures and trying closer points.

In [Figure 16](#) we add a break condition on the maximum number of iterations allowed. One could also limit the amount of times a single point fails, to limit how small the step size can get (because it shrinks exponentially).

Although this algorithm works for a one-dimensional family of orbits, it can be easily adapted to families of higher dimensions, by adding sections and tweaking the search loop [Figure 15](#) to search in multiple dimensions.

D-6 Other approaches

This work is, of course, far from the first attempt at numerically finding periodic and quasi-periodic orbits. We use a very simple approach, seeing periodic orbits as roots of the flow at a certain time, and using the Newton-Raphson method to find them; see eg. [Hénon \(1974\)](#), [Hénon \(1997\)](#) for other similar approaches.

Other ways exist. For instance, when searching for quasi-periodic orbits, [Couetdic et al. \(2010\)](#) use frequency analysis to find quasi-periodic guesses which aren't solutions of the equations of motion, iteratively converging to a solution. [Moore \(1993\)](#) numerically found the Eight by imparting initial guesses with specific spatial topologies, again not solutions of the equations of motion, and “relaxing” them to converge to a solution.

It also is possible to simplify the problem: [Jorba and Masdemont \(1999\)](#) approximate the Hamiltonian with a truncated power series near equilibria (Euler points in the restricted

three-body problem), using its dynamics as a first qualitative description of the phase space near the equilibria.

The main advantage of our algorithm is its simplicity: it is relatively fast to implement, without any assumptions about the system other than the existence of a periodic orbit with the chosen period (and the sections) – thus being generic, and applicable to a large set of potential problems. Although we designed it as a first approach, it ended up being sufficient to find all results presented in the following chapters.

Chapter E

An example: Marchal's family, from the restricted problem to the equal masses case

Εὐρηκα!

— Archimede

E-1 Preliminary results: the average circular co-orbital problem

We first try an analytical approach, to generalise the results of [Zagouras \(1985\)](#), [Marchal \(2009\)](#) to the non-restricted case. We will quickly present a simple model, the average circular co-orbital problem in the planetary limit; a more detailed approach can be found in the first part of [Prieur and Robutel \(2026\)](#).

E-1.1 The average co-orbital circular problem in the planetary limit

E-1.1.1 The co-orbital resonance

Let us consider a planetary three-body problem, with masses $(m_0, \varepsilon m_1, \varepsilon m_2)$ where ε is a small parameter and m_0, m_1, m_2 are of order 1. We use the Poincaré-like coordinates from [B-2.2](#). Our goal is to reduce the problem from four degrees of freedom (after the Jacobi reduction) to one.

We are interested in the neighborhood of the co-orbital resonance: we will limit ourselves to a neighborhood of a constant a^* of size $\mathcal{O}(\varepsilon)$:

$$a_1, a_2 = a^* + \mathcal{O}(\varepsilon) \quad (38)$$

Meaning the difference of Λ_j is of order $\sqrt{\varepsilon}$. We can replace the (Λ_j, λ_j) variables with

$$\begin{aligned} Z_1 &= \Lambda_1 - \Lambda_1^* & \zeta_1 &= \tilde{\lambda}_1 - \tilde{\lambda}_2 + \pi, \\ Z_2 &= \Lambda_1 + \Lambda_2 - (\Lambda_1^* + \Lambda_2^*), & \zeta_2 &= \tilde{\lambda}_2 \end{aligned} \quad (39)$$

where the Λ_j^* are the constants associated with a^* . Because the Z_j are of order $\sqrt{\varepsilon}$, we can develop the Hamiltonian up to $\mathcal{O}(\varepsilon)$; having only orders of order $\mathcal{O}(1)$, $\mathcal{O}(\sqrt{\varepsilon})$ or $\mathcal{O}(\varepsilon)$ remaining.

E-1.1.2 Switching to an average, circular problem

Typically, near a resonance, the system has three time scales:

- A fast time scale of order 1, the synchronous rotation of bodies around the central mass;
- A semi-fast time scale of order $\sqrt{\varepsilon}$;

- A slow, or secular, time scale, of order ε .

Averaging the system (see B-3) over the fast angle ζ_2 rids us of the fast time scale. Moreover, in the average problem, its conjugated variable Z_2 is constant, gaining us yet another degree of freedom.

Finally, because the average Hamiltonian is even in the variables $(x_j, \tilde{x}_j)_{j=1,2}$, the manifold $x_j = \tilde{x}_j = 0$ is invariant under its flow: as these variables are proportional to the eccentricities, this amounts to restricting the study to circular orbits – thus gaining two degrees of freedom. As a consequence, we are left with the study of the one parameter family of integrable Hamiltonian

$$\begin{aligned}
 F_c(Z_1, \zeta_1, s_0) &= -\alpha Z_1^2 + \varepsilon \beta F_1(\zeta_1, s_0) \\
 &= -\alpha Z_1^2 + \varepsilon \beta \left((1 - s_0^2) \cos \zeta_1 - \frac{1}{\pi} \int_0^\pi \frac{d\zeta_2}{\sqrt{U_{s_0}(\zeta)}} \right) \\
 &\quad \text{with } \begin{cases} U_{s_0}(\zeta) = 2 - 2 \cos \zeta_1 + 2s_0^2(\cos \zeta_1 - \cos(\zeta_1 + 2\zeta_2)) \\ \alpha = \frac{3}{2a^{*2}} \frac{m_1 + m_2}{m_1 m_2} \\ \beta = \frac{\mu_0}{a^*} \frac{m_1 m_2}{m_0} \end{cases} \quad (40)
 \end{aligned}$$

where we introduce the parameter $s_0 = \sin\left(\frac{J_0}{2}\right)$, J_0 being the mutual inclination of the orbits and neglect the terms in $\mathcal{O}(\varepsilon^{\frac{3}{2}})$. Note that the integral appearing in Equation 40 can be expressed using the incomplete elliptic integral of first kind as:

$$\begin{aligned}
 \frac{1}{\pi} \int_0^\pi \frac{d\zeta_2}{\sqrt{U_{s_0}(\zeta)}} &= \frac{1}{2\pi\sqrt{u}} \left[\mathcal{F}\left(\frac{\zeta_1 + \pi}{2}, \frac{s_0^2}{\sqrt{u}}\right) - \mathcal{F}\left(\frac{\zeta_1 - \pi}{2}, \frac{s_0^2}{\sqrt{u}}\right) \right] \\
 \text{where } p &= \sin\left(\frac{\zeta_1}{2}\right), u = p^2 + s_0^2 - p^2 s_0^2, \\
 \text{and } \mathcal{F}(x, y) &= \int_0^x \frac{dz}{\sqrt{1 - y^2 \sin^2 z}} \quad (41)
 \end{aligned}$$

The proof is provided in the appendix I-1.

E-1.2 $\mathcal{VF}_\mathcal{L}$ in the average circular planetary problem

For a given s_0 , the fixed points of the Hamiltonian in Equation 40 satisfy the implicit equation

$$(Z_1, \zeta_1) = (0, \zeta_0) \quad \text{with} \quad \frac{\partial F_1}{\partial \zeta_1}(\zeta_0, s_0) = 0. \quad (42)$$

It is important to note that this equation is independent of masses. The same will be true for the position of the fixed points solutions of this equation, provided that the terms in $\varepsilon^{\frac{3}{2}}$

are negligible. We will see in E-2 that when this assumption is not fulfilled, the shape of the family of fixed points, or rather periodic orbits, evolves as the “planetary” masses increase.

Although, except for $s_0 = 0$, we cannot give an explicit expression for the solutions to this equation, we can either calculate them numerically or give an analytical approximation. In the latter case, we can express ζ_0 in Taylor series of s_0 , provided that the inclination is close enough to zero. When $J_0 = s_0 = 0$, the fixed points correspond to Lagrange's equilateral equilibria at $\zeta_0 = \frac{\pi}{3}$ and $5\frac{\pi}{3}$ and to Euler colinear configuration at $\zeta_0 = \pi$. These equilibria are the starting point of fixed points families parameterized by the mutual inclination J_0 – although we could also parametrize them with the total angular momentum $\mathcal{C}$, J_0 provides a more natural and intuitive parameter. Solving Equation 42 numerically, with initial condition $\zeta = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ for $J_0 = 0$, provides a unique family of fixed points, which will be denoted $\mathcal{VF}_L$ for Vertical Family of Lagrange. This family connects the two equilateral Lagrange plane configurations while containing inclined trajectories with maximum mutual inclination $J_c \approx 2.542567\text{rad} \approx 145.678^\circ$ at $\zeta = \pi$. Since F_1 is even and 2π -periodic, $\mathcal{VF}_L$ is symmetric with respect to the line $\zeta_0 = \pi$. This line coincides with the second family of solutions of Equation 42, which will be called the Vertical Family of Euler ($\mathcal{VF}_E$). This latter structure originates from Euler's aligned configuration, in which the two planets are on both sides of the primary body, intersect $\mathcal{VF}_L$ at J_c and persist until they meet at $J = \pi$, where the two planets encounter each other. These two families, obtained by numerical resolution of the Equation 42, are displayed in Figure 17.

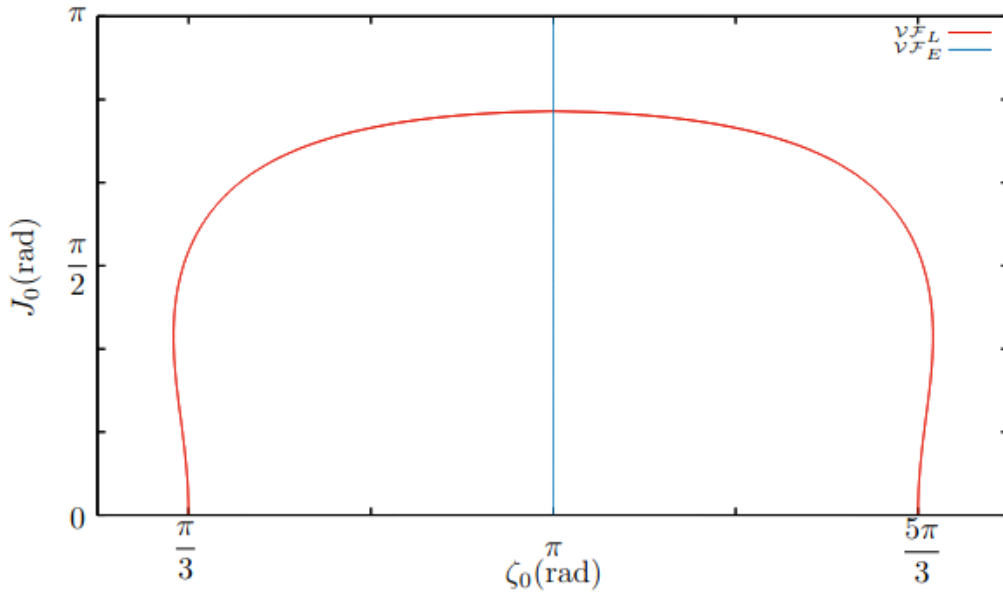


Figure 17: The two families of fixed points derived from Lagrange equilibria in the average problem. The two branches of $\mathcal{VF}_L$ are identical up to permutation of the two small bodies.

The behaviours of $\mathcal{VF}_L$ and $\mathcal{VF}_E$ shown here are similar to those in Marchal (2001), and before in Zagouras (1985) (for $\mathcal{VF}_L$) Zagouras and Kazantzis (1979) (for $\mathcal{VF}_E$) in the restricted problem, and those mentioned in Leleu (2016) in the planetary case. According to Marchal (2009), the maximal value of the mutual inclination reaches on these families is equal to 145.68° which is what we get here. As mentioned above, the agreement observed between the restricted and planetary cases is due to the fact that the solutions to Equation 42

are independent of the masses. This remains valid as long as the smallness of the planetary masses allows for the terms of order $\varepsilon^{\frac{3}{2}}$ in Equation 40 to be ignored.

The solutions of Equation 42 can also be approximated by Taylor expansion. Indeed $\mathcal{VF}_L$ can be expressed with a relative error lower than 10^{-5} up to $J_0 \leq 60^\circ$ by the twentieth degree Taylor expansion (all analytical results were found using the TRIP language Gastineau and Laskar (2011)):

$$\begin{aligned} \zeta_c(J_0) = & \frac{\pi}{3} - \sqrt{3} \left(\frac{1}{3}s_0^2 - \frac{4}{9}s_0^4 - \frac{17}{27}s_0^6 + \frac{1865}{2592}s_0^8 + \frac{13877}{9720}s_0^{10} - \frac{75553}{23328}s_0^{12} \right. \\ & \left. - \frac{63599}{10206}s_0^{14} + \frac{254313827}{17915904}s_0^{16} + \frac{384292411}{13436928}s_0^{18} - \frac{86371296841}{1209323520}s_0^{20} \right) \\ & + \mathcal{O}(s_0^{22}), \end{aligned} \quad (43)$$

where $s_0 = \sin\left(\frac{J_0}{2}\right)$. The expression of the branch originating from L_5 is expressed in the same way by taking $\zeta'_c(J_0) = 2\pi - \zeta_c(J_0)$.

Note that by expressing $\zeta_c(J_0)$ using directly J_0 rather than s_0 , we find

$$\zeta_c(J_0) \approx \frac{\pi}{3} - \frac{\sqrt{3}}{12}J_0^2 + \frac{5\sqrt{3}}{144}J_0^4 + \dots \quad (44)$$

which corresponds to the expression given by Namouni and Murray (2000) for the restricted 3-body problem.

As long as we limit ourselves to studying the dynamics restricted to the plane $x_j = 0$, the linear stability of the fixed points is given by the eigenvalues of the linearized differential system at the equilibrium these are roots of the characteristic polynomial:

$$P(X) = X^2 - 2\varepsilon\alpha\beta \frac{\partial^2 F_1}{\partial \zeta^2}(\zeta_0) = X^2 - 3\varepsilon n^{\star^2} \frac{m_1 + m_2}{m_0} \frac{\partial^2 F_1}{\partial \zeta^2}(\zeta_0) \quad (45)$$

The nature of the fixed point (elliptic/hyperbolic) is deduced from the sign of $\frac{\partial^2 F}{\partial \zeta^2}(\zeta_0)$. Below the critical mutual inclination J_c , equilibria belonging to the families $\mathcal{VF}_L$ are stable (purely imaginary eigenvalues), while these are unstable (real eigenvalues) along this part of $\mathcal{VF}_E$. These fixed points degenerate (zero eigenvalues) at the bifurcation point where the three families join together ($\mathcal{VF}_E$ and the two branches of $\mathcal{VF}_L$, symmetrical under permutation of the two small bodies). Above this critical value, $\mathcal{VF}_E$ becomes elliptic and ends with a collision orbit at $J_0 = 180^\circ$. The eigenvalues of the elliptic family $\mathcal{VF}_L$, resp. hyperbolic $\mathcal{VF}_E$ of the form $\pm i\nu_L$, resp. $\pm \nu_E$, with

$$\begin{cases} \nu_\tau = \sqrt{\varepsilon} n^{\star} \tilde{\nu}_\tau \sqrt{\frac{m_1 + m_2}{m_0}}, & \tau \in \{L, E\} \\ \tilde{\nu}_\tau = \sqrt{\gamma_\tau} (1 + \mathcal{O}(s_0^2)) \end{cases} \quad (46)$$

where $\gamma_L = \frac{27}{4}$ and $\gamma_E = \frac{21}{8}$. The dimensionless quantities $\tilde{\nu}_\tau$, for which Taylor expansions are given in the appendix I-2, are plotted in Figure 18. The range of J_0 is limited to the

interval $[0, J_c]$ as higher values are of limited relevance for the present study. $\tilde{\nu}_L$ (red) and $\tilde{\nu}_E$ (purple) both decrease until they reach zero at J_c , where the three families merge. The Jacobi reduction allowed us to eliminate the inclination and longitude of the node of each planet. As indicated in section B-2.3, these quantities can be derived from Equation 8 and Equation 9 (to within a constant for Ω_p). Thus, the ascending node of the orbits of $\mathcal{VF}_L$ precess at a constant rate, which, according to Equation 42, is given by:

$$\dot{\Omega}_p = \dot{\psi}_p = \frac{\partial F_c}{\partial \mathcal{C}}(0, \zeta_0, s_0) = -\varepsilon n^* \frac{m_1 + m_2}{2m_0} \sqrt{1 - 4s_0^2 \frac{m_1 m_2}{(m_1 + m_2)^2}} \frac{\partial F_1}{\partial s_0^2}(\zeta_0, s_0) \quad (47)$$

The function $-\frac{\partial F_1}{\partial s_0^2}(\zeta_0, s_0)$ is plotted against $s_0 = \sin\left(\frac{J_0}{2}\right)$ in Figure 18 and its Taylor expansion is provided in the appendix I-2. On the family derived from the equilateral configuration, the node line has a prograde precession motion, whereas on $\mathcal{VF}_L$, the motion is retrograde up to $s_0 \approx 0.9235$ (or $J_0 \approx 135^\circ$), then prograde for higher values.

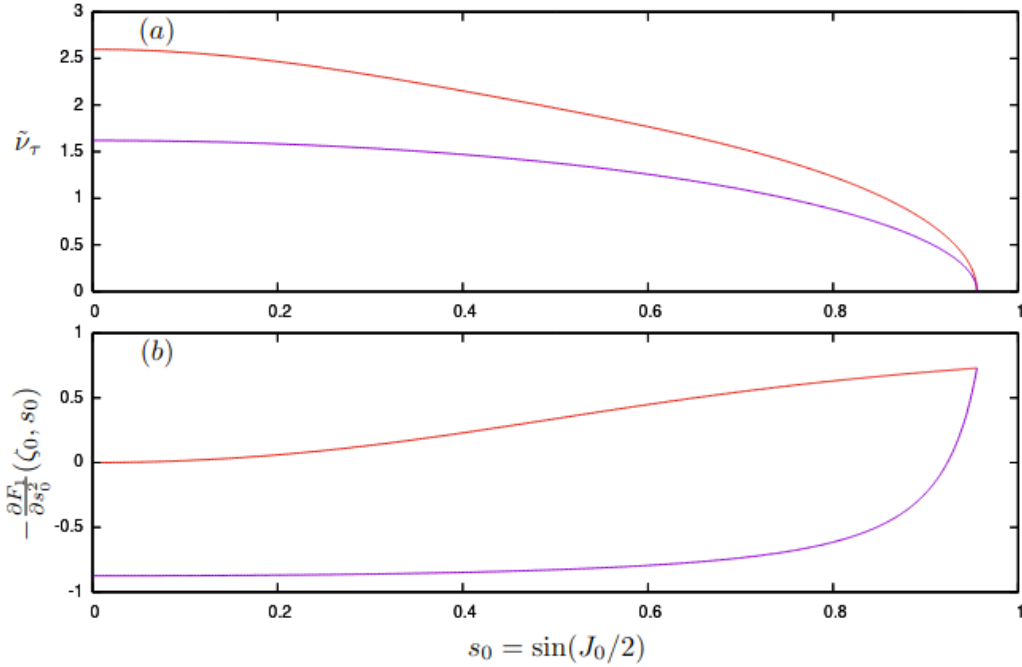


Figure 18: (a): Behavior of eigenvalues and precession frequencies of nodes along families $\mathcal{VF}_E$ (purple) and $\mathcal{VF}_L$ (red). (b): Precession of the ascending node.

E-1.3 Comparison with numerical results in the non-average problem

Let us now see how this local aspect of the geometry of the problem can be part of a global view of its dynamics, and in particular how the presence of $\mathcal{VF}_L$ can influence the stability of co-orbital regions. First, let us examine the stability map shown in the Figure 19. This map, consists of a grid of initial conditions with coordinates $(\lambda_1 - \lambda_2, J_0)$ with $\lambda_1 \in [0^\circ, 360^\circ]$ and $2I_1 = 2I_2 = J_0 \in [0^\circ, 180^\circ]$, the other initial conditions and parameters being set as follows: $a_1 = a_2 = 1$, $e_1 = e_2 = 0$, $\lambda_2 = 0$, $\Omega_1 = \Omega_2 + \pi = 0$. and $\frac{m_1}{m_0} = \frac{m_2}{m_0} = 10^{-3}$. Each point on the grid is associated with a color indicating the maximum value reached by e_1

and e_2 during a numerical integration lasting 40406 orbital revolutions¹³. As indicated on the color bar on the right-hand side, blue corresponds to maximum values below 0.05, while colors ranging from yellow to dark red correspond to values between 0.05 and 1. White is associated with systems that either left co-orbital resonance before the end of integration or collided. Here we use the maximum eccentricity as an indicator of instability. On the one hand, this is a relative indicator in the sense that it is the variations in its value that indicate changes in dynamic behavior such as the presence of resonance or chaotic regions [Zhou et al. \(2009\)](#), [Leleu \(2016\)](#). On the other hand, it also provides absolute information in the sense that it indicates the regions in which circular or quasi-circular orbits exist. The contrast between regions with regular dynamics (blue) and chaotic dynamics is very marked. The orbits that are located at the center of the two co-orbital regions (ζ close to 60° or 300°) undergo only very slight variations in eccentricity (smaller than 0.05), and over the time interval considered, are indistinguishable from quasi-periodic orbits; the measurement of the diffusion in the frequency space provided by frequency map analysis [Laskar \(1990\)](#), [Laskar \(1999\)](#) allows us to confirm this. Moving horizontally (variation of ζ) or vertically (increasing J_0) on this map, we encounter a very narrow and sharp transition zone between stable and chaotic regions. Whether, in the horizontal direction, this boundary is caused by the overlap of secondary resonances associated with commensurabilities between the libration frequency (associated with the resonant angle $\lambda_1 - \lambda_2$) and the orbital frequency common to both planets [Érdi et al. \(2007\)](#), [Robutel and Bodossian \(2009\)](#), the reason for the presence of the vertical transition zone at 60 degrees in [Figure 19](#) is not clearly identified.

Before addressing this question, let us return to our $\mathcal{VF}_L$ family. Since the members of this family are fixed points of the reduced average problem, they should correspond, in the reduced full problem (non-average and moving with the line of nodes), to periodic orbits whose frequency is equal to the mean motion common to the two planets. In a fixed reference frame, these are therefore quasi-periodic orbits whose second frequency is the precession frequency of the nodes. Let us add that the elliptic elements, which are constant in the average problem, are animated by a fast (mean motion) motion whose amplitude is of the order of the perturbation ε . In particular, as we will see in [Figure 21](#), the trajectories may no longer be circular, but their eccentricity also remains of the order of ε . For these reasons, although they are not expressed in the same coordinate system, it remains relevant to project $\mathcal{VF}_L$ (green curve) onto the stability map displayed on [Figure 19](#). As long as the mutual inclination along $\mathcal{VF}_L$ remains below 60° , the family passes through the core of resonance (most stable regions in blue); above this value, all $\mathcal{VF}_L$ trajectories become unstable¹⁴.

One of the keys to this puzzle lies in calculating the eigenvalues of the equilibrium points of the family. We briefly mentioned this issue in [E-1.2](#), but the eigenvalues calculated there only corresponded to those associated with the dynamics in the plane (ζ, Z) . We must therefore consider the dynamics in the $(x, \tilde{x})$ planes of coordinate; this would require

¹³The exact number is due to an irrational step size, to avoid artificial resonances. The order of magnitude was chosen to have secular effects, over a time span $\mathcal{O}(\frac{1}{\varepsilon}) \approx 10^3$, to happen.

¹⁴A similar observation is made for elliptical orbits in [Giuppone and Leiva \(2016\)](#), although the cause of the instability is not identified.

numerically solving the variational equations for variations in the $(x, \tilde{x})$ directions. Instead, in the following section we will directly numerically compute the periodic orbits (in rotating coordinates) in the full problem and solve their variational equations. The characteristic exponents thus calculated will provide information about the stability of the orbits.

Moreover, since we are only considering the average Hamiltonian at order 1 of the masses, Equation 42 is independent of the masses. Of course, this approximation is only valid in the context of planetary problems (small masses). Determining the upper limit of planetary masses for which this approximation is valid would require at least calculating the average Hamiltonian to the second order, which is beyond the scope of this paper. Direct numerical integration of the full, non-approximated problem will circumvent this limit, and give us some insight on the validity of the approximation: Figure 34 compares between results in the average and the full problem for various mass parameters (although the angles used in the two problems are slightly different, see E-2.2 for a more detailed discussion).

To answer both of these problems, in the following section we will directly numerically compute the periodic orbits (in rotating coordinates) in the full problem and solve their variational equations. The characteristic exponents thus calculated will provide information about the stability of the orbits.

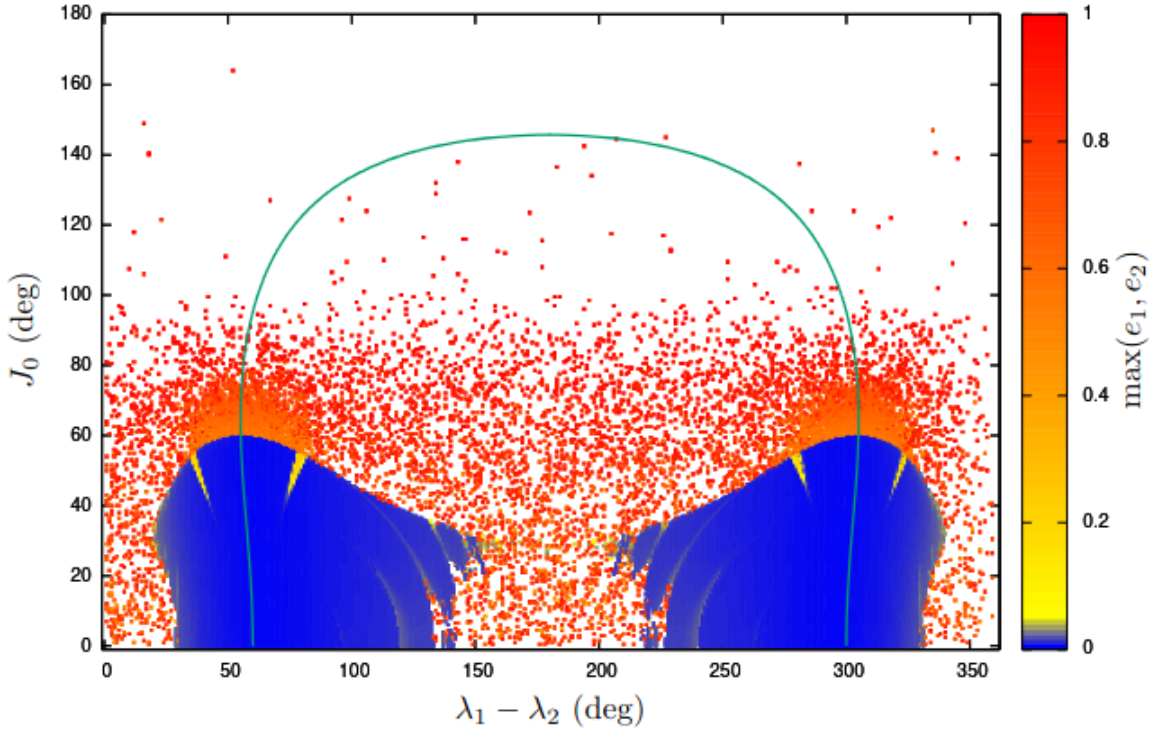


Figure 19: Stability map of the co-orbital region in the plane $(\lambda_1 - \lambda_2, J_0)$ for $m_1 = m_2 = 10^{-3}m_0$. The stability index, from blue to red, corresponds to the maximum eccentricity values reached by the two planets during their integration in the full problem, while the white is associated with orbits that either left co-orbital resonance before the end of integration or collided. The green curve is the projection of $\mathcal{VF}_{\mathcal{L}}$ (computed in the average problem) onto the stability map.

E-2 Numerical results for $\mathcal{VF}_{\mathcal{L}}$ in the full problem

E-2.1 Numerical details

As mentioned in E-1.3, after reduction of the nodes, the orbits that we are looking for are periodic. Due to the three-body problem's scale invariance, this period is arbitrary since resizing an orbit by a factor α corresponds to changing the time scale by a factor of $\alpha^{\frac{2}{3}}$. We will impose that this period T be equal to 1.

The periodic orbits that are members of the $\mathcal{VF}_{\mathcal{L}}$ family can be found out as fixed points of a first return map. Here, we consider the function f defined as follows:

$$f : \begin{cases} \mathbb{R}^9 \rightarrow \mathbb{R}^8 \\ (x; \mathcal{C}) \mapsto \Phi_T^{\mathcal{C}}(x) - x \end{cases} \quad (48)$$

where $x = (R_j, G_j, r_j, w_j)_{j=1,2}$ and $\Phi_T^{\mathcal{C}}$ is the flow at time T of the vector field associated with the Hamiltonian $H_{\mathcal{C}}$ defined in Equation 15. The Hamiltonian system possesses two independent first integrals: the energy and the total angular momentum. This diminishes the rank of f by two, meaning the roots will never be locally unique. This can be solved by adding two sections (see D-3.2): instead of finding roots of f , we search for roots of

$$(f, \sigma_1, \sigma_2) : \mathbb{R}^9 \rightarrow \mathbb{R}^{8+2} \quad (49)$$

The first section, for isolating a point on the orbit (also called a Poincaré section), is

$$\sigma_1(x) = w_1 + w_2 \quad (50)$$

to keep the symmetry between the two small bodies.

For the family section, given the prior insight about $\mathcal{VF}_{\mathcal{L}}$, we choose the mutual inclination J , which can be expressed with the reduced Hill variables as

$$\sigma_2(x; \mathcal{C}) = \cos J - \cos J_p = \frac{\mathcal{C}^2 - G_1^2 - G_2^2}{2G_1 G_2} - \cos J_p \quad (51)$$

where J_p is a parameter we will increase along the family. Note that, contrary to the average problem, J is *not* constant along the orbit; J_p is the value of J at the Poincaré section σ_1 . We will numerically see that this corresponds to the minimum of J along the orbit (cf. Figure 21). For a summary of the different J parameters used, see table Table 1.

Parameter	Problem	Definition	Property
J_0	Average problem	J at zero eccentricity for given $\mathcal{C}$	Maximum J for given $\mathcal{C}$
J_p	Full problem	J on an orbit when $w_1 + w_2 = 0$	Minimum J along the orbit

Table 1: Summary of the J parameters used in the article. J_0 is defined in E-1, and J_p in E-2.1.

We will focus on the case of two equal masses, $m_1 = m_2$, giving us the triplet $(1 - 2\varepsilon, \varepsilon, \varepsilon)$ – motivated by Equation 42, showing that if the masses are small, the periodic orbits are independent of them; see appendix III-3 for a short discussion about unequal masses.

To compute Φ_T , we use the `taylor` software (see D-2.5) in Julia via `TaylorInterface.jl` (see D-5.2). `taylor` allows automatically chooses step size and approximation order when specified both a relative and absolute maximum error; we set both to 10^{-16} .

We do the root search with the Newton-Raphson implementation described in D-5.3; we choose a tolerance of 10^{-12} . As we are searching a continuous family of orbits, we use the previous point as an initial guess for the next search (beginning with the Lagrange equilibrium), and allow for a maximum of 50 iterations (otherwise, the search fails).

E-2.2 $\mathcal{VF}_\mathcal{L}$ for $\varepsilon = 10^{-3}$

For $\varepsilon = 10^{-3}$, we find the family for $J \in [0^\circ; 179^\circ]$; we conjecture these inclined co-orbitals lead to double collisions at $J = 180^\circ$ (the two small masses follow retrograde orbits of identical semi-major axis). Figure 20 is the equivalent in the full problem of Marchal (2009, fig. 3) in the restricted problem and Figure 17 in the average problem; although the angle difference is different in each case: Figure 20 considers draconic true anomalies w_j ; Marchal (2009, fig. 3) considers mean longitudes; and Figure 17 considers draconic mean anomalies $\tilde{\lambda}_j$. However, the variations between these angles are of order of the eccentricities, which are of order ε , meaning the quantitative difference isn't visible; further comparison between w and $\tilde{\lambda}$ can be found in appendix III-2.

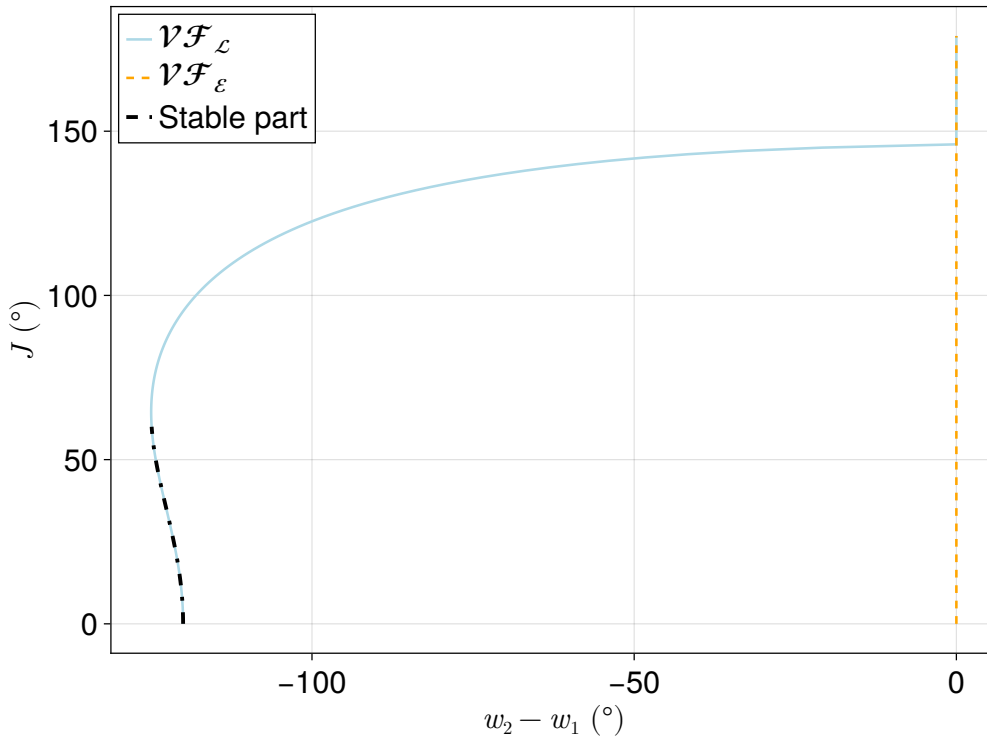


Figure 20: $\mathcal{VF}_\mathcal{L}$ and $\mathcal{VF}_\varepsilon$ for $\varepsilon = 10^{-3}$, in the $(w_1 - w_2, J)$ plane. For stability results, see Chapter F.

We find again the bifurcation described by [Marchal \(2009\)](#) and [Zagouras \(1985\)](#), the connection to $\mathcal{VF}_\varepsilon$ and to $\mathcal{VF}_\mathcal{L}$ with permuted small masses; the upwards branch continues up to the conjectured collisions.

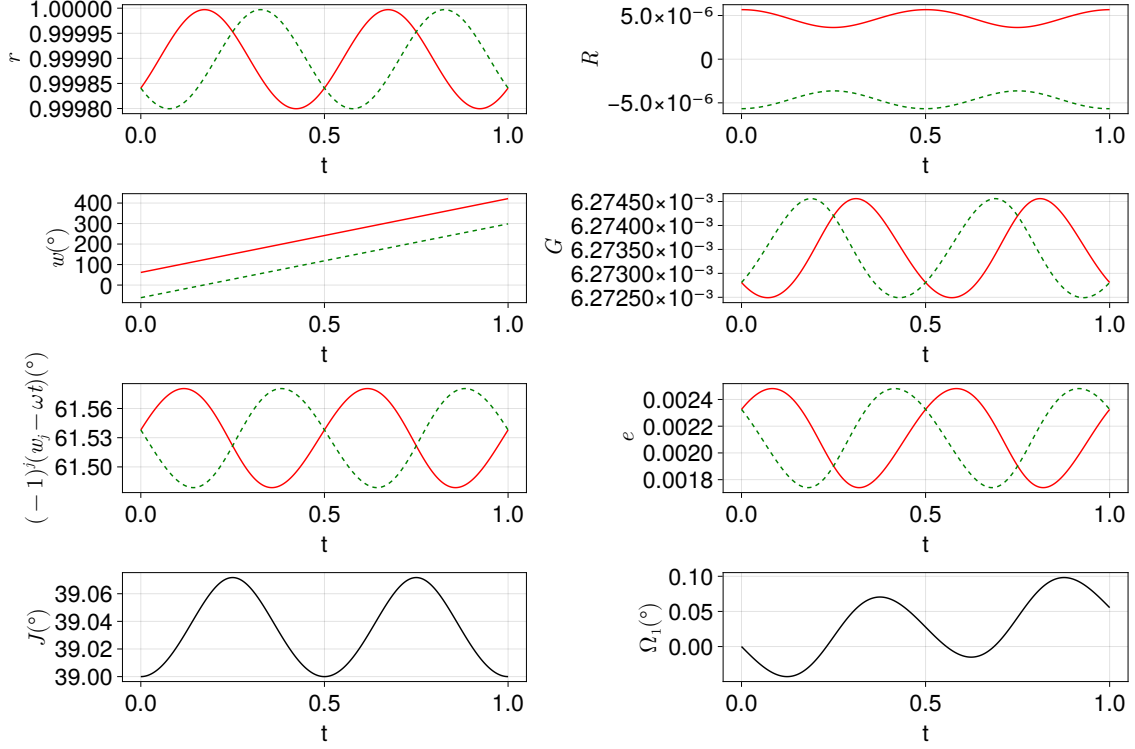


Figure 21: Coordinates for one trajectory, at $\varepsilon = 10^{-3}$ and $J = 39^\circ$. We can see that the initial J , fixed as J_p by a section of the root-finding algorithm, is indeed a minimum of J along the trajectory; we verified that this is a general property for the family. When applicable, the body 1 is in plain red, and the body 2 is in dashed green.

The time evolution of the Hill's coordinates along a single orbit of the family is shown in [Figure 21](#) for $J = 39^\circ$. We can check that the $J_p = J(t = 0)$ defined in [Equation 49](#) is indeed the minimal mutual inclination (this has been checked numerically for all orbits). The initial condition on the orbit is fixed by [Equation 49](#); choosing a different value for $w_1 + w_2$ would yield a different result. For instance, if we had chosen $\sigma_1(x) = w_1 + w_2 - \pi$ instead, we would have selected the maximal mutual inclination. However, as indicated above, along a trajectory, the variations of J remain of order ε (see [Figure 21](#) bottom left). Therefore, choosing its maximum or minimum value as an initial condition has very little influence on the results presented below.

Similarly, we confirm that the planetary eccentricities (third row, right in [Figure 21](#)) are of order of the mass ratios, around 10^{-3} . It is worth noting that all variables (apart from the non-periodic ones: $w_1 + w_2$ and Ω) are $\frac{T}{2}$ -periodic (up to numerical error). The longitude of the nodes Ω has a linear behavior associated with its precession, superposed with low-amplitude $\frac{T}{2}$ -periodic oscillations. Note that, in the full problem, the precession frequency corresponding to this trajectory is of 0.05542° per period (period of 6496 orbital revolutions), while we found 0.0555° by application of [Equation 47](#). This indicates that, at least up to this value of the small parameter ε , the average problem provides a good approximation of

the trajectories of the full problem. The analytical approximation in Equation 63 provides, for the same value of inclination, a frequency equal to $0.0523^\circ/\text{period}$, which suggests that this analytical approximation is still quite reasonable for $J = 39^\circ$, even though it was obtained using an expansion at $J = 0$.

E-3 Symmetries of $\mathcal{VF}_\mathcal{L}$

E-3.1 Initial symmetries

The algorithm as presented in E-2.1 fails at higher masses, beyond $\varepsilon \approx 0.14$ (halfway to the equal masses, at $\varepsilon = \frac{1}{3}$). Brute force, such as increasing numerical precision or decreasing the J step, can only go so far; we need to improve the algorithm. In order to do so, we can examine the properties of the family we found up until now, and more specifically its symmetries – inspired by the conjectured link with P_{12} , known for its high symmetry order.

E-3.1.1 The half-period symmetry

First, we can find numerically that the orbits of $\mathcal{VF}_\mathcal{L}$ have this symmetry: at half the period, they are reversed with respect to the origin. In equations,

$$\forall \mathbf{x} \in \mathcal{VF}_\mathcal{L}, \forall t \in \mathbb{R}, \mathbf{x}\left(t + \frac{T}{2}\right) = -\mathbf{x}(t) \quad (52)$$

E-3.1.2 The time-reversal symmetry

The orbits of $\mathcal{VF}_\mathcal{L}$ have another symmetry:

$$\forall \mathbf{x} \in \mathcal{VF}_\mathcal{L}, \forall t \in \mathbb{R}, \begin{cases} \mathbf{r}(-t) = R_y(\pi) p_{1 \leftrightarrow 2} \mathbf{r}(t) \\ \tilde{\mathbf{r}}(-t) = S_y p_{1 \leftrightarrow 2} \tilde{\mathbf{r}}(t) \end{cases} \quad (53)$$

where $R_y(\pi)$ is the rotation around the y -axis by an angle π , S_y is the symmetry with respect to the Oxz -plane and $p_{1 \leftrightarrow 2}$ is the permutation of the small bodies. This symmetry depends on the choice of initial time $t = 0$, and thus on the chosen Poincaré section – which is $w_1 + w_2 = 0$ in our case.

E-3.2 Consequences on the periodic orbit search

E-3.2.1 Halving integration time

The half-period symmetry means that instead of searching for roots of $\Phi_T - I$, we can search for roots of $\Phi_{\frac{T}{2}} + I$ (because $\Phi_{\frac{T}{2}}(-\mathbf{x}) = -\Phi_{\frac{T}{2}}(\mathbf{x})$). This has two benefits: not only does it halve integration time, making the computations faster (because integration time is a numerical bottleneck), it also excludes periodic orbits which don't have this symmetry, improving numerical stability.

E-3.2.2 Adding sections

The time-reversal symmetry can be translated in terms of additional sections for the root-finding algorithm; indeed, taking Equation 53 at $t = 0$ gives us

$$\begin{cases} \mathbf{r} = R_y(\pi) p_{1 \leftrightarrow 2} \mathbf{r} \\ \tilde{\mathbf{r}} = S_y p_{1 \leftrightarrow 2} \tilde{\mathbf{r}} \end{cases} \Rightarrow \mathbf{x} = \rho \circ p_{1 \leftrightarrow 2} \mathbf{x} \quad (54)$$

where $\rho : (\mathbf{r}, \tilde{\mathbf{r}}) \mapsto (R_y(\pi)\mathbf{r}, S_y\tilde{\mathbf{r}})$. We still have to translate this symmetry, and especially the rotation part, in terms of reduced Hill coordinates. To do this, we explicit the effect of ρ in Hill coordinates, by reconstructing the Hill coordinates and propagating the effect. We write $\mathbf{r}_j$ the positions and $\tilde{\mathbf{r}}_j$ the momenta in cartesian coordinates, in the rotating heliocentric frame where the symmetry happens. The positions are multiplied by the square matrix of diagonal $(-1, +1, -1)$, which we condense as $\mathbf{diag} \begin{pmatrix} -1 \\ +1 \\ -1 \end{pmatrix}$; the momenta are multiplied by $\mathbf{diag} \begin{pmatrix} +1 \\ -1 \\ +1 \end{pmatrix}$.

Radii	$r_j = \mathbf{q}_j \Rightarrow \boxed{r_j \mapsto_{\rho} r_j}$
Radial velocities	$R_j = \frac{\mathbf{r}_j \cdot \tilde{\mathbf{r}}_j}{r_j} \Rightarrow \boxed{R_j \mapsto_{\rho} -R_j}$
Angular momenta	$\mathbf{G}_j = \mathbf{q}_j \times \mathbf{p}_j \Rightarrow \mathbf{G}_j \mapsto_{\rho} \mathbf{diag} \begin{pmatrix} +1 \\ -1 \\ +1 \end{pmatrix} \cdot \mathbf{G}_j$
	$G_j = \mathbf{G}_j \Rightarrow \boxed{G_j \mapsto_{\rho} G_j}$
Inclination	$\cos(I_j) = \mathbf{G}_j \cdot \mathbf{e}_z \Rightarrow \boxed{I_j \mapsto_{\rho} I_j}$ as $I_j \in [0, \pi]$
Ascending node vector	$\boldsymbol{\Omega}_j = \mathbf{e}_z \times \frac{\mathbf{G}_j}{G_j} \Rightarrow \boldsymbol{\Omega}_j \mapsto_{\rho} \mathbf{diag} \begin{pmatrix} -1 \\ +1 \\ 0 \end{pmatrix} \cdot \boldsymbol{\Omega}_j$
	$\Omega_j = \text{angle}(\mathbf{e}_x, \boldsymbol{\Omega}_j) \Rightarrow \boxed{\Omega_j \mapsto_{\rho} \pi - \Omega_j} \quad (55)$

Thus, we end up with three more sections:

$$\begin{cases} R_1 + R_2 = 0 \\ G_1 - G_2 = 0 \\ r_1 - r_2 = 0 \end{cases} \quad (56)$$

These sections exclude families without such a symmetry, helping with potential bifurcations, on top of increasing numerical stability.

E-4 A sister family: $\mathcal{VF}_\varepsilon$

Although it isn't the focus of our study, we present here a few findings about the family $\mathcal{VF}_\varepsilon$ meets at high inclination: $\mathcal{VF}_\varepsilon$, the vertical family emerging from the Euler relative equilibrium. Similarly to the Euler equilibrium, all orbits are linearly unstable for any mass ratios. A remarkable feature is the central body's movement: it always is confined to the vertical axis, again for all mass ratios; for two equal masses, the shape is an isoscele triangle rotating and pulsating non-isotropically.

As J increases, the two orbiting bodies have increasingly close encounters; for two equal masses, the family ends at double collisions for $J = 180^\circ$.

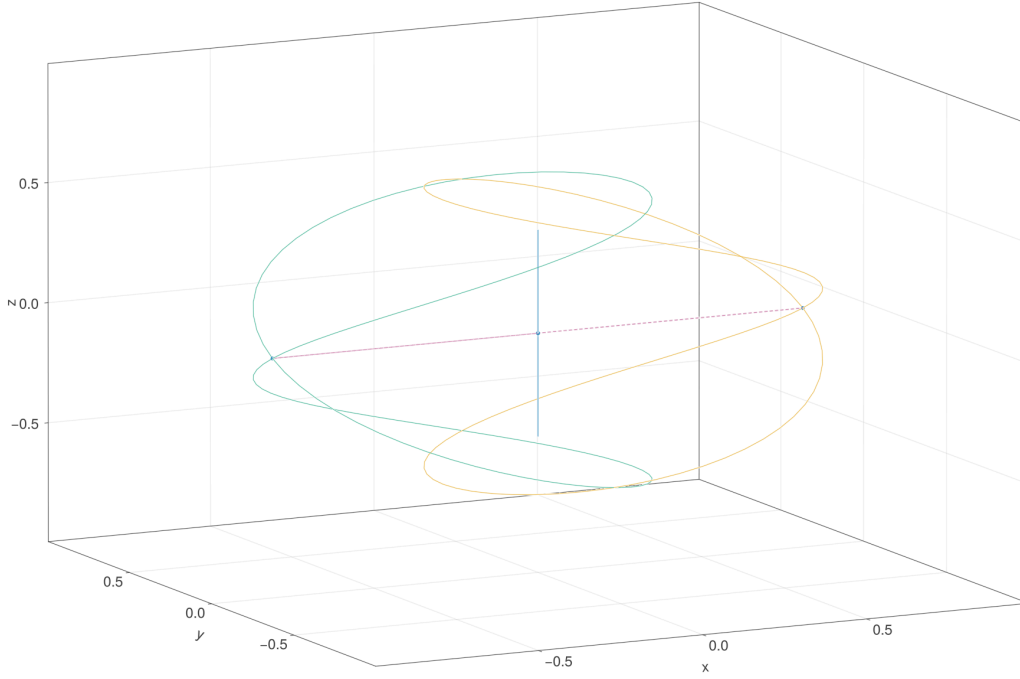


Figure 22: Orbit of $\mathcal{VF}_\varepsilon$, at $\varepsilon = 0.21$, $J = 130^\circ$, in the rotating heliocentric frame.

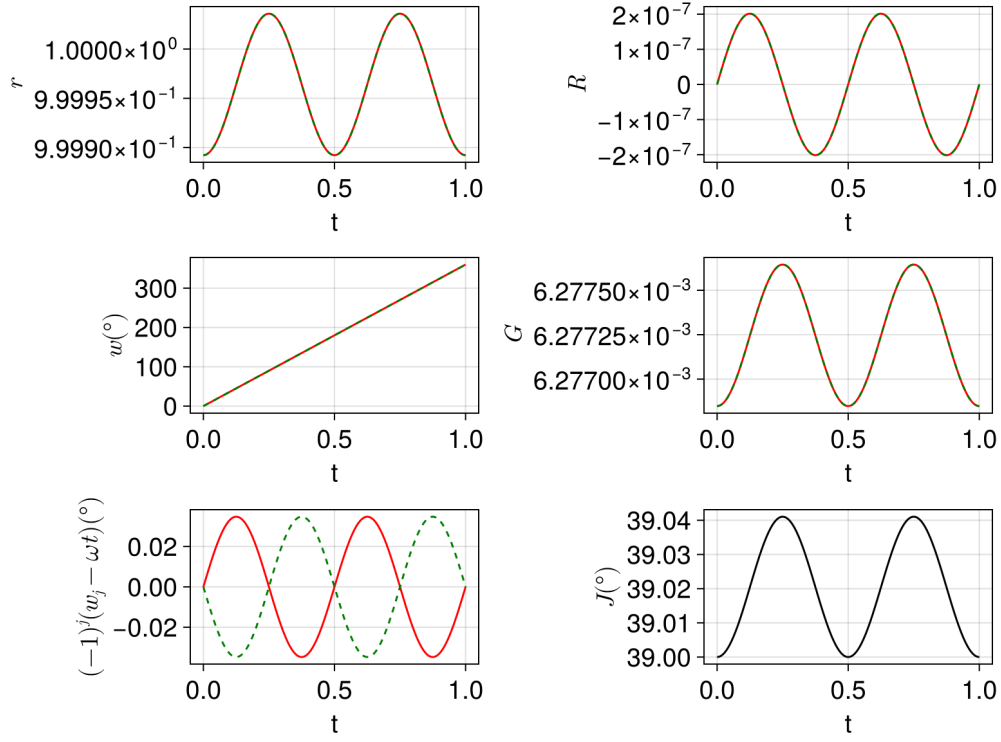


Figure 23: Coordinates along one period of the $\mathcal{VF}_\varepsilon$ orbit at $\varepsilon = 10^{-3}$, $J = 39^\circ$.

Chapter F

Stability along $\mathcal{VF}_\mathcal{L}$

There is nothing stable in the world; uproar's your only music.

— John Keats

F-1 Stability at low masses, and stable remnants beyond Gascheau

F-1.1 Stability in the planetary limit

The family is stable from Lagrange ($J = 0^\circ$) up until about 60° ; the branch starting from Euler, and everything above the 60° threshold is unstable. Figure 24 shows the eigenvalue trajectories in the complex plane around the stable $\rightarrow$ unstable change, with four eigenvalues colliding in pairs and leaving the unit circle. This stability change corroborates the frequency analysis results shown in Figure 19 and Leleu (2016); a long-term-stable region surrounds the linearly stable part of the family.

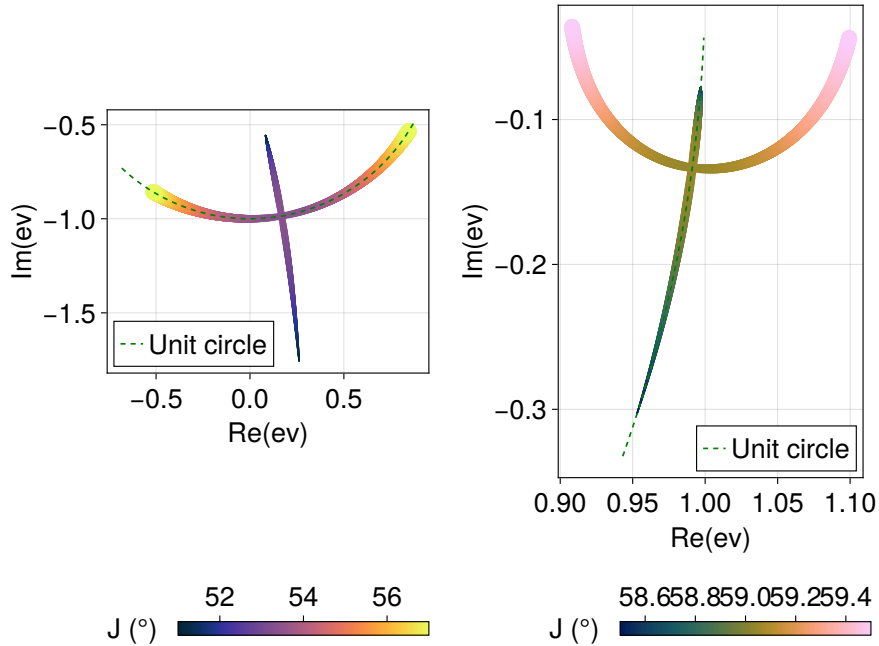


Figure 24: Trajectories of two pairs of monodromy matrix eigenvalues in the complex plane, along the stability changes of $\mathcal{VF}_\mathcal{L}$. Here, $\varepsilon = 0.04$. The collisions are mirrored at positive imaginary parts, as eigenvalues come in pairs of complex conjugates (because we use symplectic coordinates in a Hamiltonian system); thus, four eigenvalues (out of the total eight) collide in two double collisions. Curve width increases with J .

After this stability change, two other bifurcations happen shortly after in the family: the four eigenvalues collapse on the real axis; then, two of them quickly fall back on the unit circle, colliding at 1 and separating afterwards. Figure 25 shows the trajectories including them.

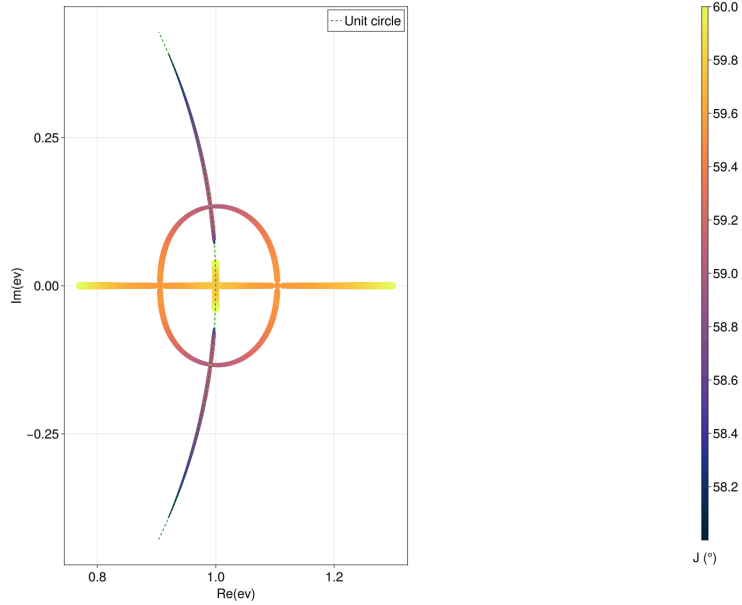


Figure 25: Eigenvalue trajectories at $\varepsilon = 0.04$; shortly after the bifurcation seen in Figure 24 right, two other bifurcations happen on the real axis.

F-1.2 $\mathcal{VF}_{\mathcal{L}}$ above Gascheau's masses

Since the $\mathcal{VF}_{\mathcal{L}}$ family appears to play a central role in the stability of the co-orbital regions surrounding equilateral equilibria, it is important to know whether its stability, at least partial (we have seen that this is the case for mutual inclinations ranging from 0° to approximately 60°), persists for planetary masses higher than those considered so far, and to explore the consequences for the whole co-orbital regions. Although Lagrange's equilateral (circular) configurations become unstable above the Gascheau value (in our case, when $\varepsilon \geq \frac{3-2\sqrt{2}}{9} \approx 0.0191$) it has been shown that, in the elliptical case, stability can persist beyond this value Danby (1964), Roberts (2002b), Nauenberg (2002). It therefore seems relevant to ask whether a similar phenomenon exists for inclined quasi-circular orbits.

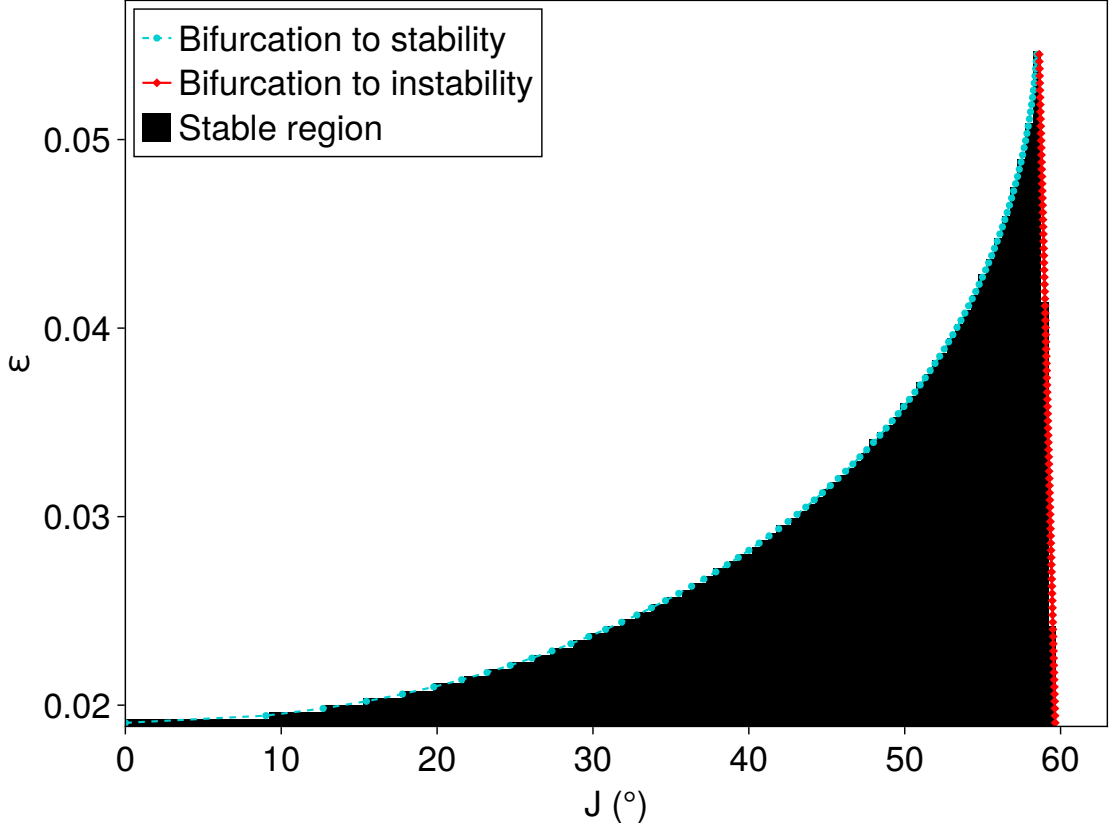


Figure 26: Linear stability of $\mathcal{VF}_{\mathcal{L}}$ orbits in the (J, ε) plane, with bifurcations up to 10^{-4} on J . In the black region, the orbits are stable; in the white region, they are unstable. The blue and red curves show the bifurcations delimiting the stable region (respectively unstable $\rightarrow$ stable and stable $\rightarrow$ unstable, for increasing J).

In order to answer this question, we repeat the numerical analysis from E-2 for various mass parameters ε ; results are shown in Figure 26. Each pixel is a point of the family at a given mutual inclination J_p and mass parameter ε ; its color is black if the orbit is linearly stable (the maximum distance of characteristic exponents to the unit circle is less than 10^{-6}), and white otherwise. We confirm a stable region above Gascheau's value (which is at the figure's bottom): at a certain J_p value (given by the green curve in Figure 26), the four eigenvalues from Equation 20, which leave the unit circle above Gascheau, collide back in pairs, rejoining the circle along the family; the bifurcation is shown in the left plot of Figure 24 for $\varepsilon = 0.04$. The linear stability limit, where $\mathcal{VF}_{\mathcal{L}}$ becomes unstable – delimited by the red curve in Figure 26 – is the continuation of the bifurcation discussed in E-2.2; the mutual inclination J_p where it happens slowly decreases with ε . The stability remnant decreases in width with ε , eventually disappearing for $\varepsilon \approx 0.058$.

F-2 Influence on the co-orbital region

F-2.1 Chimney equivalent in 3D

As we did in E-1.3, it is important to examine the possible links between the stability of $\mathcal{VF}_{\mathcal{L}}$ and that of co-orbital regions for different values of planetary masses. For each

value of ε in $\{0.02, 0.03, 0.04, 0.05\}$, we integrate, over 10 000 orbital periods in the full problem, a grid of initial conditions chosen as follows. We consider a discrete set $(r_j(J_p), R_j(J_p), w_j(J_p), G_j(J_p))$ of $\mathcal{VF}_\mathcal{L}$ initial conditions parameterized by $J_p = k_J \delta_J$ with $k_J \in \{0, \dots, 320\}$ and $\delta_J = 0.2^\circ$. In order to build a 2D initial conditions grid, we consider curves parallel to $\mathcal{VF}_\mathcal{L}$ for which only w_2 is translated as $w_2 = w_2(J_2) + k_w \delta_w$ with $k_w \in \{-40, \dots, 40\}$ and $\delta_w = 0.5^\circ$. The results of these simulations are plotted in Figure 27, in the coordinates system $(J, \Delta w, \varepsilon)$, where $\Delta w = w_1 - w_2$. The four blue horizontal layers correspond to regions of long-term stability that group together trajectories for which the relative diffusion in orbital frequency is less than 10^{-6} , thus ensuring their stability [Robutel and Laskar \(2001\)](#). As is the case for $\varepsilon = 0.001$ (see Figure 19), the linearly stable section of the family $\mathcal{VF}_\mathcal{L}$ occupies the center of these regions, whose boundary in the J direction coincides with the red curves marking the stable-unstable transitions (solid lines for the lower bound and dotted lines for the upper bound) of $\mathcal{VF}_\mathcal{L}$ reproduced in Figure 24. As mentioned in E-1.3, the contours of stability domains are sculpted by secondary resonances, the most notable being the 2 : 1 resonance between the libration frequency of the angle $w_1 - w_2$ and the mean motion. Note that the resonance 3 : 1 between the same frequencies across the co-orbital zone for $\varepsilon = 0.01$ (the figure is not reproduced here).

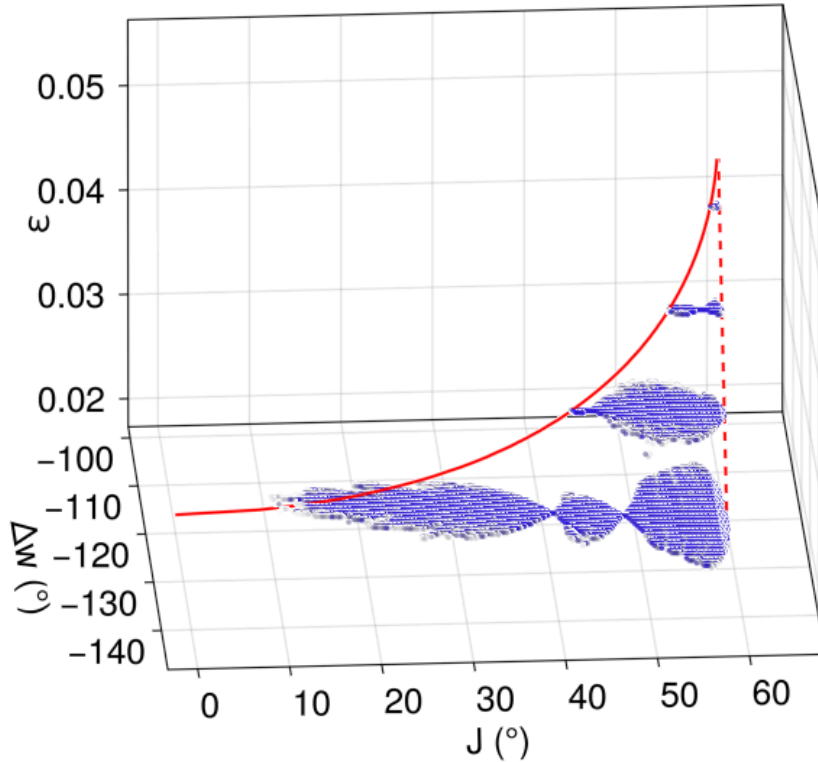


Figure 27: Stability maps (in blue) for $\varepsilon = 0.02, 0.03, 0.04, 0.05$. The spectral stability limits are represented by red lines (see the text for more information).

F-3 Astronomical consequences?

Inspired by [Marchal \(2009\)](#), the stability change of $\mathcal{VF}_L$ could explain the distribution of Jupiter Trojans in inclination; we would predict that the maximum stable inclination being around 60° , no Trojan could have a long-term inclination above it. However, we see in [Figure 28](#) that the population density falls around 10° before.

Indeed, there are many other instabilities before the $\mathcal{VF}_L$ stability change intervenes, due to the resonances with the fundamental frequencies of the Solar System (see eg. [Robutel and Gabern \(2006\)](#)). However, [Qi and de-Ruiter \(2020\)](#) mentions a topology change around $i = 60^\circ$ which we conjecture to be linked to $\mathcal{VF}_L$'s stability change.

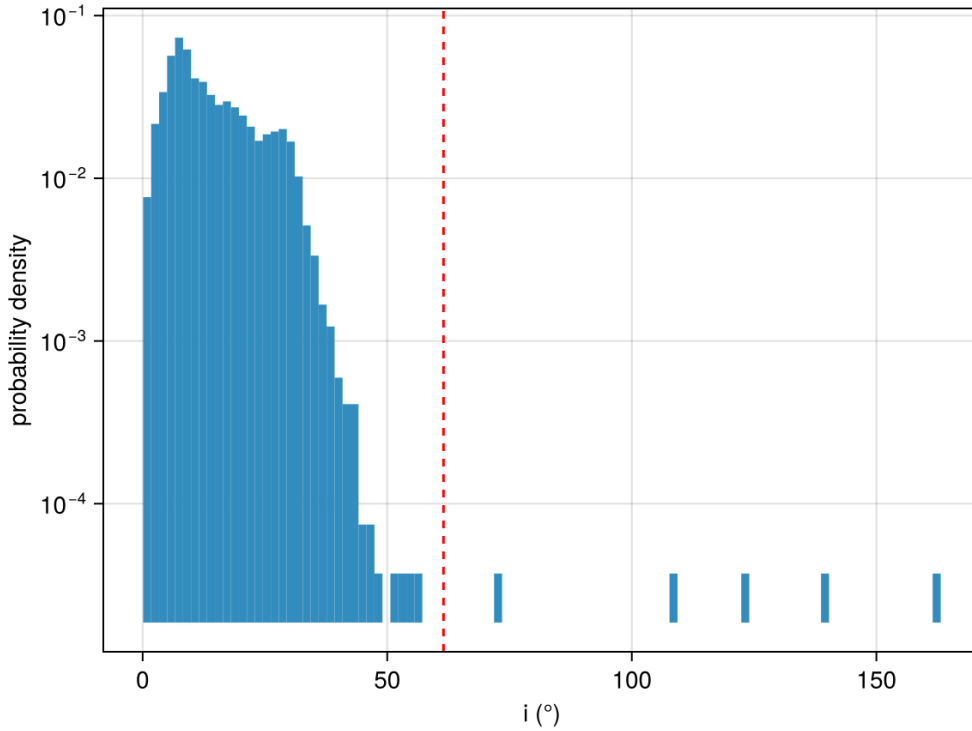


Figure 28: Probability density function of inclination among the 16494 Trojan asteroids. The dashed red line represents the stability change in $\mathcal{VF}_L$ asteroid density falls around 10° beforehand, the low bins each representing single asteroids. Data retrieved on 22/06/2026 from [SSODNET](#)¹⁵, a service of the LTE.

¹⁵<https://ssp.imcce.fr/webservices/ssodnet/>

Chapter G

Linking $\mathcal{VF}_{\mathcal{L}}$ to P_{12}

“The last, utter sweep which has no confines and which outreaches fancy and mathematics alike.”

— H. P. Lovecraft, “Through the Gates of the Silver Key” - written with E. Hoffman Price, October 1932 - April 1933; first published in *Weird Tales*, Vol. 24, No. 1 (July 1934)

G-1 A few words about rotating coordinates

Until now, we called orbits periodic if, after a finite non-zero amount of time, they came back to their initial state: positions and momenta of all three bodies. In [Marchal \(2001\)](#), Marchal calls these *absolute* periodic orbits, distinct from *relative* periodic orbits, where the bodies come back to their *relative positions and momenta* compared to the other bodies. He claims:

It was traditional to discriminate between “absolute periodic orbits” [...], and “relative periodic orbits” [...]. This subtle and artificial distinction is generally neglected today and, if necessary, it is specified that the orbit is “periodic in a suitable uniformly rotating set of axes”.

— [Marchal \(2001\)](#)

While the distinction remains important for numerical purposes, the orbits of $\mathcal{VF}_{\mathcal{L}}$ are indeed relative periodic orbits, and classically periodic in a rotating frame. The draconic frame (defined in [B-2.3](#)) rotates non-uniformly (see eg. bottom right of [Figure 21](#): the nodes, which the frame is following, do not rotate at a linear rate). However, if an orbit is periodic of period T in a frame rotating at a non-uniform rate $\Omega(t)$, it will be periodic in a frame rotating at a uniform rate $\frac{\Omega(T)-2\pi}{T}$. We show the precession rate along part the family in [Figure 29](#), as an example.

Another uniformly rotating frame is of interest: the orbits of P_{12} are choreographies (meaning all three bodies follow the same trajectory) in a frame rotating *opposite* to the initial Lagrange triangle’s rotation. In this frame, the Lagrange triangle rotates *twice* per period; we call this frame, rotating at $\frac{\Omega(T)-2\pi}{T}$, the *anti-rotating* frame. An example $\mathcal{VF}_{\mathcal{L}}$ trajectory (at non-equal masses) and a P_{12} trajectory are shown in [Figure 30](#); we can recognize the P_{12} “scallop” trajectories in $\mathcal{VF}_{\mathcal{L}}$, suggesting a link.

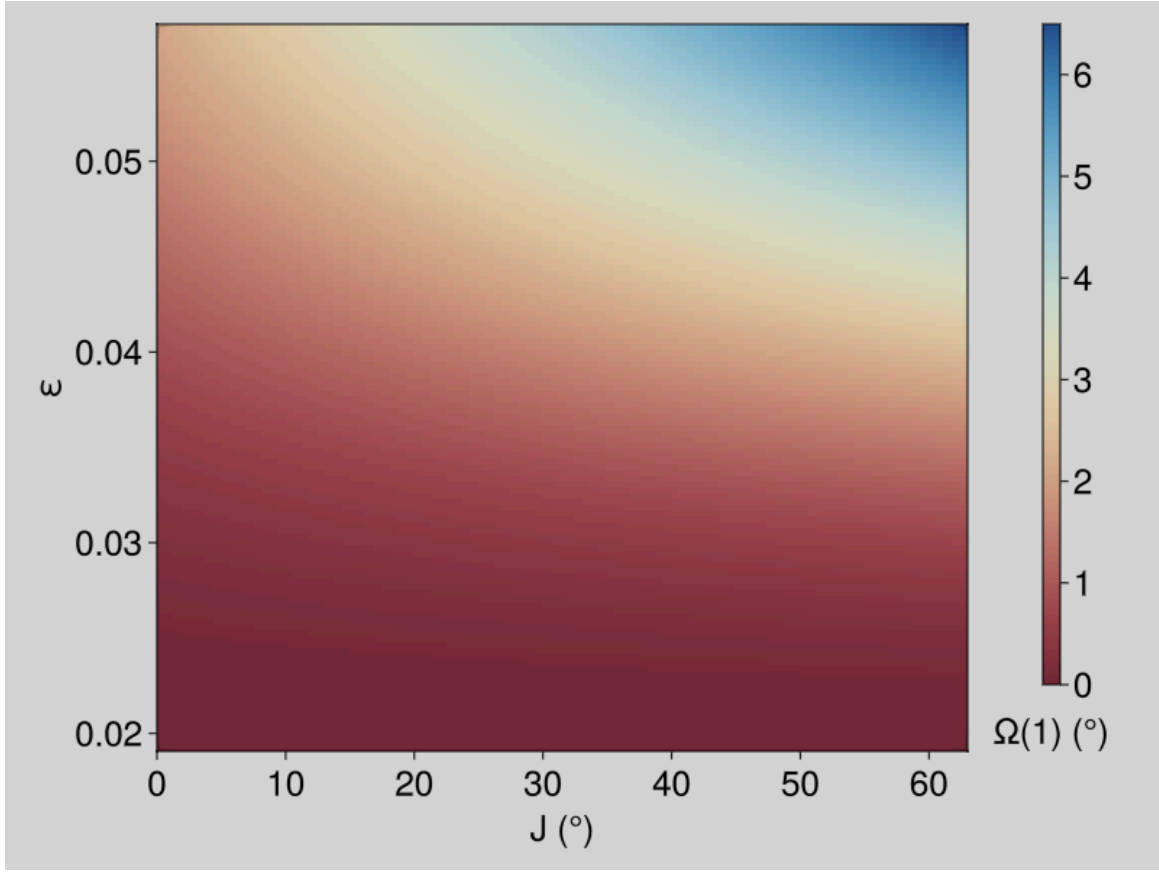


Figure 29: $\Omega_1(T)$ in the (J, ε) plane; as a reminder, we take T as a constant along the family.

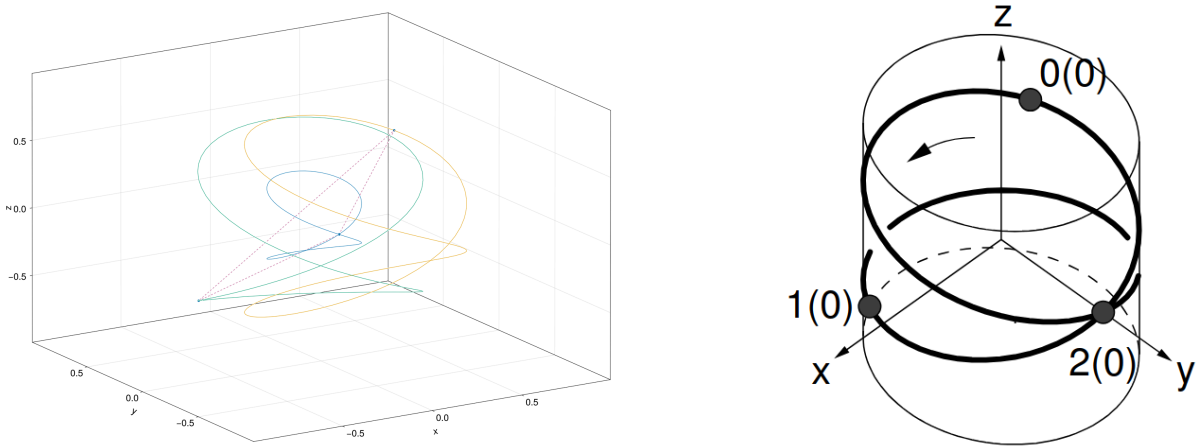


Figure 30: Left: $\mathcal{VF}_{\mathcal{L}}$ trajectory in the anti-rotating frame, at $\varepsilon = 0.21$, $J = 106^\circ$. The scallop-shaped orbits of P_{12} are recognisable. Right: P_{12} orbit in the anti-rotating frame, dubbed the “scallop”. Source: figure 1 of [Chenciner and Fejoz \(2008\)](#).

G-2 Attaining the equal masses

G-2.1 P_{12} and $\mathcal{VF}_{\mathcal{L}}$

When increasing ε up to $\frac{1}{3}$, we see a sharp change of behaviour at the equal masses, indicative of a bifurcation. While at near equal masses, the algorithm outputs the continuation of $\mathcal{VF}_{\mathcal{L}}$, ending at $\Delta w = 0$ with double collisions, at equal masses the family in the $(\Delta w, J)$ plane changes completely after $J \approx 90^\circ$, ending at the Eight figure – see [Figure 31](#) for an

example in the $(\Delta w, J)$ plane. In 3D space, this translates to a symmetry break: under 90° , $\mathcal{VF}_{\mathcal{L}}$ at near equal masses is a quasi-choreography in a suitably rotating frame (P_{12} orbits are such choreographies), with very close orbits; after the critical value, the orbits significantly distance each other, converging to a $\mathcal{VF}_{\mathcal{E}}$ orbit at $J \approx 120^\circ$. While we didn't prove $\mathcal{VF}_{\mathcal{E}}$ orbits aren't a choreography in a suitable frame, the central body's starkly different orbit (one-dimensional) leads us to believe they aren't.

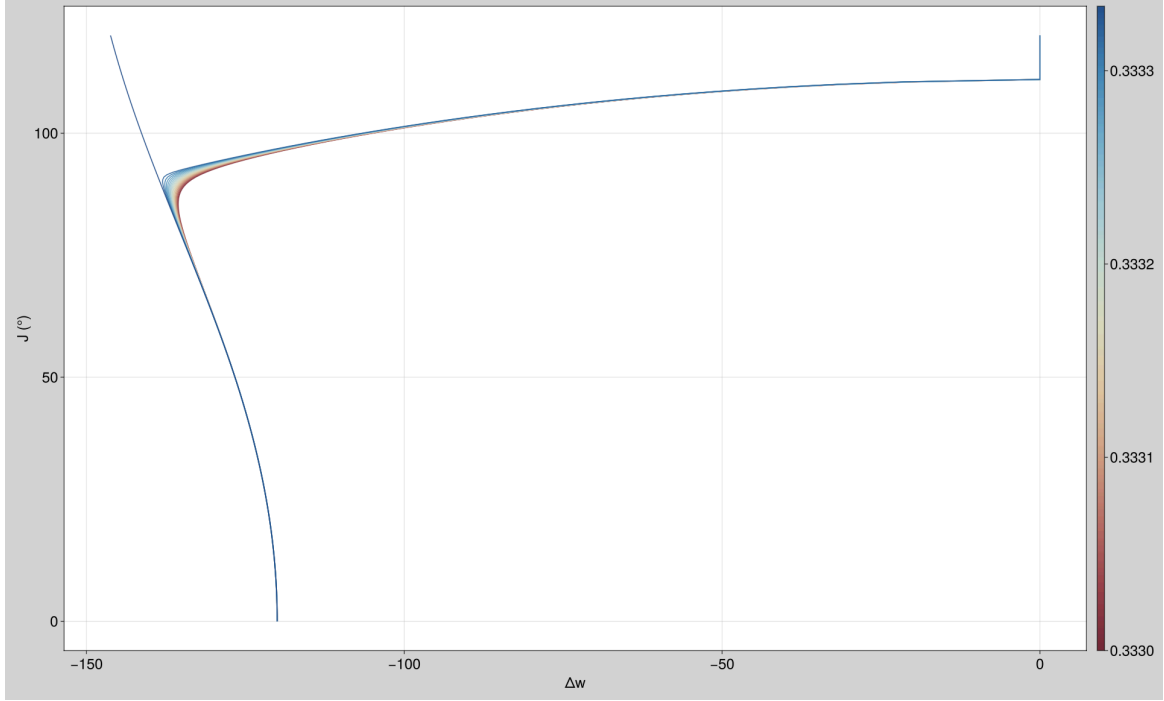


Figure 31: $\mathcal{VF}_{\mathcal{L}}$ in the $(\Delta w, J)$ plane for $\epsilon \in [0.333, \frac{1}{3}]$. At equal masses ($\epsilon = \frac{1}{3}$), the family found by the algorithm changes significantly.

Indeed, when taking a point further along $\mathcal{VF}_{\mathcal{L}}$ as an initial guess for the Newton-Raphson algorithm allows us to confirm that the $\mathcal{VF}_{\mathcal{L}}$ branch still exists at equal masses, confirming a bifurcation between P_{12} and $\mathcal{VF}_{\mathcal{L}}$ at $J_p = 91.54^\circ$. This prompted us to look for other bifurcations along P_{12} . We find a total of 3; the second is the bifurcation between P_{12} and $\mathcal{VF}_{\mathcal{L}}$, the other two are detailed below.

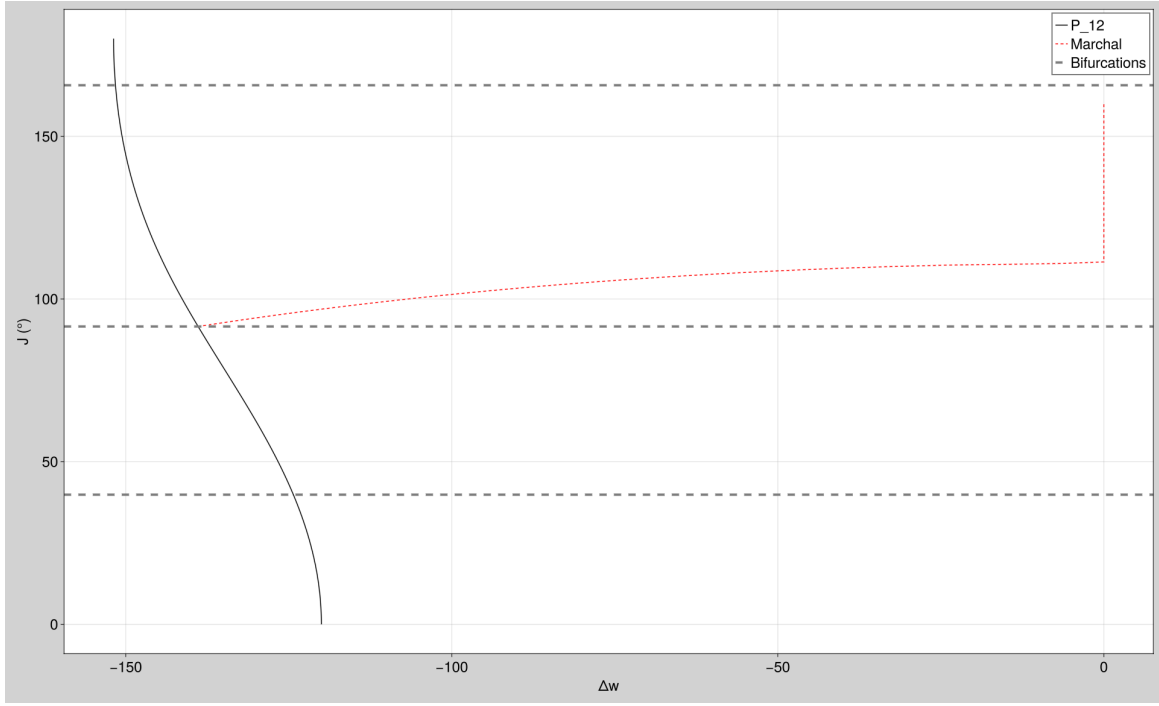


Figure 32: $\mathcal{VF}_{\mathcal{L}}$ and P_{12} families at equal masses, in the $(\Delta w, J)$ plane.

G-2.2 Bifurcations of P_{12}

G-2.2.1 The first bifurcation: turn-around to a “lazy Eight”

The first bifurcation, at $J = 39.83^\circ$, is currently explored by Hénot, Fejoz & Chenciner. It was first discovered by [Calleja et al. \(2021\)](#), who seem to have followed the wrong branch up to another Eight than P_{12} ’s: a “lazy” Eight, horizontal instead of vertical.

This bifurcation also exists in the non-equal masses case; we confirmed its existence at $\varepsilon = 10^{-3}$. We conjecture it is the one [Zagouras \(1985\)](#) found in the restricted problem. [Doedel et al. \(2007\)](#) followed $\mathcal{VF}_{\mathcal{L}}$ (which they call V4 and V5, emerging from L_4 and L_5) in the circular restricted problem, and also find a single bifurcation on each family before they merge with $\mathcal{VF}_{\mathcal{E}}$ (called V3 there), confirming [Zagouras \(1985\)](#)’s results. At the bifurcation points, V41 and V51, V4/V5 bifurcate with the families W4/W5, which merge at another bifurcation.

G-2.2.2 The second bifurcation: between $\mathcal{VF}_{\mathcal{L}}$ and P_{12}

The bifurcation has been described above; we also have found it at non-equal masses, eg. $\varepsilon = 10^{-3}$. This lets us conjecture a continuation of P_{12} exists beyond the equal masses; we have found the branch for $\varepsilon = 10^{-3} - 10^{-5}$. However, we haven’t yet developed the tools to follow bifurcating families, which would allow us to find this conjectured continuation of P_{12} beyond; at the moment, everything relies on which branch the Newton-Raphson algorithm will converge to.

[Doedel et al. \(2003\)](#) have continued the Eight for non-equal masses $(1 - 2\varepsilon, \varepsilon, \varepsilon)$, but with $\varepsilon > \frac{1}{3}$, the case opposite of ours (see [Figure 33](#) for example orbits); these orbits could be the end point of the conjectured P_{12} continuation.

G-2.2.3 The third bifurcation: unexplored

The third bifurcation is close to the Eight, at $J = 165.70^\circ$. As far as we know, it hasn't been described yet. Finding P_{12} 's conjectured continuation would allow us to confirm or deny its existence beyond the equal masses case.

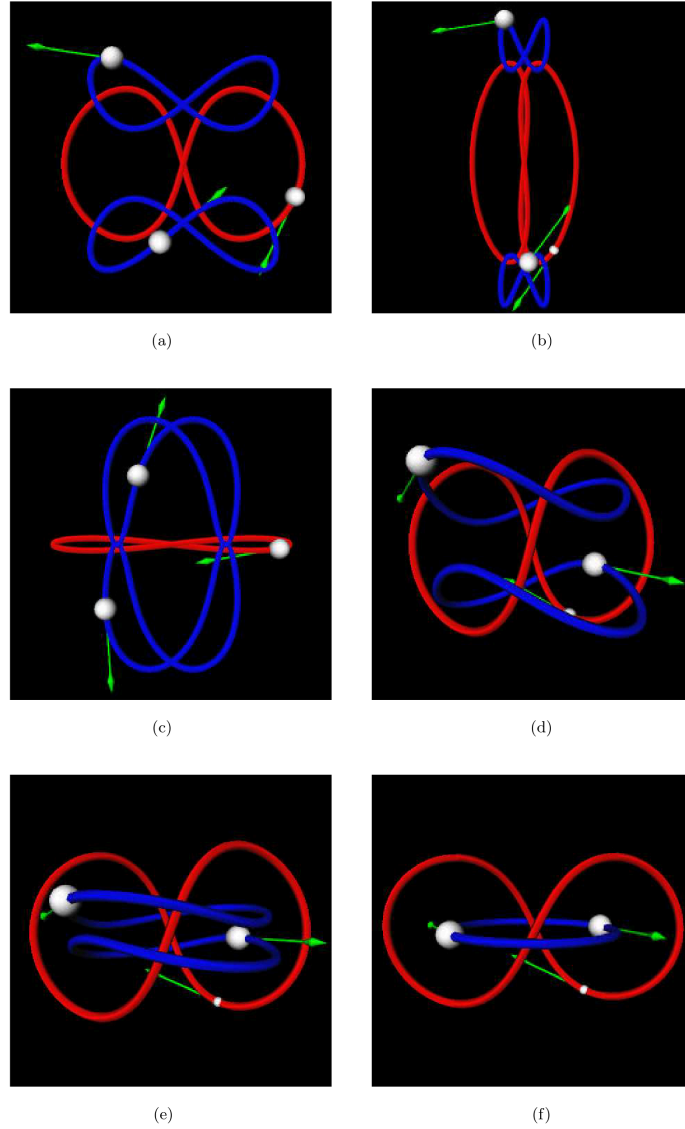


Figure 33: Example orbits of continuations of the Eight for non-equal masses, for varying mass ratios. Source: figure 19 of [Doedel et al. \(2003\)](#).

Chapter H

Conclusion

[...]

Η Ιθάκη σ' έδωσε τ' ωραίο ταξίδι.

Χωρίς αυτήν δεν θα' βγαίνες στον δρόμο.

Άλλα δεν έχει να σε δώσει πια.

Κι αν πτωχική την βρεις, η Ιθάκη δεν σε γέλασε.

Έτσι σοφός που έγινες, με τόση πείρα,

ήδη θα το κατάλαβες οι Ιθάκες τι σημαίνουν.

[...]

Ithaca gave you the marvelous journey.

Without her you wouldn't have set out.

She has nothing left to give you now.

And if you find her poor, Ithaca won't have fooled you.

Wise as you will have become, so full of experience,

you'll have understood by then what these Ithakas mean.

— *Ιθάκη*, Constantine P. Cavafy, 1910

Ithaca, translated by Edmund Keeley and Philip Sherrard.

This thesis was conducted from September 2023 to September 2026 by Alexandre Prieur, under the supervision of Philippe Robutel and Jacques Fejoz, at the Astronomy and Dynamical Systems team in the LTE lab, of the Observatoire de Paris. Results up to [Chapter F](#) were published in [Prieur and Robutel \(2026\)](#), and the results of [Chapter G](#) are in an article in preparation.

H-1 A new numerical framework for recurring dynamics exploration

We showcased new numerical tools devoted to local search for periodic orbits, by leveraging the `taylor` software in Julia, a modern programming language – through [TaylorInterface.jl](#)¹⁶, an open-source publicly available package. We integrated it in a root-finding algorithm, with lifting of degeneracies through sections. The framework is general and can be applied to any dynamical system, although we focused on the three-body problem; the version associated with [Prieur and Robutel \(2026\)](#) is available at the Zenodo archive [Prieur \(2025\)](#).

H-2 $\mathcal{VF}_{\mathcal{L}}$ and P_{12}

After retracing the state of the art about the vertical family we call $\mathcal{VF}_{\mathcal{L}}$, we extended the results from the restricted to the general case, showing the family exists, and joins with the vertical family $\mathcal{VF}_{\mathcal{E}}$ emerging from the Euler colinear solution at high inclination. We found the orbits are stable up to a mutual inclination of around 60° , retaining a stable part even when the Lagrange triangle becomes unstable – for masses up to three times above Gascheau's limit, and showed that the family governs the co-orbital region's stability. Finally, we shed light on Marchal's generalised conjecture, discovering the bifurcation between P_{12} and $\mathcal{VF}_{\mathcal{L}}$ at equal masses; we also linked this bifurcation to the one detected in the restricted case, and found another new bifurcation along P_{12} .

¹⁶<https://github.com/Alseidon/TaylorInterface.jl>

H-3 Future prospects

BIFURCATIONS OF P_{12} We found three bifurcations along P_{12} . While the first is currently studied by Hénot, Fejoz & Chenciner, the second one, between P_{12} and $\mathcal{VF}_{\mathcal{L}}$ has yet to be continued, and the third one is completely unknown.

SYMMETRIES The symmetries of $\mathcal{VF}_{\mathcal{L}}$, their relation to those of P_{12} , and the symmetry breaks at the bifurcations, have yet to be fully explored.

FOLLOWING BIFURCATING FAMILIES While we are able to detect bifurcations, without prior knowledge search for other families has yet to be implemented.

OTHER USE CASES OF THE FRAMEWORK Other use cases of the framework are currently envisioned, eg. in the resonant planetary four-body problem with Jérémy Couturier.

IMPROVE ROOTFINDING We currently use a simple Newton-Raphson root-finding algorithm; variations and alternatives could improve the rate of success with numerically unstable functions, such as close approaches in the N-body problem.

INVESTIGATE NUMERICAL INTEGRATION ALTERNATIVES While `taylor` is a powerful integrator, the fact that it is written in C prevents the intricate integration with other tools that pure Julia code allows. State-of-the-art alternatives with support for higher order are being developed in Julia – eg. [Tan et al. \(2026\)](#) for an alternative Taylor integrator implementation, or [Pérez et al. \(2026\)](#) for a Taylor integrator inspired by `taylor`.

H-4 A note on public outreach: my experience at the Cité des Sciences et de l’Industrie

H-4.1 Sensibilising, not teaching: the role of outreach

One of my deepest convictions is that the societal impact of the academic world isn’t restricted to pure research results. Researchers can play a crucial role in sensibilising the general public to science, its methods, its promises and its possibilities, beyond the ranks of university – through scientific outreach.

I had the chance of spending a hundred days, during my PhD, at the *Cité des Sciences et de l’Industrie*, a museum dedicated to science and its impact on society. During this time, I learned how to interact with an audience, designed and led animations in numerous formats: shows, debates, astronomy nights, live experiments, “pétanque” stands...

The objective never was to impart specific knowledge, but to spark an interest and a mindset of curiosity and wonder; to touch kids and parents alike, and to make them learn, or remember, the sense of marvel towards our world.

H-4.2 Experimenting new methods: a tabletop roleplaying game... about celestial mechanics

Amongst my best outreach memories lie the multiple sessions of an animation I designed, of a new kind: communicator colleagues asked me to create a tabletop roleplaying game,

for outreach! Even though tabletop roleplaying (abridged as TTRPG) usually is linked with dragon slaying, swords clashing and guns blazing, at its core it simply is people sitting around a table, forgetting their age and manners, and playing pretend for a couple of hours.

The idea was to create a one-shot, lasting one hour, accessible to anybody without prior knowledge of space or TTRPGs, around my thesis subject. Players are sent

In two years, together with a PhD colleague, we led 50 people into “Mission Lagrange: in the Sun’s shadow”, a one-year trip to the Sun-Earth L_3 point, investigating a strange signal coming from there; featuring players from 10 to 70 years, each session was a unique and memorable moment of laughs, excitements, discovery and tension – teaching people about space travel, its challenges and dangers, but also about letting go and having fun like kids again. The experience was unforgettable for them, but in a way even more for me, because sharing my passion for space and roleplaying games is miles better than experiencing it.

Appendix I

More on $\mathcal{VF}_{\mathcal{L}}$ in the planetary average circular co-orbital problem

I-1 Expression of the average circular Hamiltonian using elliptic integrals

Setting $p = \sin\left(\frac{\zeta_1}{2}\right)$ and $u = p^2 + s_0^2 - p^2 s_0^2$, the function U_{s_0} , defined in [E-1](#), can be rewritten as:

$$\begin{aligned}
 U_{s_0}(\zeta_1, \zeta_2) &= 2 - 2 \cos \zeta_1 + 2s_0^2(\cos \zeta_1 - \cos(\zeta_1 + 2\zeta_2)) \\
 &= 4 \left[p^2(1 - s_0^2) + s_0^2 \sin^2 \left(\frac{\zeta_1}{2} + \zeta_2 \right) \right] \\
 &= 4u \left[1 - u^{-1} s_0^2 \cos^2 \left(\frac{\zeta_1}{2} + \zeta_2 \right) \right] \\
 &= 4u \left[1 - u^{-1} s_0^2 \sin^2 \left(\frac{\zeta_1}{2} + \zeta_2 - \frac{\pi}{2} \right) \right]
 \end{aligned} \tag{57}$$

It follows that:

$$\begin{aligned}
 \frac{1}{\pi} \int_0^\pi \frac{d\zeta_2}{\sqrt{U_{s_0}(\zeta_1, \zeta_2)}} &= \frac{1}{2\pi\sqrt{u}} \int_{\frac{\zeta_1-\pi}{2}}^{\frac{\zeta_1+\pi}{2}} \frac{d\theta}{\sqrt{1 - u^{-1} s_0^2 \sin^2 \theta}} \\
 &= \frac{1}{2\pi\sqrt{u}} \left[\mathcal{F} \left(\frac{\zeta_1 + \pi}{2}, \frac{s_0}{\sqrt{u}} \right) - \mathcal{F} \left(\frac{\zeta_1 - \pi}{2}, \frac{s_0}{\sqrt{u}} \right) \right]
 \end{aligned} \tag{58}$$

where $\mathcal{F}$ is the incomplete elliptic integral of first kind defined by:

$$\mathcal{F}(x, y) = \int_0^x \frac{dz}{\sqrt{1 - y^2 \sin^2 z}} \tag{59}$$

I-2 Taylor expansion of the frequencies

In this appendix, the expansions in power series of $s_0 = \sin\left(\frac{J_0}{2}\right)$ for the libration frequencies ν_L and the precession frequency of the node $\dot{\Omega}_p$ are given along $\mathcal{VF}_{\mathcal{L}}$. These expansions are limited to degree 20, which provides a good approximation of the exact values for inclinations ranging from 0° to approximately 60° . According to [Equation 46](#), ν_L reads:

$$\nu_L = \sqrt{\varepsilon} n^* \tilde{\nu}_L \sqrt{\frac{m_1 + m_2}{m_0}} \tag{60}$$

with

$$\begin{aligned} \tilde{\nu}_L \sqrt{\frac{4}{27}} = & 1 - \frac{4}{3}s_0^2 + \frac{67}{36}s_0^4 - \frac{23}{27}s_0^6 - \frac{24721}{5184}s_0^8 + \frac{14653}{3888}s_0^{10} + \frac{1165859}{62208}s_0^{12} \\ & - \frac{914219}{46656}s_0^{14} - \frac{3195892471}{35831808}s_0^{16} + \frac{8594100529}{80621568}s_0^{18} + \frac{1796301904997}{3869835264}s_0^{20} \\ & + \mathcal{O}(s_0^{22}) \end{aligned} \quad (61)$$

From Equation 47, the precession frequency of the ascending node is given by

$$\dot{\Omega}_p = \varepsilon n^* \frac{m_1 + m_2}{m_0} \sqrt{1 - 4\gamma s_0^2} F_s, \quad \text{with} \quad \gamma = \frac{m_1 m_2}{(m_1 + m_2)^2} \quad (62)$$

and where F_s satisfies the expansion:

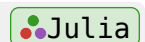
$$\begin{aligned} F_s = & \frac{3}{4}s_0^2 - \frac{3}{2}\gamma s_0^4 - \left(\frac{45}{32} + \frac{3}{2}\gamma^2\right)s_0^6 + \left(\frac{45}{16}\gamma - 3\gamma^3\right)s_0^8 \\ & + \left(\frac{645}{128} + \frac{45}{16}\gamma^2 - \frac{15}{2}\gamma^4\right)s_0^{10} + \left(-\frac{645}{64}\gamma + \frac{45}{8}\gamma^3 - 21\gamma^5\right)s_0^{12} \\ & + \left(-\frac{91325}{4096} - \frac{645}{64}\gamma^2 + \frac{225}{16}\gamma^4 - 63\gamma^6\right)s_0^{14} \\ & + \left(\frac{91325}{2048}\gamma - \frac{645}{32}\gamma^3 + \frac{315}{8}\gamma^5 - \gamma^7\right)s_0^{16} \\ & + \left(\frac{3602205}{32768} + \frac{91325}{2048}\gamma^2 - \frac{3225}{64}\gamma^4 + \frac{945}{8}\gamma^6 - \frac{1287}{2}\gamma^8\right)s_0^{18} \\ & + \left(-\frac{3602205}{16384}\gamma + \frac{91325}{1024}\gamma^3 - \frac{4515}{32}\gamma^5 + \frac{1485}{4}\gamma^7 - 214\gamma^9\right)s_0^{20} \\ & + \mathcal{O}(s_0^{22}) \end{aligned} \quad (63)$$

Appendix II

Benchmarks of TaylorInterface.jl

We compare TaylorInterface.jl, our Julia bridge to the taylor integrator, and TaylorIntegration.jl, a pure Julia implementation inspired by taylor. Integration is done on a 13th Gen Intel i7-1365U CPU, with 32 GB of RAM, on Ubuntu 25.10. Results were gathered with the @benchmark macro of BenchmarkTools.jl. We used Julia 1.12.16, TaylorInterface.jl v1.0.1 and TaylorIntegration.jl v0.18.15.

```
1 @benchmark flow($handler, $x_lag, $(50T_lag))
2 @benchmark taylorinteg(
3     $equations_motion_3BP!, $x_lag, 0, $(50T_lag),
4     25, 1e-16, $(equal_masses); maxsteps=1000)
5
6 @benchmark flow($handler, $x_eight, $(10T_eight))
7 @benchmark taylorinteg(
8     equations_motion_3BP!, x_eight, 0, 10T_eight,
9     25, 1e-16, equal_masses; maxsteps=1000)
```



Listing 8: Benchmark commands for results in Table 2. Value interpolation (with the \$ sign) is used to avoid evaluating the parameters for each sample, biasing the results.

	Lagrange over 50 periods	Eight over 10 periods
TaylorInterface.jl	1.518 ms $\pm$ 55.257 μ s (3294 samples)	3.749 ms $\pm$ 100.181 μ s (1334 samples)
TaylorIntegration.jl	116.716 ms $\pm$ 156.535 ms (43 samples)	281.723 ms $\pm$ 231.550 ms (18 samples)
Ratio of mean times	101	75

Table 2: Integration benchmarks.

The @taylorize macro of TaylorIntegration.jl could significantly reduce this difference, by parsing the function with the equations of motion beforehand, but it is still in an experimental state, and as of 11/07/2026 (Julia 1.12.6, TaylorIntegration.jl v0.18.15), returns an error¹⁷. Tinkering to make it work by simplifying the expression graph is probably possible, albeit cumbersome for high dimension values.

¹⁷LoadError: MethodError: no method matching parse! (::Espresso.ExGraph, ::Nothing)

Appendix III

$\mathcal{VF}_{\mathcal{L}}$ at planetary masses

III-1 Lagrange equilateral relative equilibrium in Hill's coordinates

Although neither w_j nor Ω_j are geometrically defined in the plane, Lagrange equilibria remain well defined in Hill coordinates. According to [Equation 18](#), all coordinates are constant except for true anomalies, which satisfy the differential equation

$$\dot{w}_j = \omega \left(a + b \sin \left(2w_j + (-1)^j \frac{\pi}{3} \right) + c \cos \left(2w_j + (-1)^j \frac{\pi}{3} \right) \right) \quad (64)$$

where the coefficients a , b and c are given by:

$$\begin{aligned} a &= (4m_0^4 + 6\sigma_0 m_0^3 + (4\sigma_0^2 + p_0)m_0^2 + 5p_0\sigma_0 m_0 + 2p_0^2)d^{-1}, \\ b &= -\sqrt{3}p(m_1 - m_2)m_0 d^{-1}, \\ c &= -p(3\sigma_0 m_0 + 4m_0^2 + 2p_0)d^{-1}, \\ d &= m_0(2m_0 + m_1)(2m_0 + m_2)(m_0 + m_1 + m_2), \\ p &= m_0 m_1 + m_0 m_2 + m_1 m_2, \\ p_0 &= m_1 m_2, \quad \sigma_0 = m_1 + m_2 \end{aligned} \quad (65)$$

where a straightforward calculation leads the equality $a^2 - b^2 - c^2 = 1$. The general solutions to this differential equation can be expressed as: $w_j(t) = w_j^0 + \omega t + F(\omega t)$, where F is 2π -periodic.

The dominant term of the 3 coefficients is given by:

$$\begin{aligned} a &= 1 + \varepsilon^2 a_2 + \mathcal{O}(\varepsilon^3) \quad \text{with} \quad a_2 = \frac{(m_1 + m_2)^2}{2m_0^2}, \\ b &= \varepsilon^2 b_2 + \mathcal{O}(\varepsilon^3) \quad \text{with} \quad b_2 = \frac{\sqrt{3}(m_2^2 - m_1^2)}{4m_0^2}, \\ c &= \varepsilon c_1 + \varepsilon^2 c_2 + \mathcal{O}(\varepsilon^3) \quad \text{with} \\ c_1 &= -\frac{(m_1 + m_2)}{m_0} \quad \text{and} \quad c_2 = \frac{(3m_1^2 + 2m_1 m_2 + 3m_2^2)}{4m_0^2} \end{aligned} \quad (66)$$

By neglecting terms in ε^3 and more, the solution of [Equation 64](#) reads:

$$w_j(t) = w_j^0 + \omega t + \varepsilon \frac{c_1}{2} \sin \theta_j + \frac{\varepsilon^2}{8} (4c_2 \sin \theta_j - 4b_2 \cos \theta + c_1^2 \sin 2\theta_j) + \mathcal{O}(\varepsilon^2) \quad (67)$$

where $\theta_j = 2\omega t + (-1)^j \frac{\pi}{3}$. As mentioned above, in the planar case, the angular position of the planets is identified by true longitude $l_j = w_j + \Omega_j$. Since Lagrange's relative equilibrium is a periodic solution with frequency ω , the longitudes of the nodes Ω_j remain defined and are equal to: $\Omega_j(t) = \Omega_j^0 - F(\omega t)$. These latter quantities therefore undergo only ω -periodic oscillations.

III-2 Difference between draconic mean anomalies in the average problem and draconic true anomalies in the full problem

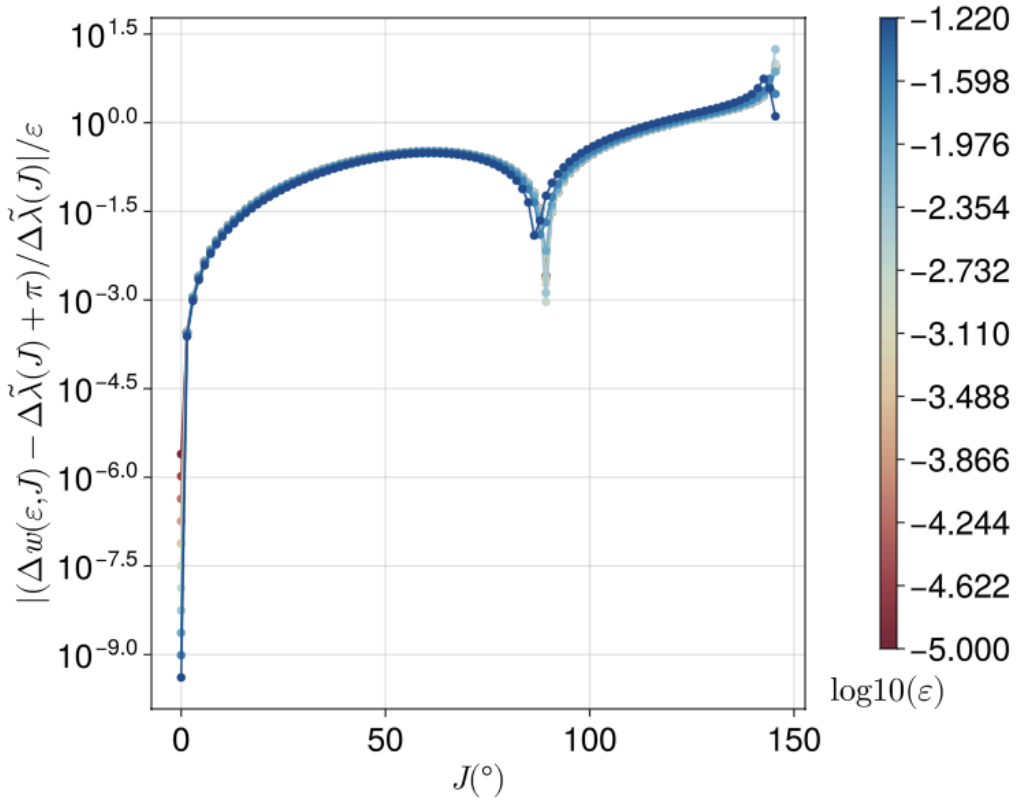


Figure 34: Relative variation of $\Delta\lambda$ and Δw along $\mathcal{VF}_\mathcal{L}$ for various masses, divided by the mass parameter ε . The addition of π is due to the offset of the $\tilde{\lambda}_j$. The drop near $J = 100^\circ$ is due to the change of sign of the difference $\Delta w - \Delta\lambda$.

[E-1](#) uses draconic mean anomalies, noted $\tilde{\lambda}$ in the first-order circular average problem, whereas [E-2](#) uses draconic true anomalies in the full problem. [Figure 34](#) shows the relative difference between $\Delta\tilde{\lambda}$ and Δw , used as a family coordinate in figures (e.g. [Figure 17](#) and [Figure 20](#)), along $\mathcal{VF}_\mathcal{L}$, for mass parameters ε ranging from 10^{-5} to 0.06. As [Equation \[eq:equil_circ\]](#) is independent of the masses, only Δw is changing. The curves for the various masses are closely overlapping when divided by ε ; thus, the relative difference is mostly proportional to ε , with a coefficient less than 1, up to high mutual inclinations (around $J = 120^\circ$). This gives an indication of the validity range of [Equation 42](#).

III-3 $\mathcal{VF}_\mathcal{L}$ for unequal planetary masses

We find $\mathcal{VF}_\mathcal{L}$ for unequal planetary masses, with a ratio of small masses going up to 10^4 – keeping the sum of the two small masses at 10^{-3} , and the sum of all three masses at 1. Results are shown in Figure 35 for the variation in Δw and the position of the stability change, showing that qualitatively nothing is significantly changed, apart from the symmetry break between the two small masses. In both cases, results seem to converge to a limit, which one could imagine to be the restricted case, although as mentioned in B-2.4, we cannot attain it with the Jacobi reduction.

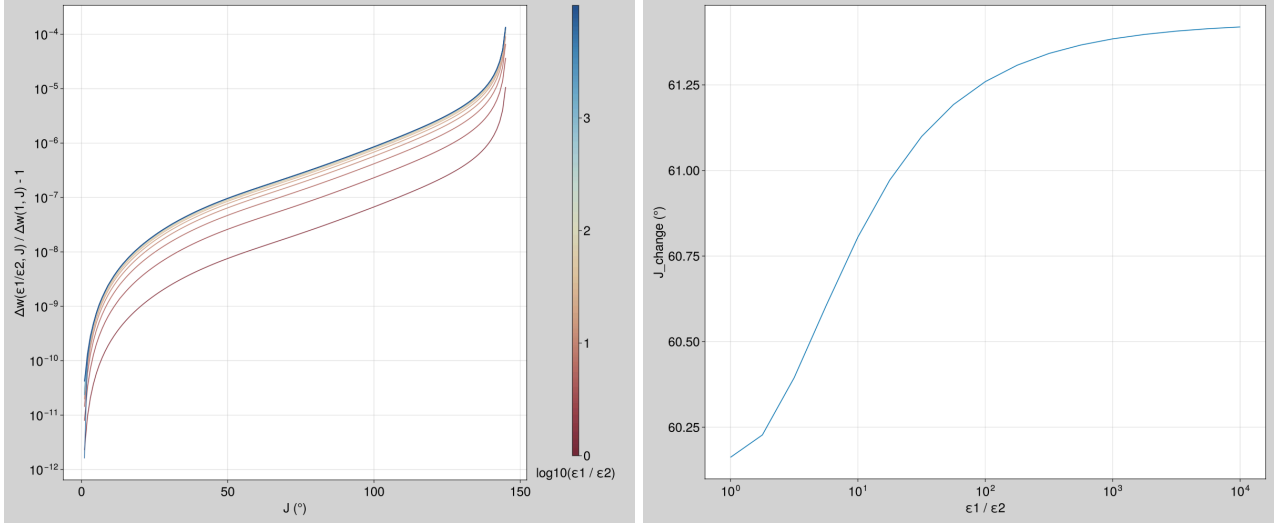


Figure 35: Left: Relative difference of Δw between the unequal and equal masses case, for various aspect ratios, along $\mathcal{VF}_\mathcal{L}$. Right: Value of the mutual inclination for the stability change in $\mathcal{VF}_\mathcal{L}$ for different mass ratios. In both cases, the sum of the two small masses is kept at 10^{-3} , and the sum of all three masses at 1.

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RÉSUMÉ

La mécanique céleste décrit les trajectoires dans l'espace des objets célestes, des satellites artificiels et astéroïdes jusqu'aux galaxies. Dans beaucoup de cas, à courte échelle de temps les trajectoires peuvent être approximées par des orbites quasi-périodiques, justifiant un intérêt particulier pour la dynamique récurrente. Dans cette thèse, nous présentons des nouveaux outils numériques pour étudier la dynamique récurrente :

- `TaylorInterface.jl`, une librairie open-source pour utiliser l'intégrateur numérique `taylor` depuis le langage de programmation Julia;
- un algorithme de recherche d'orbites périodiques levant les dégénérescences locales;
- des outils d'étude de la stabilité et des bifurcations des familles d'orbites quasi-périodiques.

En utilisant ces méthodes, nous trouvons la famille verticale d'orbites quasi-périodiques émergeant de l'équilibre relatif de Lagrange dans le problème à trois corps, que nous appelons $\mathcal{VF}_L$. Nous montrons que cette famille conserve une partie stable au-delà de la limite de Gascheau-Routh. Nous apportons des preuves numériques pour la conjecture généralisée de Marchal sur le lien entre $\mathcal{VF}_L$ et la famille P_{12} à masses égales, en montrant qu'elles bifurquent à une inclination mutuelle d'environ 91° . Nous trouvons deux autres bifurcations, non étudiées jusqu'à présent, le long de P_{12} .

MOTS-CLÉS

Systèmes dynamiques, Dynamique des systèmes planétaires, Mécanique céleste, Physique numérique

ABSTRACT

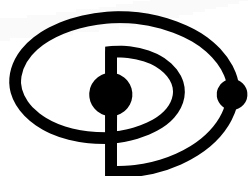
Celestial mechanics describe the trajectories in space of celestial objects, from artificial satellites and asteroids to galaxies. Recurring dynamics are of special interest, as a significant amount of body trajectories can be approximated at short time scales by slowly evolving quasi-periodic orbits. In this thesis, we present new numerical tools to study recurring dynamics:

- `TaylorInterface.jl`, a package to call the `taylor` numerical integrator from the Julia programming language, bundled in an open-source package;
- a search algorithm to find periodic orbits and lift local degeneracies through sections.
- tools to study the stability, and bifurcations of families of periodic orbits.

We apply these tools to find the vertical family of quasi-periodic orbits emerging from the Lagrange relative equilibrium of the three-body problem, we call $\mathcal{VF}_L$. We show this family retains a stable part at medium inclination beyond Gascheau-Routh's mass limit. We shine light upon Marchal's generalised conjecture linking $\mathcal{VF}_L$ to the P_{12} family in the equal masses case, showing they bifurcate at a mutual inclination near 91° . We find two other bifurcations along P_{12} , currently unstudied.

KEYWORDS

Dynamical systems, Dynamics of planetary systems, Celestial mechanics, Numerical physics



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